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How to Memorize Cards, GUIDs, Hashes, Public Keys, and More — Revision 19

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1 Change log

This is very much a work-in-progress and nowhere near clean enough for publication and distribution. However, it is useful, and it's FREE. Use at your own risk. Better yet, submit pull requests that fix problems or TODO items.

- Revision 17
  - Introduce PAO in Appendix 15.10; Minor maintenance
- Revision 16
  - Work a little on pegs for the 400's
- Revision 15
  - Add caching to the org-babel code.
  - new pegs to 1000
- Revision 12
  - Simplify the 3-digit pegs; adding new technique.
- Revision 11
  - Add section on people's names, a new application of the technique, never before published to our knowledge.
- Revision 10
  - All Pegs now use babomorph for typesetting.
- Revision 7
  - non-Alphabetic ASCII
- Revision 6
  - Automate production of tables with Clojure program.
- Revision 5
  - Include Tarot images.
- Revision 4
  - Include remarks about passwords and mental hash functions from JDC.
- Revision 3
  - Include some sound-alikes and peg candidates from JDC, plus a hint that science fiction is a fount of sound-alikes.
  - Begin promoting the first row in each table to "preferred sound alikes."

2 Prologue

Brian is the curator and primary author of this document. He wrote the first version before enlisting John and Ondrej to contribute. Any use of the first person "I," "we," and so on, in the document is historical and refers to Brian. Brian is responsible for all mistakes.

2.1 How to Modify this Document

The PDF file is generated from `.org` source via Emacs. The org source contains blocks of Clojure code, specifically calls of the `babomorph` function, which extracts strings from the following Clojure file:

`src/how-to-memorize/core.clj`.

If you want to change the peg words, the ordering of the words, the capitalization of the words, the formatting of the tables in the PDF document, or anything else about this code, modify that `core.clj` file, `cider jack-in`, and evaluate the code via `Ctrl-C Ctrl-C` while Emacs is visiting the file `core.clj`. If you want to change prose or other aspects of the document, modify the `.org` file and you won't have to bother with `core.clj`.

The results of running all the Clojure code is cached in the `.org` file. If you change the Clojure code, evaluate `core.clj` via `Ctrl-C Ctrl-C`, then manually run affected blocks in the `.org` file by removing the `:results:` blocks and keyboarding `Ctrl-C Ctrl-C` (same command, just while visiting the `.org` file) with your cursor somewhere inside the `begin_src` or `babomorph` block.

For example, if you change the table for 00 through 09 in the `core.clj` file, `Ctrl-C Ctrl-C` in `core.clj`, then remove the results drawer and its hash on that table here in the `.org` file—it looks like this:

```
#+RESULTS[7ecf5cbc3481e60410646e89bc93d7140a6cad17]:
:results:
|----+----------+---------+----------+----------+--------...
| 00 | SAUCE    | OASIS   | icehouse | sissy    | sass   ...
|    | assess   | assize  | houses   | sassy    | say-so ...
|    | seize    | size    |          |          |        ...
...
:end:
```

Delete all the results through the `:end:`.

Then keyboard `Ctrl-C Ctrl-C` on the babomorph block, which looks like this:

```
#+begin_src clojure :cache yes :exports results :results ...
  (babomorph 0 10)
#+end_src
```

and you'll see the table regenerated.

Note that the `babomorph` commands are off-by-one-hundred, so `(babomorph`

`510 520)` builds the table for pegs 410 through 419. This is so we can accommodate pegs that begin with zero.

To produce the PDF, you need a full LaTeXinstallation, Emacs 29.1 or later, Clojure 1.10.3 or better (see `project.clj`), and CIDER installed. There may be other ways to build the PDF, but that's what I use. Manually `cider-jack in` to a CIDER session to warm up the Clojure environment. Run

```
org-latex-export-to-pdf
```

while visiting the `.org` file in an Emacs buffer to produce a `.tex` file and possibly a `.pdf` file.

Something went wrong in my recent `tex` integration, so I run

```
pdflatex -shell-escape how-to-memorize-hex-r36-r64.tex
```

twice, manually in a side terminal, after running

```
org-latex-export-to-pdf
```

but that's not a big roadblock. Your installation's of LaTeXand Emacs may go more smoothly.

The `.org` file contains lots of stuff that isn't output to the PDF file. That stuff represents either work-in-progress or variations on the main theme, variations that aren't good enough to commit to print.

3 Introduction

Here is a word that you may never have encountered before:
Definition 1. A *phantasmagoria* is

1. a vivid mental image like Superman or Bilbo Baggins

2. a mental movie clip like Rhett Butler carrying Scarlett O'Hara up the staircase, two steps at a time; or Legolas the elf in "Lord of the Rings" firing off arrows

3. a sensation like the taste of cherry pie with vanilla ice cream or the smell of burnt motor oil in an old engine block

4. a musical riff like Clapton's "Down to the Crossroads" or Bach's "Toccata and Fugue in D Minor"

5. anything else that's vivid and easy to recall without effort

The premise of everything in this document is
Premise 1. *Almost everyone can recall phantasmagorias without effort. Let's trick our minds into memorizing numerical or symbolic data via permanent associations between numbers and phantasmagorias.* ■

We'll find that we can memorize long sequences of numbers, long lists of arbitrary items, whole texts verbatim, packs of shuffled cards, etc., then recall them in any order by making up phantasmagorias, or *phants* for short. It's fun, easy, and empowering. It requires some effort up front, but the effort is incremental. It takes less than 30 minutes for anyone to learn how to memorize lists of up to 10 items effortlessly and without writing anything down. In a couple of weeks, anyone can learn to memorize lists of up to 100 items effortlessly. That's good enough for a lifetime of practical application. With a twist, PAO (Appendix 15.10), our list of 100 pegs becomes 300, The seriously committed can spend a few months learning to memorize up to 1110 items effortlessly. All these methods and tricks are based on one idea: the *peg*.

4 The Peg System, Contrasted

This document is primarily about the *Mnemonic-Major* or *Peg* system,[1, 2] which makes *direct* and permanent associations between numbers and permanent phantasmagorias — the pegs — then temporary phantasmagorias for the memory task at hand. For example, in my personal version of the Major system, 1 is always and forever the peg TEA, 2 is NOAH, 3 is ME, and so on. To remember the temporary shopping list "broccoli, grapes, lettuce," I picture a big, Disneyland teacup overflowing with broccoli, Noah squishing grapes into his beard, and me in the mirror with a head of lettuce instead of my actual head, and so on. When I'm done with the shopping list, I mentally discard it, but not the pegs. I reuse the pegs later for the next task. The pegs are forever.

The famous Roman-Room system[3], also called *Memory Palace* or *Method of Loci*, assigns temporary phantasmagorias for the memory task at hand to permanent mental images of favorite places that you normally visit in sequence, such as bedroom, bathroom, kitchen. When you want to remember the shopping list "broccoli, grapes, lettuce," picture the broccoli in the bedroom, grapes in the bathroom, lettuce in the kitchen, and so on. If you have a big sequence of favorite places or *loci*, as they're called, you can memorize big lists this way.

For me, the pegs act just like a Memory Palace. I don't really have a sequence of places that I visit frequently whose images I can recall without effort. I *have* practiced the pegs so much, however, that I can recall them in sequence or at random without effort. The pegs *are* my memory palace.

That last point highlights the fact that these memory systems are highly personal. I will show you the foundational pegs that thousands or millions of people share, but you will be encouraged to personalize it as much as necessary to make it strong and instantaneous *for you*.

The equally famous Person-Action-Object (PAO) system [4] amplifies my realization of the Major system, as explained in Appendix 15.10.

My primary sources for the peg system are Krill[5] and Furst.[6]. Krill calls his system *Pseudonumerologic-VIP*, which is a clever way of remembering the fundamental pegs 0123456789 by sound: `SDNMRLJKVP`.

These methods are based on tables of

pegs permanent, immutable mental images in one-to-one correspondence with numbers. For instance, I pegged the number 68 to CHEF. Whenever I see a chef or think of one, I smell the kitchen and hear the sounds of boiling and sizzling phantasmagorically, and instantly think of the number 68. Likewise, whenever I see the number 68, I instantly visualize a chef, with his signature

[1] `https://forum.artofmemory.com/t/the-mnemonic-major-system-and-gregg-shorthand-have-the-same-underl`
`26339`

[2] `https://en.wikipedia.org/wiki/Mnemonic_major_system`

[3] `https://artofmemory.com/blog/roman-rooms/`

[4] `https://artofmemory.com/blog/pao-system/`

[5] `https://folk.ntnu.no/krill/pseudonumerology.htm`

[6] `https://a.co/d/fv1uRmv/`

chef's hat and colored necktie, turning pancakes over. That's 1-to-1 correspondence.

sound-alikes fluid, mutable lists of mental-images that are similar in their consonant sounds to the pegs and in many-to-one correspondence to the numbers. 68's CHEF sounds like CHIEF, SHEAF, JAVA, CHAFF, amongst others. While 68 always means CHEF, recalled instantly, a chef might not fit well into an ongoing phantasmagoria. For instance, if we're memorizing 276872, the pegs are NECK, CHEF, and CAN. That sequence does not invoke an obvious story or mental movie, so we start looking for sound-alikes. How about INCA CHIEF COIN? Just picture a coin with an Inca chief embossed on it, and we're done. It took a few seconds to derive INCA CHIEF COIN from NECK CHEF CAN, but the other direction is almost instantaneous. Sound-alikes are many-to-one because there are many words that map to each number. Typically, you make them up on-the-fly to devise a mental movie.

The big point is this: you can instantly go from a number to a peg, and you can, almost instantly, go from a sound-alike to a number. It might take a little while to go from a peg to a sound-alike, but if doing so helps memorize a big list, then it's worth the effort.

With PAO, the sequence 276872 becomes, more quickly, NEFERTIRI TURNS a CIGAR, easy to visualize and hard to forget. PAO helps you memorize long strings of numbers six at a time without devising sound-alikes. Both methods work well, and I encourage you to internalize them both, but in time. Master the pegs. PAO is built on them.

To memorize data other than decimal numbers, like names, hexadecimals, base-36 numbers, stories and speeches, we will develop new sub-systems based on the pegs and on PAO. That is the meat of this book. The key to *everything*, however, is to master the pegs. Make them one of your memory palaces. For me, they are my only memory palace. Some champion memorizers have many memory palaces. I'm not in their league, but my system works amazingly well for memorizing all kinds of data.

5 Basic Pegs

The pegs for the decimal digits are the foundation for the entire system. All longer pegs, like CHEF for 68, are built up from these. Here are my basic pegs, with some mnemonic hints:

digit	peg	mnemonic	sound-alike rules
0	saw	circular saw, sawed log; round like 0	s, z
1	tea	giant Disneyland teacup; T looks like 1	d, th, dh (them)
2	Noah	… and the Ark; N has 2 legs (2 = ng for Krill only, nk is always 27)	ng (Krill only — "Krill" has 2 "l"s)
3	me	… in the mirror tying a bow tie; M has 3 legs	
4	ray	laser pointer, flashlight, light saber no simple mnemonic to 4	
5	law	5 fingers as the cop holds up his hand the hand is L-shaped as you look at it	
6	jaw	6 looks like Lurch's lantern jaw	sh, ch, zh (pleasure)
7	key	7 looks like an Allen key (7 = ng for Furst and me, nk is always 27)	g, ng (Furst only)
8	fee	8 looks like a dollar sign	f, v
9	pea	The Princess and the Pea: 9 mattresses	p, b

Memorize this table, or one personalized to you, by whatever means, by brute force if necessary. The rest of the system relies on it. If you don't like these specific pegs, pick some others out of the rows for 0 through 9 in Section 16. Pick your own peg for each digit, make your own table, memorize it, and stick with it forever. NEVER, EVER CHANGE ANYTHING IN IT. Brian did that once more than 50 years ago, and continues to regret it. This table must become an internal, effortless vocabulary like your native language.

Make up pegs and sound-alikes for longer numbers by combining consonant sounds. For example, here is the peg and sound-alikes for 32:

32	MAN	MOON	mane	hymen	mine	minnow
	amen	ammonia	human	main	mana	mania
	manna	many	mean	mien	moan	money
	omen	woman	yeoman			

Notes on such tables:

- the peg is in all-caps and in first position; other terms in all-caps are recommended alternative pegs; use them if they work better for you
- the first line contains preferred sound-alikes in preference order
- the remaining sound-alikes appear in alphabetical order

MAN is the peg: it is *equal* to 32 in your mind. I visualize Superman flying out of the phone booth, impossible to forget for me.

5.1 Old 3-Digit System

We go up to three digits in this document. That means that every number from 00 through 099 and 0 through 1000 has a peg and a list of sound-alikes. That's 1100 pegs by my count.

It is a very serious commitment to memorize 1100 pegs. My dad had it, and John has done it. It will take months and it must be refreshed frequently. I think it's more of an intellectual challenge than a utility for memory, especially compared against the PAO system (Appendix 15.10), which lets me easily and quickly come up with phantasmagorias for any triple or sextuple of digits base just on my 110 pegs for 0 through 9 and 00 through 99). However, I present the 1100-peg 3-digit system for the adventurous mental mountaineers out there.

The 3-digit system comes in an old form and a new form. The old form is based on Krill and Furst, like the 2-digit system. The new system is from John and is presented below.

In the old system, the peg and sound-alikes for the 3-digit number, xyz, are formed from 3 hard consonant sound alikes, a for x, b for y, and c for z. For instance, here is the list for $xyz = 352$:

352	MELON	Himalayans	hemline	semolina	milliner	homeland
	malign	melancholy	millennial	millennium	million	

When making up pegs and sound-alikes, weak consonant sounds like H, WH, Y, and W are free. For example, ignore the H in HEMLINE and think of HEMLINE as MLN for 352.

Vowel sounds are also free — use as many or as few as you like.

From this point on, when we write "sound" or "sounds," we mean "hard consonant sound" or "hard consonant sounds."

Ignore trailing sounds beyond the first three: MILLENNIAL and MILLENIUM both encode 352.

5.1.1 Leading S and Z sounds

Furst's tables are ambiguous for 3-digit pegs with leading 0. For example, in his original, SEMOLINA appears under both 352 and 035. We can't have ambiguity, however, so in our tables, a leading S or Z sound encodes a number that begins with a 0.

Parenthetically, there is an alternative: *if the number of sounds is four or more, then ignore the leading S or Z.* SEMOLINA would be 352 under this rule. We found, however, that this is too cumbersome in practice. So we back off of that and commit to fixing Furst's tables.

TODO it is an arduous process to correct Furst's tables to move all words with leading S or Z sounds into the tables for numbers with leading zeros. As of Nov 2025, it's done through 360. I (Brian) will complete the process as time permits. For

now, be tolerant and interpret any words with leading S or Z sounds, perhaps with leading soft constants and vowels, as belonging to numbers with leading zeros.

5.2 TODO *Composite Pegs*: New 3-Digit System

(John's idea)

It can be challenging to come up with 1100 3-digit pegs out of the blue. Perhaps we could get to 1100 more easily by combining 110 prefix pegs with 10 fixed, suffixes. For example:

740 Creamy Strawberry Pie
741 Creamy Treacle
742 Creamy Nougat
743 Creamy Marshmallow
744 Creamy Raspberry Pie
745 Creamy Lemon Meringue Pie
746 Creamy Cherry Pie
747 Creamy Coconut Pie
748 Creamy Vanilla Pie
749 Creamy Blackberry Pie

and

840 Fresh-baked Strawberry Pie
841 Fresh-baked Treacle
842 Fresh-baked Nougat
843 Fresh-baked Marshmallow
844 Fresh-baked Raspberry Pie
845 Fresh-baked Lemon Meringue Pie
846 Fresh-baked Cherry Pie
847 Fresh-baked Coconut Pie
848 Fresh-baked Vanilla Pie
849 Fresh-baked Blackberry Pie

We only need the first 64 for radices.

010 A Sodden Strawberry Pie
011 B Sodden Treacle
012 C Sodden Nougat
013 D Sodden Marshmallow
014 E Sodden Raspberry Pie
015 F Sodden Lemon Meringue Pie
016 G Sodden Cherry Pie
017 H Sodden Coconut Pie
018 I Sodden Vanilla Pie
019 J Sodden Blackberry Pie

020	K	Sun-baked Strawberry Pie
021	L	Sun-baked Treacle
022	M	Sun-baked Nougat
023	N	Sun-baked Marshmallow
024	O	Sun-baked Raspberry Pie
025	P	Sun-baked Lemon Meringue Pie
026	Q	Sun-baked Cherry Pie
027	R	Sun-baked Coconut Pie
028	S	Sun-baked Vanilla Pie
029	Th	Sun-baked Blackberry Pie
030	U	Semi-soft Strawberry Pie
031	V	Semi-soft Treacle
032	W	Semi-soft Nougat
033	X	Semi-soft Marshmallow
034	Y	Semi-soft Raspberry Pie
035	Z	Semi-soft Lemon Meringue Pie
036	a	Semi-soft Cherry Pie
037	b	Semi-soft Coconut Pie
038	c	Semi-soft Vanilla Pie
039	dh	Semi-soft Blackberry Pie
040	e	Sorbet and Strawberry Pie
041	f	Sorbet and Treacle
042	g	Sorbet and Nougat
043	h	Sorbet and Marshmallow
044	i	Sorbet and Raspberry Pie
045	j	Sorbet and Lemon Meringue Pie
046	k	Sorbet and Cherry Pie
047	l	Sorbet and Coconut Pie
048	m	Sorbet and Vanilla Pie
049	nh	Sorbet and Blackberry Pie
050	o	Salad and Strawberry Pie
051	p	Salad and Treacle
052	q	Salad and Nougat
053	r	Salad and Marshmallow
054	s	Salad and Raspberry Pie
055	t	Salad and Lemon Meringue Pie
056	u	Salad and Cherry Pie
057	v	Salad and Coconut Pie
058	w	Salad and Vanilla Pie
059	xh	Salad and Blackberry Pie
060	y	Sushi and Strawberry Pie
061	z	Sushi and Treacle
062	+	Sushi and Nougat
063	,/	Sushi and Marshmallow

Here are some candidates for a hundred prefixes.

000 Saucy Strawberry Pie
010 Sodden Strawberry Pie
020 Sun-baked Strawberry Pie
030 Semi-soft Strawberry Pie
040 Sorbet and Strawberry Pie
050 Salad and Strawberry Pie
060 Sushi and Strawberry Pie
070 Saccharine and Strawberry Pie
080 Savory Strawberry Pie
090 Soup and Strawberry Pie

100 Tossed Strawberry Pie
110 Toothy Strawberry Pie
120 Thin-crusted Strawberry Pie
130 Thumbnail Strawberry Pie
140 Dairy-based Strawberry Pie
150 Tallow-free Strawberry Pie
160 Dutch-chocolate Strawberry Pie
170 Thick-crusted Strawberry Pie
180 Taffy-iced Strawberry Pie
190 Deep-dish Strawberry Pie

200 Anice-laced Strawberry Pie
210 Nut-covered Strawberry Pie
220 Non-fattening Strawberry Pie
230 Anemic Strawberry Pie
240 Near-perfect Strawberry Pie
250 Newly-eaten Strawberry Pie
260 Unshared Strawberry Pie
270 Uncanned Strawberry Pie
280 Unflavored Strawberry Pie
290 Unboxed Strawberry Pie

TODO: finish

5.3 Brian's 2-Digit-Plus List

Brian's basic 2-digit peg list appears below, extended from 99 to 127 for ASCII and with nine extra pegs for court cards (Section 15.8). Brian knows this table by heart.

00	sauce	10	toes	20	nose	30	mass
01	sod	11	tot	21	net	31	mat
02	sun	12	tan	22	nun	32	man
03	seam	13	tam	23	name	33	mama
04	soar	14	tar	24	Nero	34	mare
05	sail	15	tail	25	nail	35	mail
06	sash	16	tissue	26	niche	36	match
07	sack	17	tack	27	neck	37	mike
08	safe	18	taffy	28	navy	38	muff
09	sap	19	tap	29	nap	39	map
40	race	50	lace	60	chess	70	case
41	rat	51	lot	61	jet	71	cat
42	rain	52	lane	62	chain	72	can
43	ram	53	lime	63	chime	73	cam
44	rear	54	lair	64	chair	74	car
45	rail	55	lily	65	jail	75	coal
46	rash	56	lash	66	judge	76	cage
47	rack	57	lake	67	chalk	77	cake
48	reef	58	leaf	68	chef	78	cuff
49	rope	59	lap	69	chip	79	cap
80	face	90	base	100	thesis	110	tights
81	fat	91	bat	101	toast	111	teetotaler
82	fan	92	bean	102	dozen	112	titan
83	fame	93	beam	103	twosome	113	totem
84	fare	94	bar	104	dowser	114	Tatar
85	fall	95	ball	105	diesel	115	title
86	fish	96	badge	106	wood-sage	116	deathwatch
87	fig	97	back	107	tusk	117	hot dog
88	fife	98	beef	108	adhesive	118	auto-da-fe
89	fob	99	pipe	109	teaspoon	119	teddy bear
120	dance	911	potato	411	rat-a-tat	711	cadet
121	donut	912	baton	412	rattan	712	katana
122	Athenian	913	podium	413	radium	713	catamaran
123	tenement						
124	tenor		spades		hearts		clubs
125	toenail		court		court		court
126	tension		cards		cards		cards
127	tank						

5.3.1 Re-Deriving a Forgotten Peg

In case you find yourself forgetting a peg, there is an easy way to re-derive it: go down the vowel sounds, short and long, in order — a, e, i, o, u — and try sandwiching each vowel between the bracketing consonants — *nan*, *nane*, *nen*, *nene*, *nin*, *nine*, *non*, *known*, *nun*. Eventually, you'll hit the right one. For the ambiguous consonant clusters like 6 = J, SH, CH, you may have to try a few combinations of consonants as well.

6 Credit-Card Numbers

Consider the following, fake VISA number from

`https://www.getcreditcardnumbers.com`

`4716 5752 8659 3845`

From the peg list, we get

4716	RACK TISSUE
5752	LAKE LANE
8659	FISH LAP
3845	MUFF RAIL

There isn't an obvious narrative, here, but we get better by starting with "fish" (picturesque and smelly) and working sound-alikes backwards:

4716	RACK dash	Take a fishing rod off the RACK and DASH
5752	LAKE LANE	down to the LAKE via our favorite LANE.
8659	FISH leap	The FISH are LEAPing today, so we
3845	move reel	MOVE the REEL slowly (on the fishing rod).

I can recall this credit-card number weeks after devising the story, without much refreshing in between. Devising it took less than a minute. I memorized it while devising it without extra effort because I know the 2-digit peg table by heart.

In PAO, I'd break it up like this

471657	My friend YAIR DABS a lego-model ISLAND with a tissue
528659	6 OF CUPS (tarot) DISTRIBUTES LAPs (??)
3845	BARBRA STREISAND SIGNALS detonation of an H-Bomb

It's debatable whether that's easier to remember than the fish story, but it doesn't require thinking. PAO doesn't use sound-alikes.

7 Hexadecimal: GUID & SHA-1

There are several ways we might approach memorizing numbers that include alphabetical glyphs or letters, such as numbers in base 16, 36, and 64. One way would be to mentally picture things that go along with the letters, such as an A-frame house like a ski chalet for A, and a honey bee for B, and so on. Feel free to proceed this way should you like it. I recommend, however, that we use the existing 2-digit codes for the ASCII values of the letters. For example, capital A has ASCII code 65, with peg JAIL, and little A has ASCII code 97, with peg as BACK. We get two phantasmagorias: a jail cell and a back, say the back of a prison guard. We'll need the capital-uncial distinction for radix-64 but not for hex or radix-36.

There is a price to pay for this choice: we can no longer encode 65 as JAIL or 97 as BACK, depending on the radix we're working in. We'll have to do something else for them. The examples will illustrate the technique.

For mapping *hexits* (hexadecimal digits), extend the table with the ASCII codes for A-F.

hex	NATO	ASCII		PHANTASMAGORIA
A	ALPHA	JAIL	65	the alpha-male lion is in jail
B	BAKER	JUDGE	66	passing judgment on the lion
C	CHARLIE	CHALK	67	hopscotch on the sidewalk
D	DELTA	CHEF	68	day job of a PhD-thesis student
E	ECHO	CHIP	69	a poker chip in a policeman's (LAW:5) hand
F	FOXTROT	CASE	70	filled with a dozen pink roses

and their sound-alikes:

65	JAIL	CHILI	jelly	jewel	shell	shawl
A	Chile	July	agile	chill	jolly	jowl
	shall	shallow	shill	shoal	shyly	usual
66	JUDGE	HASHISH	juju	Jewish	cha-cha	choochoo
B	chichi	chowchow	jaygee	josh	shoji	shush
	wishy-washy					
67	CHALK	JUG	jack	sheikh	chick	cheek
C	Django	aging	check	chic	choke	chuck
	chug	etching	itching	jingo	jock	jockey
	joke	shack	shag	shake	shako	shock
	shuck	washing	watching	wishing		
68	CHEF	CHEVY	chief	sheaf	chaff	chive
D	Java	Jehovah	achieve	chafe	jayvee	jiffy
	jive	shave	shove	wash-off		
69	CHIP	JEEP	jib	sheep	ship	Job
E	GP	chap	chapeau	cheap	chop	chubby
	gyp	jab	jibe	job	shabby	shape
	ship	shop				
70	CASE	GOOSE	gas	kiss	ox	eggs
F	accuse	acquiesce	ax	cause	causeway	chaos
	cozy	cuss	gauze	gaze	guess	guise
	guys	hex	hoax	quasi	wax	

Remember that 7 can be *ng* in Furst's system, the one we're following, explaining why we get so many . . . *ing* words for 67.

As noted, we lose the six pegs for 65 through 70. You can't memorize a 65 in the text by picturing JAIL because JAIL means A in hex. We need some other encoding or grouping of the digits. For example, seeing a `265A`, we might think NICHE(26), LAW(5), JAIL(A), or ANGEL(265), JAIL(A).

7.1 Example GUID:

Hex is only a little harder than decimal. Consider this random GUID generated by Java:

`5D00E7AA-C15C-44A5-93FA-A4DE6750B28A`

Often, in real-world application, you don't need to memorize the whole thing, but we'll do that for practice. Start with the pegs:

`5D00 E7AA`	LAW CHEF SAUCE CHIP KEY JAIL JAIL
`C15C`	CHALK TAIL CHALK
`44A5`	REAR JAIL LAW
`93FA`	BEAM CASE JAIL
`A4DE 675`	JAIL RAY CHEF CHIP CHUCKLE
`0B28 A`	SAW JUDGE NAVY JAIL

Note:

- Encode 6750 as CHUCKLE SAW, not 67 50 CHALK LACE. CHALK decodes as C, not 67.

In the credit-card example, we worked backwards from the picturesque peg FISH 86. Here we work backwards from the seductive sound-alike NAIVE 28. The seductive emotion will make it easier to memorize. Hallucinate a HELL-CHEF SAUCE, a recipe for a naive victim.

`5D00 E7AA`	HELL-chef SAUCE: CHOP, HACK, and CHILL the JELLY.
`C15C`	SHAKE with TALLOW, CHECK
`44A5`	the RARE CHILI OIL.
`93FA`	"BE-MY HAGGIS, JILL!"
`A4DE 675`	CHILL for an HOUR, SHAVE, CHOP, CHUCKLE.
`0B28 A`	"ASSAY, JUDGE!" NAIVE JILL.

7.2 Example SHA-1 Hash

From `https://onlinehashtools.com`, consider the following, 160-bit hash: For git, it usually suffices to remember the first 8 hexits. Let's do them all as a workout:

`A638 1B63`	JAIL SHAM-FRUIT TEA JUDGE CHIME
`D532 69BA`	CHEF LEMON JAW PEA JUDGE JAIL
`418E 5FB1`	ARTFUL CHIP LAW CASE JUDGE TEA
`BD50 3950`	JUDGE CHEF WHOLESOME PILLOWS
`860D D679`	FISHES CHEF CHEF CHECKUP

Notes

- CHIME for 63 is ok in hex and R36, but not ok in R64, where it decodes as "/".
- Don't encode 69 as CHIP, rather as JAW PEA, because CHIP decodes as E.
- An alternative to WHOLESOME PILLOWS for 50 3950 is LACE MAP LACE, but it's clumsy.

Let's bribe a judge and put him to sleep with hashish (puns help the memory), then summon middle-Eastern heavenly dreams.

`A638 1B63`	with SHAWL and SHAM-FRUIT, TEA for the JUDGE, JIM.
`D532 69BA`	SHAVE a LEMON WEDGE, PAY the JUDGE to CHILL.
`418E 5FB1`	an ARTFUL SHAPE of this LAW CASE, SHUSH THEE!
`BD50 3950`	HASHISH ACHIEVES ELYSIUM BLESSINGS
`860D D679`	FETCHES the CHIEF, JEHOVAH for JACOB!

8 Radix 36

Radix-36 is more difficult to find "in the wild" than is either hex or Radix-64. I most often see it in randomly generated file and directory names because some file systems are not case-sensitive and neither is R36. R36 is also sometimes found in URLs for encoding queries or other data. Despite the fact that URLs, except for the domain name, are case-sensitive, some sites, especially older ones, and even some browsers and query-generators, don't adhere to the standard properly. Radix-64 is overkill in such circumstances and uses up alternatives, but radix-36 is immune to the commonplace nonconformance of the internet.

For encoding *rixits* in R36 (or biscuits in base-36?), I use ASCII pegs for the capital letters beyond hex. Find their sound-alikes in Section 16. Because R36 is case-insensitive, we need only one set of pegs. For Radix-64, we need both upper and lower case, so two pegs per letter.

The stars in the table below mark pegs for the lower case. They help for remembering the ASCII pair — upper and lower case — but they're only pertinent for R64 (Section 9). The numbers in parentheses refer to the ordinal number of the letter in the alphabet: K is the eleventh letter in the alphabet, and 11 has peg *tot*, whom I visualize as Alice of Wonderland. Further associations with Hebrew keywords like *mayim* (water) and *nakhash* (snake) or with playing cards like the club jack are given, as explained in section 11.

	NATO		ASCII	PHANTASMAGORIA; ASCII-pair refs marked with *
G	GOLF	71	cat	on the golf cart with my twosome*
H	HOTEL	72	can	ashtray outside in the smoking area (smell)
I	INDIA	73	cam	diesel* Rolls-Royce in a junkyard in India
J	JULIET	74	car	Juliet walks a wood-sage* to his NASCAR* ride
K	KILO	75	coal	Alice(11) gives coal to a tusked* troll
L	LIMA	76	cage	containing tanned(12) beach-goers
M	MIKE	77	cake	an orca (Mayim:13) flips a cake on its nose
N	NOVEMBER	78	cuff	a snake (Nakhash:14) no arms, cuffs on shirt
O	OSCAR	79	cap	a teetotaler wearing a Carrie-Nation cap
P	PAPA	80	face	Dad cries (tissue:16) at Liz Taylor's face
Q	QUEBEC	81	fat	did Pierre Trudeau get fat?
R	ROMEO	82	fan	Romeo hands a fan to a Tatar* beauty
S	SIERRA	83	fame	on stage at my book-signing, title* "Sierra"
T	TANGO	84	fare	a death-watch* soldier gets on the bus
U	UNIFORM	85	fall	a cadet(club jack) falls into a net(21)
V	VICTOR	86	fish	the Christian symbol, nun(22), auto-da-fe*
W	WHISKEY	87	fig	a Few Figs from Thistles; reading it with a dram
X	X-RAY	88	fife	Nero(24) plays the fife, dances* while Rome burns
Y	YANKEE	89	fob	a watch-fob hanging from a nail(25) on a US map
Z	ZULU	90	base	an Athenian* ball player at bat on home base

8.1 Example: Unix Time

I found an example here:

`https://medium.com/analytics-vidhya/base-36-why-how-its-important-a7b084b02548kg4cebk9`

Pegs, first, converting to upper-case:

`KG4`	COAL CAT RAY
`CEB`	CHALK CHIP JUDGE
`K9`	COAL PEA

Do some forensics on a dead pet.

`KG4`	KILL the CAT with a RAY.
`CEB`	CHECK for a CHIP, lest we be JUDGED.
`K9`	COOL! it's UP.

8.2 Example from Spack

Spack (`https://spack.io/`) is a dependency solver for software build systems. It produces directory names in R36 that won't (statistically) collide with SHA-1 hashes and work fine on case-insensitive file systems. Here's an example

`[+] /home/blahblah/yackyack/spack_packages/scripts/tmp-spack2/spack/opt/spack/linux-ubuntu18.04-haswell/gcc-7.5.0/gdbm-1.19-rdt674nck5k5hbawjaywb4mv4r72osnq`

Pegs first, except cheat on pronounceable subsequences or others that appeal to your personal memory!

`RDT6 74NC`	REDDIT JAW KEY RAY CUFF CHALK
`K5K5 HBAW`	COAL LAW COAL LAW, CAN JUDGE JAIL FIG
`JAYW B4MV`	JAY FIG, BEFORE MOVIE (JUDGE RAY CAKE FISH)
`4R72 OSNQ`	RAY FAN KEY NOAH, FACE FAME CUFF FAT.

How about a tale of canceling on social media?

`RDT6 74NC`	REDDIT is AHUSH, a COY WAR, GOOFY and SHOCKING.
`K5K5 HBAW`	KILL ILL, UGLY WILL; AGONY SHUSH the SHILL, the FAKE
`JAYW B4MV`	JAY-bird FAKE, BEFORE MOVIE
`4R72 OSNQ`	OUR FAN GUY NOAH, with VOICE FAIR, GIVE FIGHT!

Notes

- CAR for 74 is not ok because it decodes as J. Peg it as KEY RAY, then try sound-alikes.
- FISH is V in R36, not 86. If we had 86, peg it as FEE JAW, then try sound-alikes.
- 72 can't be CAN, because that's H in R36. Peg it as KEY NOAH then try sound-alikes.

9 Radix 64

For encoding rixits in R64, add the lower-case letters plus special cases for "+", 62, and "/", 63. Stars in the table below mark pegs for the upper-case ASCII codes. They help for remembering the ASCII pair together.

	NATO		ASCII	PHANTASMAGORIA; ASCII-pair refs marked with *
a	alpha	97	back	the jailer* has her back turned to the lion
b	baker	98	beef	the judge* eats a beef dinner in court
c	charlie	99	pipe	smoking while chalking* the hopscotch grid
d	delta	100	thesis	the chef* hasn't finished the thesis yet
e	echo	101	toast	the cop(5) has toast between his fingers
f	foxtrot	102	dozen	a dozen pink (Vered:6) roses in a case*
g	golf	103	twosome	on the golf cart with the cat*
h	hotel	104	dowser	done smoking outside (ash-can*), dowsing inside
i	india	105	diesel	holding a camshaft* from diesel Rolls-Royce
j	juliet	106	wood-sage	Juliet escorts a druid to its NASCAR ride
k	kilo	107	tusk	Alice(11) gives coal* to a tusked troll
l	lima	108	adhesive	dangle a fly-paper into the cage*
m	mike	109	teaspoon	drip icing onto the cake* in the water (Mayim:13)
n	november	110	tights	the snake (Nakhash:14) has no legs, wears tights
o	oscar	111	teetotaler	wearing the Carrie Nation cap*
p	papa	112	Titan	protecting Liz Taylor
q	quebec	113	totem	a fat* Pierre Trudeau kneels before the totem
r	romeo	114	Tatar	Romeo hands a fan* to a Tatar beauty
s	sierra	115	title	on stage receiving fame* for my book titled Sierra
t	tango	116	death-watch	a soldier gets on the bus paying fare* and dancing
u	uniform	117	hotdog	a cadet falls* into a net(21) holding a hotdog
v	victor	118	auto-da-fe	a Christian* nun(22) overlooks the ritual
w	whiskey	119	teddy bear	while reading a Few Figs* from Thistles
x	x-ray	120	dance	Nero(24) plays fife*, looks through dancers' clothes
7	yankee	121	donut	hangs on the nail(25) next to the fob*
z	zulu	122	Athenian	on home base* at bat
+		62	chain	
'/		63	chime	

9.1 Example Public Key

Don't bother encoding AAAA, which precede and follow a canonical prefix of 16 rixits for (supposedly) a machine-specific part of the key. With the remaining 44 rixits and 8 for the two AAAAs, we get the required 68 rixits. We have 60 rixits to encode.

```
ssh-ed25519
AAAA C3Nz aC1l ZDI1 NTE5 AAAA
IFi4 CpLS hSK/ uUux BxM3 6WwK
C6RD 1/Qn tXQg BV3m pUZq
```

Pegs first, then try sound-alikes:

`C3Nz aC1l`	CHALK ME CUFF ATHENIAN, BACK CHALK TEA ADHESIVE
`ZDI1 NTE5`	BASE CHEF CAM TEA, CUFF FARE CHIP LAW
`IFi4 CpLS`	CAM CASE DIESEL RAY, CHALK TITAN CAGE FAME
`hSK/ uUux`	DOWSER FAME COAL CHIME, HOTDOG FALL HOTDOG DANCE
`BxM3 6WwK`	JUDGE DANCE CAKE ME, JAW FIG TEDDY-BEAR COAL
`C6RD 1/Qn`	CHALK JAW FAN CHEF, TEA CHIME FAT TIGHTS
`tXQg BV3m`	DEATHWATCH FIFE FAT TWOSOME, JUDGE FISH ME TEASPOON
`pUZq`	TITAN FALL BASE TOTEM

A tale of overeating:

`C3Nz aC1l`	CHALK picture, ME adjusting CUFFS, looking down my ATHENIAN nose. Turn my BACK, another CHALK picture; my TIE is flypaper ADHESIVE.
`ZDI1 NTE5`	On BASE, now. A CHEF flips burgers with a CAMSHAFT and brews TEA. COFFEE, FAIR-traded! SHOPPED WELL.
`IFi4 CpLS`	The CAMSHAFT stirs EGGS, a DAZZLING omelet, ROE and JACK cheese. I'll be a TWO-TON and KEDGE to bed: HEAVE-HIM!
`hSK/ uUux`	DESSERTS? WHIFF-EM! CALL a dentist, JIM! my TOOTHACHE JIM! My TOOTHACHE FEELS like a TOOTHACHE, DUNCE!
`BxM3 6WwK`	CHOW-CHOW! DUNCE CAKE for ME. JAW a FIG and a TOOTHPICK, COOL!
`C6RD 1/Qn`	CHECK my JAW OFTEN, FAWN CHEF. TO SHAME FOOD TADS.
`tXQg BV3m`	DEATH-WISH? VIVA my FAT TWOSOME. SHUSH, VETCH, MY TEASPIT.
`pUZq`	I'm a TWO-TON, I'll FALL BESIDE the DAYTIME.

10 Passwords

Don't make passwords easy for bots to guess. Random, long passwords are best, but hard to remember, even with our systems. They're even more difficult to associate uniquely with a web site.

Modern web sites don't store your passwords in the clear. Virtually all, nowadays, hash your password with a one-way hash (and usually some salt and a nonce challenge). That means you expose nothing by re-using part of a password, so long as it's different from site-to-site.

Pick a difficult prefix, but memorable to you. Mix in a special character and a numeral because many sites insist on such. Start with something picturesque:

An Octopus has eight legs with suckers.

Now, partially encode and obfuscate it; this is your personal salt prefix:

`2Octopu$Ha$8Leg$With0740`

Finally, append the site name, perhaps obfuscated:

`2Octopu$Ha$8Leg$With0740EBAY`

`2Octopu$Ha$8Leg$With0740APPLE`

`2Octopu$Ha$8Leg$With0740OVERLEAF`

Anyone who knows your personal salt prefix and your algorithm will break you. But bots will just see a random hash, completely different for each site you visit. Remember that hashes are uniformly distributed. The hash of any input is equally likely to be anywhere in the space of hashes. There is no way to tell how similar two input texts A and A' are from their hashes, even if A and A' differ by no more than one bit. Bots will have no way to know that you are even using an algorithm to construct your passwords.

11 Daily Practice: ASCII, Cards, NATO, Hebrew, Dates

Table 1 exploits the coincidence that there are 52 glyphs in the ASCII Latin alphabet, 52 cards in a standard pack, and 52 weeks of the year. There are only 22 Hebrew letters, so the last four rows of each 26 are empty in the Hebrew column.

Table 1 is in groups of 13 corresponding to the suits of playing cards. Each group also contains dates in a quarterly month-week pattern of 5-4-4: 5 weeks in Jan, 4 in Feb, 4 in Mar, etc. The first days of the months are in the easy pattern 155, 274, 263, 153. You may memorize this as a honeymoon tale of TAILLIGHTS, NIAGRA, UNASHAMED, HOTELMAN. Table 2 focuses on some phantasmagorias specifically for dates.

These are for non-leap years only, so not great for computing day-of-the-week. I have another method for that (the method of Furst, debugged) in Section 15.9.

My daily practice includes another $156 = 3 \times 52 = 2 \times 78$ three columns of phantasmagorias for the 156 faces of a standard Tarot pack, obverse and reverse. See Section 13. These are a source of vivid mental imagery.

I go down table 1 in my mind when I need to fall asleep. I usually get four or five rows before drifting off, so next time through, I start up where I left off. Sometimes, I recite it backwards just to make sure the end of the table doesn't get soft from lack of repetition. Here are the "mental movies," expanded from the hints in Sections 7, 8, and 9. I include sense of smell because it works for me. It might not work for you.

1. ALPHA — The ALPHA LION (ARYEH) is in the giant, Disneyland TEACUP. He and the teacup are in JAIL and the jailer has her BACK turned to us. A female jailer, being unusual, is easier to remember. Smell the dank musty odor of the jail.
2. BAKER — is serving a BEEF meal to a JUDGE, who is in full magistrate's dress including a powdered wig. The meal is taking place in the little BOOTH (BAYIT) on top of NOAH's Ark. Smell the food.
3. CHARLIE — is riding a CAMEL (GAMAL), looking at himself in the ME (3, mirror) having a hell of a time tying his bow tie while bouncing around on the back of the camel and smoking a PIPE. The camel is dragging ropes with colored sidewalk CHALK behind him making marks on the ground. Smell the pipe tobacco smoke mixed with stinky camel fur and road dust.
4. DELTA — I am pointing a flashlight (RAY) at a giant, brass-and-wooden DOOR (DALET), looking for a keyhole in the rain and the dark (smell mold and mildew and moss and grass and old machine-oil grease on the door's hardware). I open it up and there is a PhD student who never finished his THESIS so must work as a CHEF. I see, hear, and smell the food he is cooking. Alternatively, a laser pointer is etching the blue lines around the margins of the THESIS paper (such thesis paper with blue lines was common in my day).
5. ECHO — Moses is delivering the LAW on mount Sinai. A cop at the bottom of the hill (HAR) is holding up his hand: "stop!" In between his fingers

there are poker CHIPS and pieces of Cinnamon TOAST (smell it). ECHO begins with E, which is Greek Eta and looks like the letter H of *hill* and *HAR*. (*Hill* and *har* are obvious English-Hebrew cognates by a shift of the liquid consonants *l* and *r*.) Smell the desert sands.

6. FOXTROT — Lurch is clenching a pink (VERED) rose in his teeth and lantern JAW. He has an open CASE on his lap filled with a DOZEN more pink roses. Smell the roses.
7. GOLF — I have a golden (ZAHAV) KEY to my golf cart, where my black CAT is waiting with my golf TWOSOME. Smell the fuel of the golf cart mixed with the grass smell of the golf course.
8. HOTEL — I am in the lobby of the beautiful Bellagio hotel with my smelly donkey (CHAMOR) and a #10 coffee CAN full of stinky cigar butts. I have a forked DOWSER rod in my hands. I am waiting in line to pay the exorbitant FEE. The dowser is vibrating as it detects the nearness of water, or gold, because I am going to pay my fee in gold.
9. INDIA — I am in a junkyard full of old Rolls-Royce DIESEL (smell it) automobiles with their camshafts poking out of their hoods (remember via the trivium that there are more Rolls-Royces in India than in Britain). In the cockpit of one Rolls-Royce is a pilot (TAYYAS) up to his neck in PEA soup.
10. JULIET — with her painted TOES (10) and fingernails, is leading a WOOD-SAGE (dwarf druid) by the hand (YAD) to his brightly colored racing CAR parked in a clearing in the woods. Smell the forest.
11. KILO — Alice of Wonderland (a TOT) comes off her throne (KISEH) to deliver a kilo of COAL to a troll with huge TUSKS. Smell the coal and the body odor of the filthy troll king.
12. LIMA — A girl is sunning herself and holding a silver platter under her chin to ensure it gets TANNED. On the tray is a loaf of bread (LECHEM) with flies buzzing around it (smell it because it's extra hot, baking in the sunshine). She takes a strip of ADHESIVE flypaper and dangles it into a CAGE to trap the flies.
13. MIKE — A pod of orca whales are flipping a Tam-j'Shanter (beanie hat with a red pom-pom) in the water (MAYIM). I take a TEASPOON of water and drizzle it on a gigantic chocolate CAKE, the size of a five-gallon cylindrical paint barrel. Smell the seawater.
14. NOVEMBER — A snake (NACHASH) is sizzling on a patch of hot TAR asphalt (smell it). It has neither arms nor legs, so, of course, it is wearing a shirt with CUFFS, plus TIGHTS instead of trousers (absolute absurdity helps the memory).
15. OSCAR — A TEETOTALER wearing a Carrie-Nation CAP is holding an Oscar cinema award. A tiger with a striped TAIL is clenching a book (SEFER) in its jaws and bringing it to the teetotaler. Smell the tiger: a "zoo" smell.
16. PAPA — is wiping tears from his eyes (AYIN) with a TISSUE. He is overcomes with emotion looking at the beautiful face of Liz Taylor as Cleopatra,

whose picture is held by a giant TITAN robot. Smell Liz's perfume.

17. QUEBEC — A very FAT premier of Quebec is lying face-down on a bed of TACKS and turning his face (PANIM) to a TOTEM to pray. Smell the wood and paint of the totem.
18. ROMEO — has a sparrow (TZIPPOR) with her feet stuck in a patch of sticky TAFFY (smell it) on his shoulder. He brings her to a TATAR princess, who is being FANNED by several slave girls (or boys!).
19. SIERRA — I finally get some FAME from publishing my first book, TITLED "Sierra." I celebrate with a beer pulled from a TAP by a monkey (QUF). Smell the beer.
20. TANGO — A DEATHWATCH soldier with four arms and three swords is carrying Goliath's head (ROSH), which has a giant NOSE. Smell the blood. He tangos awkwardly toward a bus, where he mounts the steps in a palsy and pays his FARE inside.
21. UNIFORM — A skeleton (SHELED) holding a butterfly net FALLS into a bigger NET along with a man-sized HOT-DOG. Smell it.
22. VICTOR — A NUN with a Christian FISH pendant presides over a victory, a burning at the stake (AUTO-DA-FE). She is holding a pet crocodile (TANIN) by a leash. Smell everything.
23. WHISKEY — A TEDDY-BEAR rests on a bed with a book of Edna St. Vincent Millay's poems: "A few FIGS from Thistles." A small NAMEPLATE rests against a tumbler of WHISKEY on the nightstand (smell the whiskey).
24. X-RAY — NERO is playing his FIFE (not a fiddle) while watching Rome burn. He uses his X-RAY vision to look through the clothes of a bunch of DANC-ING people in the yard.
25. YANKEE — An American flag is NAILED to the wall. Two nails stick out. A DOUGHNUT with icing and sprinkles (smell it) dangles from one nail and a watch-FOB is looped over the other nail.
26. ZULU — An ATHENIAN comes up to BASE (baseball). Next to him is a NICHE with a bust of Athena dressed in Zulu regalia in it.

It is a small matter to add the date phantasmagorias to the mental movies we already have for the 52 glyphs. It takes only a little more effort to add the "Day-of-Year Phantasmagorias."

Table 1: ASCII, Playing cards, NATO, Hebrew, Dates

	NATO	ASCII	HEBREW	TRANSLATION	SUIT + PIP	DATE	DOY
A	ALPHA	jail	ARYEH	lion	SA	JAN 01	1
B	BAKER	judge	BAYIT	booth	S2	JAN 08	8
C	CHARLIE	chalk	GAMAL	camel	S3	JAN 15	15
D	DELTA	chef	DALET	door	S4	JAN 22	22
E	ECHO	chip	HAR	hill	S5	JAN 29	29
F	FOXTROT	case	VERED	pink	S6	FEB 05	36
G	GOLF	cat	ZAHAV	gold	S7	FEB 12	43
H	HOTEL	can	CHAMOR	mule	S8	FEB 19	50
I	INDIA	cam	TAYAS	pilot	S9	FEB 26	57
J	JULIET	car	YAD	hand	ST	MAR 05	64
K	KILO	coal	KISEH	throne	SJ	MAR 12	71
L	LIMA	cage	LECHEM	bread	SQ	MAR 19	78
M	MIKE	cake	MAYIM	water	SK	MAR 26	85
N	NOVEMBER	cuff	NACHASH	snake	HA	APR 02	92
O	OSCAR	cap	SEFER	book	H2	APR 09	99
P	PAPA	face	AYIN	eye	H3	APR 16	106
Q	QUEBEC	fat	PANIM	face	H4	APR 23	113
R	ROMEO	fan	TZIPPOR	sparrow	H5	APR 30	120
S	SIERRA	fame	QUF	monkey	H6	MAY 07	127
T	TANGO	fare	ROSH	head	H7	MAY 14	134
U	UNIFORM	fall	SHELED	skeleton	H8	MAY 21	141
V	VICTOR	fish	TANIN	crocodile	H9	MAY 28	148
W	WHISKEY	fig			HT	JUN 04	155
X	X-RAY	fife			HJ	JUN 11	162
Y	YANKEE	fob			HQ	JUN 18	169
Z	ZULU	base			HK	JUN 25	176

Table 2: ASCII, Playing cards, NATO, Hebrew, Dates (cont.)

	NATO	ASCII	HEBREW	TRANSLATION	SUIT + PIP	DATE	DOY
a	alpha	back	aryeh	lion	CA	JUL 02	183
b	baker	beef	bayit	booth	C2	JUL 09	190
c	charlie	pipe	gamal	camel	C3	JUL 16	197
d	delta	thesis	dalet	door	C4	JUL 23	204
e	echo	toast	har	hill	C5	JUL 30	211
f	foxtrot	dozen	vered	pink	C6	AUG 06	218
g	golf	twosome	zahav	gold	C7	AUG 13	225
h	hotel	dowser	chamor	mule	C8	AUG 20	232
i	india	diesel	tayas	pilot	C9	AUG 27	239
j	juliet	wood-sage	yad	hand	CT	SEP 03	246
k	kilo	tusk	kiseh	throne	CJ	SEP 10	253
l	lima	adhesive	lechem	bread	CQ	SEP 17	260
m	mike	teaspoon	mayim	water	CK	SEP 24	267
n	november	tights	nachash	snake	DA	OCT 01	274
o	oscar	teetotaler	sefer	book	D2	OCT 08	281
p	papa	titan	ayin	eye	D3	OCT 15	288
q	quebec	totem	panim	face	D4	OCT 22	295
r	romeo	tatar	tzippor	sparrow	D5	OCT 29	302
s	sierra	title	quf	monkey	D6	NOV 05	309
t	tango	death-watch	rosh	head	D7	NOV 12	316
u	uniform	hot-dog	sheled	skeleton	D8	NOV 19	323
v	victor	auto-da-fe	tanin	crocodile	D9	NOV 26	330
w	whiskey	teddy-bear			DT	DEC 03	337
x	x-ray	dance			DJ	DEC 10	344
y	yankee	doughnut			DQ	DEC 17	351
z	zulu	Athenian			DK	DEC 24	348

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Table 3: Date Phantasmagorias

	NATO	ASCII PHANT	SUIT + PIP	DATE	DATE PHANT	DOY	DOY PHANT
A	ALPHA	jail	SA	JAN 01	New tie	1	tie
B	BAKER	judge	S2	JAN 08	New view	8	view
C	CHARLIE	chalk	S3	JAN 15	New deal	15	deal
D	DELTA	chef	S4	JAN 22	New onion	22	onion
E	ECHO	chip	S5	JAN 29	New nib	29	nib
F	FOXTROT	case	S6	FEB 05	Love lie	36	image
G	GOLF	cat	S7	FEB 12	Love tune	43	room
H	HOTEL	can	S8	FEB 19	Love tap	50	loose
I	INDIA	cam	S9	FEB 26	Love nudge	57	leak
J	JULIET	car	ST	MAR 05	Irish lie	64	cheer
K	KILO	coal	SJ	MAR 12	Irish tune	71	cad
L	LIMA	cage	SQ	MAR 19	Irish tap	78	cuff
M	MIKE	cake	SK	MAR 26	Irish nudge	85	evil
N	NOVEMBER	cuff	HA	APR 02	Easter wine	92	bun
O	OSCAR	cap	H2	APR 09	Easter pie	99	pub
P	PAPA	face	H3	APR 16	Easter dosh	106	dosage
Q	QUEBEC	fat	H4	APR 23	Easter gnome	113	tedium
R	ROMEO	fan	H5	APR 30	Easter Mass	120	dunce
S	SIERRA	fame	H6	MAY 07	National ache	127	tank
T	TANGO	fare	H7	MAY 14	National tear	134	timer
U	UNIFORM	fall	H8	MAY 21	National note	141	tread
V	VICTOR	fish	H9	MAY 28	National navy	148	adrift
W	WHISKEY	fig	HT	JUN 04	Altar rue	155	taillight
X	X-RAY	fife	HJ	JUN 11	Altar teat	162	tuition
Y	YANKEE	fob	HQ	JUN 18	Altar dive	169	dodgeball
Z	ZULU	base	HK	JUN 25	Altar kneel	176	education

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Table 4: Date Phantasmagorias (cont.)

	NATO	ASCII PHANT	SUIT + PIP	DATE	DATE PHANT	DOY	DOY PHANT
a	alpha	back	CA	JUL 02	Summer wine	183	deaf-mute
b	baker	beef	C2	JUL 09	Summer pie	190	tapas
c	charlie	pipe	C3	JUL 16	Summer dosh	197	top-K
d	delta	thesis	C4	JUL 23	Summer gnome	204	answer
e	echo	toast	C5	JUL 30	Summer Mass	211	noted
f	foxtrot	dozen	C6	AUG 06	Moon witch	218	not-if
g	golf	twosome	C7	AUG 13	Moon time	225	union hall
h	hotel	dowser	C8	AUG 20	Moon noose	232	honeymoon
i	india	diesel	C9	AUG 27	Moon Nike	239	number
j	juliet	wood-sage	CT	SEP 03	Job wham	246	enrage
k	kilo	tusk	CJ	SEP 10	Job dice	253	annulment
l	lima	adhesive	CQ	SEP 17	Job tyke	260	nachos
m	mike	teaspoon	CK	SEP 24	Job near	267	unshucked
n	november	tights	DA	OCT 01	Travel tie	274	nacre
o	oscar	teetotaler	D2	OCT 08	Travel view	281	unfit
p	papa	titan	D3	OCT 15	Travel deal	288	navy wife
q	quebec	totem	D4	OCT 22	Travel onion	295	napalm
r	romeo	tatar	D5	OCT 29	travel nib	302	muezzin
s	sierra	title	D6	NOV 05	Restaurant lie	309	mess-up
t	tango	death-watch	D7	NOV 12	Restaurant tune	316	midshipman
u	uniform	hot-dog	D8	NOV 19	Restaurant tap	323	minimum
v	victor	auto-da-fe	D9	NOV 26	Restaurant nudge	330	memos
w	whiskey	teddy-bear	DT	DEC 03	Church wham	337	mimic
x	x-ray	dance	DJ	DEC 10	Church dice	344	mirror
y	yankee	doughnut	DQ	DEC 17	Church tyke	351	malt
z	zulu	Athenian	DK	DEC 24	Church near	348	Morpheus

12 Non-Alphabetic Printable ASCII Codes

Use the ordinary pegs for these, because we want to recall them in any order.

Space		MAN	32	spaceman
Bang	!	MAMA	33	with a handgun
Dirk	"	MARE	34	looks like horse's ears
Hash	#	MAIL	35	postmarks on your postage stamps
Cash	$	MATCH	36	set your cash on fire
Grapes	%	MIKE	37	on stage with Bacchus / Dionysis
Snowman	&	MUFF	38	sleeps until winter time
Tick	'	MAP	39	ticks (bugs) crawling on a map
Frown	(	RACE	40	NASCAR event begins with frown
Smile	)	RAT	41	Rat Fink (hotrodder) lurid grin
Star	*	RAIN	42	star rubies litter rain-soaked beach
Plus	+	RAM	43	cloak caparison on a ram has plus signs
Tadpole	,	REAR	44	backside of my Corvette has tadpoles
Flag	-	RAIL	45	looks like a footrail in a bar
Stop	.	RASH	46	my rash looks like measles' dots
Press	/	RACK	47	my rack of antlers is twisted rightwards
EFES	0	REEF	48	Easy if you know the Hebrew
...	..	...	..	numbers
TEYSHA	9	LAKE	57	
Snake-Eyes	:	LEAF	58	four-leaf clover, snake-eyes on dice
Hybrid	;	LAP	59	cat on my lap with one eye open
Bra	<	CHESS	60	the Queen on giant chessboard wears a bra
Gym	=	JET	61	parallel bars inside a jet airliner
Ket	>	CHAIN	62	the WC chain pulls the ket from the bra
Hook	?	CHIME	63	when the chime sounds, it's question time
Snail	@	CHAIR	64	snails covering your favorite easy-chair
Box	[	BAT	91	catch the bat with bait in a wooden box
Slope	\	BEAN	92	beans as bait behind the sloped trap door
Top	]	BEAM	93	the lid has a wooden beam for strength
Hat	^	BAR	94	go to your bar (pub) in fishing hat
Score	_	BALL	95	piano player in bar drinks a highball
Blip	`	BADGE	96	radar blips detect metallic badges
Embrace	{	DYNAMITE	123	you're embracing dynamite under your coat
Pipe	\vert	TENOR	124	singer has great pipes
Bracelet	}	TUNNEL	125	you lost your bracelet in a taxi in a tunnel
Worm	~	DUNGEON	126	you slink into dungeon like a worm

13 Tarot

Tarot cards have vivid imagery and powerful, emotional, words, great for memory. If you subscribe to the spiritual or arcane meanings of the cards, all the better for memory. But they're great as a Memory Palace even if you think all that is hooey.

There are 78 Tarot cards in a standard Rider-Waite pack. That's not divisible by 52, but if we consider the reverse images (upside-down cards) as distinct to the obverse, and most cartomancers do, then we have 156 image-word combinations, and that is divisible by 52. Thus, we get an additional three columns of picture words for daily practice (11), where we associate tarot cards with standard playing cards, the NATO alphabet, the ASCII codes, weeks of the year, day of the year, and Hebrew words, most of which have 52 realizations.

The Tarot suits are wands, clubs, swords, and pentacles, obviously related to the suits of a standard pack, but different enough that we'll just keep those two sets of suits distinct in our minds. There are 14 cards in each suit, but only 22 trumps, aka majors. Let's take the Knights out of each suit and append them to the majors so that the minor suits have 13 cards each, including three court cards — page (like "jack"), queen, and king. We want clusters of 13 because 13 divides 52 — we get four clusters for each of the three runs of 52 items.

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NATO	TAROT MAJOR OBV	INCLUDING KNIGHTS OBVERSE	TAROT MINOR REV	ALL REVERSE	TAROT MINOR OBV	ALL OBVERSE
ALPHA	MAG	willpower	wA	boredom	WA	inspiration
BRAVO	PST	intuition	w2	hesitation	W2	independence
CHARLIE	EMS	motherhood	w3	obstacles	W3	expansion
DELTA	EMP	fatherhood	w4	transience	W4	home
ECHO	HEI	tradition	w5	dodge	W5	competition
FOXTROT	LOV	union	w6	ignored	W6	recognition
GOLF	CHA	drive	w7	give up	W7	protect
HOTEL	STR	strength	w8	panic	W8	rapid
INDIA	HRM	contemplation	w9	exhaustion	W9	grit
JULIET	WHL	luck	wT	micromanage	WT	responsibility
KILO	JST	causation	wJ	procrastination	WJ	excitement
LIMA	HNG	perspective	wQ	selfishness	WQ	courage
MIKE	DTH	cycle	wK	overbearing	WK	overcome
NOVEMBER	TMP	patience	cA	blocked	CA	spirituality
OSCAR	DVL	addiction	c2	tension	C2	partnership
PAPA	TWR	destruction	c3	gossip	C3	friendship
QUEBEC	STA	faith	c4	acceptance	C4	apathy
ROMEO	LUN	misdirection	c5	peace	C5	disappointment
SIERRA	SUN	joy	c6	leaving home	C6	return home
TANGO	JDG	reckoning	c7	diversion	C7	daydreaming
UNIFORM	WLD	completion	c8	avoidance	C8	disillusioned
VICTOR	FOO	innocence	c9	smugness	C9	luxury
WHISKEY	KNW	fearless	cT	broken family	CT	fulfillment
X-RAY	KNC	idealist	cJ	immaturity	CJ	dreamer
YANKEE	KNS	defender	cQ	self-sacrifice	CQ	compassion
ZULU	KNP	efficient	cK	cold	CK	emot control

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NATO	TAROT MAJOR REV	INCLUDING KNIGHTS REVERSE	TAROT MINOR REV	ALL REVERSE	TAROT MINOR	ALL OBVERSE
alpha	mag	illusions	sA	chaos	SA	breakthrough
bravo	pst	repression	s2	hobson's	S2	indecision
charlie	ems	barrenness	s3	forgiveness	S3	heartbreak
delta	emp	tyranny	s4	restless	S4	rest
echo	hei	rebellion	s5	resentment	S5	conan
foxtrot	lov	one-sided	s6	resistence	S6	transition
golf	cha	directionless	s7	conscience	S7	cunning
hotel	str	self-doubt	s8	freedom	S8	trapped
india	hrm	isolation	s9	anxiety	S9	hope
juliet	whl	loss	sT	nadir	ST	betrayal
kilo	jst	dishonesty	sJ	deception	SJ	curiosity
lima	hng	stalling	sQ	critical	SQ	clarity
mike	dth	stagnation	sK	cruel	SK	discipline
november	tmp	excess	pA	bad investment	PA	venture
oscar	dvl	recovery	p2	overwhelmed	P2	priorities
papa	twr	aversion	p3	group conflict	P3	teamwork
quebec	sta	insecurity	p4	stingy	P4	frugality
romeo	lun	confusion	p5	receive charity	P5	poverty
sierra	sun	sadness	p6	strings attached	P6	charity
tango	jdg	doubt	p7	wasted work	P7	perseverence
uniform	wld	emptiness	p8	no motivation	P8	high standards
victor	foo	naivety	p9	spendthrift	P9	fruits of labor
whiskey	knw	reckless	pT	fleeting success	PT	legacy
x-ray	knc	moody	pJ	uncommitted	PJ	diligence
yankee	kns	feckless	pQ	jealous	PQ	abundance
zulu	knp	workaholic	pK	indulgence	PK	prosperity

14 People's names

Take the name, convert to a number using the usual rules. Memorize the number as usual using the memorization technique. When recalling, you can use pinfruit.com or other means, enter the number and it will print all possibilities. There will be more than one, but usually you will remember at this point which one is the correct one. And if not, all the other ones are sound-alike, so you will not make a big mistake. Either way, it is much better than not knowing the name at all.

Examples.

Jordan -> 6412 -> Jordan, Sheridan (memorize, e.g., as *sure tone*)

Brian -> 942 -> Brian, Brain, Bryan, Brown (memorize, e.g., as *hebrew now*)

Ondrej -> 214 -> Andrew, Andrea, Andrei (memorize, e.g., as *winter*)

Pick words and create a movie related to something that you know about the person (for example they like WINTER sports like hockey, or they are learning HEBREW NOW, or they use SURE TONE when singing).

15 APPENDICES

15.1 Pegs and Phantasmagorias

I blogged out the basics of my peg method here:

`https://rebcabin.wordpress.com/2010/11/21/memory-pegs/`

15.2 Sound-Alikes

Finding a sound-alike is "computing" or "calculating" a picture word. For example, the peg for 37 is MIKE. Picture a microphone in your hand and feel the electrical tingling. Adding synaesthesia or emotion to pegs helps my memory. It may not help other people. Do what is best for you!

To calculate some nearby picture words, I get

mica Glittering stone; feel grit on your fingers.

mug A big cold beer.

muck Smell the barnyard.

Mecca Picture the kaaba and the pilgrims circulating.

hammock Take a snooze.

15.3 Building and Revising Chains

Take time building phantasmagorias so that ideas flow. It's bad to have a break part-way-through, causing a pause or failure in recall. You must be *fast* and *accurate*. A break or even a soft link in a chain is cause for starting over and building a new chain.

15.4 One Peg Set, Forever

Choose pegs that work for you, then ***NEVER CHANGE THEM***. Recall must be instantaneous. Whenever you see or think of a number, you must recall the peg without effort.

I regret changing a few pegs once. I did it for teaching purposes because my originals got dated. For example, 3 for *May* became *me* because no one today knows what a May-pole is. Every time I see 3, however, I picture a May-pole instead of myself in a mirror. I learned May-pole first (from Furst and from my Dad). The switch slows me down even though I made the switch long, long ago.

DON'T CHANGE PEGS AFTER LOCKING THEM IN

15.5 Alternatives for Hex

15.5.1 Hex Method 1

One way to encode hex is to map A-F to 010-015. Permanent pegs are in all caps.

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010	SAWDUST	SWEETS	seducer	hostess	citizen	cities
A	South-Sea	assets	assiduous	sideways	stays	
011	STATUE	STEED	studio	estate	iced-tea	wasted
B	astute	hesitate	sedate	sedateness	sedative	sit-down
	stadium	state	state-line	statesman	statistical	status
	steadiness	steady	stout	stoutness	stud	student
	studious	stutter				
012	SATAN	STEIN	sedan	stain	satin	stone
C	Estonia	Sweden	astound	hesitant	pseudonym	sadden
	sadness	satanic	satisfaction	sit-in	sodden	stainless
	stand	standard	stench	stenographer	stint	stoneblind
	stun	stunt	sudden	sweeten		
013	SEDIMENT	STEAM	sodium	stem	esteem	wisdom
D	abstemious	estimate	stamina	stammer	stamp	stampede
	steamer	steamfitter	steamship	stimulant	stomach	stump
014	SATYR	STAR	Saturn	stair	cider	cedar
E	Austria	Easter	Saturday	astir	astray	astronaut
	austere	historian	historical	history	hysteria	ouster
	oyster	satire	setter	staircase	star	starch
	stare	stark	starling	start	steer	sterile
	sterling	stir	stirrup	storage	store	storehouse
	storm	story	strafe	straight	strain	strait
	strap	stratagem	straw	strawberry	streak	stream
	streamer	street	strength	stress	stretch	strict
	stride	strife	strike	string	strip	stripe
	stroke	stroll	strong	structure	struggle	strut
	sturdy	sturgeon	suitor	swatter	sweater	yesterday
015	SADDLE	STOOL	still	stall	steel	style
F	Stalin	Steeler	hostile	sadly	settle	settlement
	settler	stale	stalemate	steal	stealth	stellar
	still-life	stillness	styleless	stylish		

Here is the GUID, again.

`5D00E7AA-C15C-44A5-93FA-A4DE6750B28A`

How about a pseudo-poem about wildlife:

`5D00E`	OWL of WISDOM and SASSY STARING eyes.
`7AA`	HAWK, look SIDEWAYS left and SIDEWAYS right.
`C15C44`	STAND TALL, SUDDEN HARRIER
`A5`	and ASSIDUOUS OWL!
`93FA`	PUMA, STEALthy CITIZEN,
`DE6750`	STAMP! STRIKE! SHAKE, LASSIE!
`B28A`	Don't HESITATE. ENOUGH, CITIZEN!

15.5.2 Hex Method 2

Another technique for hex is to exclusively use the single-word pegs TOES-TISSUE [10..15] for the hexits [A..F]. For decimal digits that spell in 10-15, use two 1-digit pegs. For example, don't spell 15 as TAIL; that's for F exclusively. Spell it as TOE EEL or some such.

Here we go, pegs, first:

`5D00E7AA`	LAW TAM SAUCE TAR KEY TOES TOES
`C15C`	TAN TEA TAIL TAN
`44A5`	REAR TOES LAW
`93FA`	BEAM TAIL TOES
`A4DE6750B28A`	TOES RAY TAM TAR CHALK LACE TOT REEF TOES

Ignore 'ing' in the following, wherein we are assaulted by a Scottish traffic cop:

`5D00E7AA`	A Scottish policeman (LAW) wearing a TAM o'Shanter gives me SASS. Pulls me over on the TARmac and demands my KEY. I TOSS it on his TOES.
`C15C`	"DONE IT WELL, laddie!"
`44A5 93FA A4`	THEN ROARING, "'TIS LOW, you BUM! 'TILL he EATS TAZER!'"
`DE6750B28A`	"DAMN TAR! SHAKE LESS! TIGHT ENOUGH on my TOES!"

Method 1 produces more compact phantasmagorias and extends to higher radices.

15.5.3 SHA-1 by Hex Method 1

```
a638 1b63 d532 69ba 418e
5fb1 bd50 3950 860d d679
```

Using Method 1, ignore common grammatical suffixes like "-ing" and "-ful;" prepositions like "to" and "of;" pronouns and relative adjective,words like "this," and "that;" forms of the verb "to be," and other obvious connector words like "the" and "a."

Let's take a look at a statue in the artist's studio:

A638	SIDEWAYS CHAMFER?
1B6E	the TOE of this STATUE is a SHAM
D532	SODIUM is not ALUMINUM!
69BA	its SHAPE in the STUDIO is SIDEWAYS
418E	ARTFUL HISTORY
5FB1	LAY a STOOL by the STATUE'S TOE
BD50	STATUE is STAMPED "ALICE"
3950	MAYBE ALICE VOUCHES for the STAMP
D678	STAMP of a JACOBIAN?

15.6 Alternatives for Radix-64

15.6.1 Methods 1 and 2, Again

A method-1 (15.5.1) encoding of rixits, reserving 010 through 064, follows:

```
0       8 010   016     024     032     040     048     056     063
v       v v     v       v       v       v       v       v       v
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz+/
```

1. Advantages of this encoding:
   - There are so many picturesque words in English that begin with S that we have a deep well to draw from.
   - The code for a letter is its numerical value in R64: $A = 10$, $O = 24$, etc.
     - Disadvantage of this advantage: we're working on memory, not on evaluation, so maybe this feature isn't so interesting.
2. Problems with this encoding:
   (a) Off-By-One Mental Arithmetic

   Notice that 024 is O, but you may remember that O is also the 15-th capital glyph of the alphabet. Adding 10 for the first ten digits of the encoding, $15 + A = 15 + 10 = 25$, not 24. This is an opportunity for error at speed. But we can deal with it.

   To encode O, remembered as the 15-th letter of the alphabet, first subtract 1 to correct the off-by-one error, getting 14. Then add 10 to account for the leading ten digits 0..9 of the encoding, getting 24. Prepend the leading $s = 0$ for the radix to get 024, say *sunrise*.

   Encode: subtract 1, add 10, i.e., add 9.

   Here's a slogan:

   Encode: enhance nine.

   O $\rightarrow$ 15 $\rightarrow$ 24 $\rightarrow$ 024 $\rightarrow$ sunrise.

   To decode sunrise $=$ 024, remove the leading $s = 0$ to get 24. Subtract 10 to get 14. Add 1 to get 15, the 15-th capital letter.

   Decode: add 1, subtract 10, i.e., subtract 9.

   Decode: deduct nine.

   sunrise $\rightarrow$ 024 $\rightarrow$ 24 $\rightarrow$ 15 $\rightarrow$ O.

   For small letters, add and subtract 26.

   o $\rightarrow$ 15 $\rightarrow$ 41 $\rightarrow$ 50 $\rightarrow$ 050 $\rightarrow$ solstice.

   solstice $\rightarrow$ 050 $\rightarrow$ 50 $\rightarrow$ 24 $\rightarrow$ 15 in lower case.

   It's a lot of mental arithmetic, but that's fun, too. Balance the fun factor against the need for recall speed and accuracy.

(b) ASCII Popups

If you've memorized ASCII, you already have pegs for A-Z and a-z. They will pop up at speed. You can either force them away; use them as with Method 2 above, accepting the trade-off risk of collisions and ambiguity, or go with ASCII in my first method.

15.6.2 Structure in Public Keys

Generating another public key on my machine, notice that the first 6 quads, 24 Rixits, 64 bits each, 1536 bits, are the same in the two keys:

```
ssh-ed25519
AAAA C3Nz aC1l ZDI1 NTE5 AAAA  <~~~ for the machine?
IKWf n2B6 AjfK MNwX +Ekt d7mx
nhwi GALP JfEO oEwk 2X/u
```

By induction, assume without confirmation or authority that the first 24 Rixits are for the machine and $11 \times 4 = 44$ Rixits for the rest. That's 2816 bits.

The total is 4352 bits, or 68 Rixits, consistent with this article:

`https://medium.com/risan/upgrade-your-ssh-key-to-ed25519-c6e8d60d3c54`

That's it for cross-checking; on to memorization.

Let's tell a tale of poisoning an indiscreet lover.

15.6.3 The Supposed Machine Prefix.

Don't bother with the leading and trailing `AAAA`, assuming they're always present (TODO: check that!), leaving `C3Nz aC1l ZDI1 NTE5`:

`C3Nz`	SATIN HEM in a CINEMAtic SASH-WEIGHT
`aC11`	SMOOCH - STAINED, a HOT OATH
`ZDI1`	a SMALL STEAMing from the STOVE, with TEA brewing
`NTE5`	CYANAMIDE for a SNOBbish SUITOR's ALE

15.6.4 The Unique Parts

Now in groups of 3 or 4 quads:

`IFi4`	STUFF the STILL for his ZERO-HOUR HURRAH
`CpLS`	SWEETEN the SALT with a SCENT of SNUFF
`hSK/`	a SERUM SINFUL and SINISTER for his SASHIMI
`uUux`	a SOLUTION SEMISWEET, no SLUDGE in this SLOP
`BxMe`	SATED, he SLIPS into ASSONANT WHIMsies
`6WwK`	HUSHed SOMNOLENCE, SOULFUL SNOOZing
`C6RD`	SATIN HUSH, SNUG STOMACH
`1Qn/`	DIE, SWITCHMAN! SNATCH CERBERUS
`tXQg`	SLOWLY SEMIMOBILE, SNITCH ! SURRENDER!
`BV3m`	SEDATED, to the CEMETERY HOME, SERF!
`pUZq`	SLIDE into OSMOSIS, ASSIMILATE SLAIN!

15.6.5 Old Old Method

I blogged out a more trivial method, similar to method 2 for GUIDs some time ago:

`https://rebcabin.wordpress.com/2011/03/06/memorizing-in-base-64/`

This method maps the rixits directly to the pegs. The mental arithmetic is the same as above: computing a peg number from characters A..Z and a..z by adding 9 and-or-26. We just lose the leading S, so we lose some richness to the vocabulary.

15.6.6 Radix-64 Method 3

Another method, which I have not tried, is to map the alphabetic letters to their S-enhanced letter numbers, as in

A	01
B	02
...	...
Z	026
a	027
b	028
...	...
z	052

The decimal digits go to their regular pegs, 0..9. This is a variation of the old-old-method (15.6.5).

15.7 Memorizing Dates

For most kids, history class is torture because of the need to memorize dates. But that's a snap with the peg system. I still remember from 11-th grade that

RAY TAN CHAT

was the beginning day of the Civil War, because you tanned me with a RAY gun and that ended the peace talks. April 12, 1861.

15.8 Memorizing a Shuffled Card Pack

I blogged this out in December, 2010, starting here:

`https://rebcabin.wordpress.com/2010/12/08/take-the-diamonds/`

Here is a scrape of the relevant blog posts. Note that the encoding of the cards is totally unrelated to that in the [daily-practice] section (Section [BROKEN LINK: daily-practive]. Don't get them confused, and pick only one if you find it difficult to store two encodings.

15.8.1 Take the Diamonds

Do you smell machine oil the instant you see the Diamond Queen from a pack of cards? How about cigarette smoke when you see the Diamond Six, or baby powder when you see the Diamond Ace? Read on . . .

Take the entire suit of Diamonds out of a pack of cards and lay them out in order, from Ace to King, in front of you on a table. Notice that the word "diamond" begins with a D, and recall that the consonant sound D stands for the number 1 – or for the digit "1" in combination with other digits, as we have been using it so far. Think of the suit, diamonds, as the tens digit in combination with the plain digit of each card. Then the diamond Ace will be 11 – 1 for the tens digit from "diamonds" and 1 for the Ace itself, since it's the first card in the suit. The Ace through the Nine come out with no problem at all and they hook up immediately with our memorized, permanent pegs. As you go through the list below, really take some time to stare at the card and visualize the mental picture right on the face of the card. If the deck is one you can write on or draw on, draw a little cartoon or write the picture word on the face of the card. You'll be using these cards to help you lock in the pegs:

Card	*Num*	*Peg*	*Mental Picture*
D A	11	tot	Alice from Alice in Wonderland. Smell baby-powder in the air.
D 2	12	tan	Someone tanning herself holding a reflective tray under her chin. Smell tanning oil.
D 3	13	tam	A tam-o-shanter tartan cap with red pom-pom on the head of a Scottish warrior in full kilts. Smell heather.
D 4	14	tar	A sizzling hot stretch of deserted highway, so hot the tar is sticky. Smell the tar, too.
D 5	15	tail	The orange-and-black striped tail of an enormous tiger. Imagine a zoo smell.
D 6	16	tissue	Someone wiping tears from her eyes after remembering an emotional event. Smell cigarette smoke and think of the song "Smoke Gets in Your Eyes."
D 7	17	tack	A huge thumbtack or map-pin holding something on a big cork board. Smell fresh ink and paper.
D 8	18	taffy	A big blob of hot, sweet-smelling pink and green taffy candy, stuck to your fingers getting all over the card.
D 9	19	tap	A beer tap spilling beer all over the card, splashing on your clothes and making a smelly mess.

Notice I've added smell to the associations, and even a song. We can really take this idea far and involve taste and touch over all the pegs, and we may do that in the future — it will be an interesting exercise. This is one of the things that Eran Katz suggests to do in his books (highly recommended), and since smell works very well for my personal memory, I'm adding it to my permanent inventory and inviting you along for this addition.

We have only four cards to go. The diamond 10 is pretty easy – I'm just going to pick the peg "toes" and picture the woman painting her toenails all different colors. And the smell? Toenail Polish! Easy.

And for the last three, let's promote the 1 to the hundreds digit and sneak in some three-digit pegs (from the third layer – a preview). Notice we may have some extra consonant sounds on the ends of the three-digit pegs. They're throw-aways, because the system doesn't go to four or more digits at a time. 1,110 permanent pegs is quite enough!

Card	*Num*	*Peg*	*Mental Picture*
D J	111	teetotaler	Picture a prissy little man with ill-fitting clothes, a crooked bow tie, thick eyeglasses and a pocket protector pontificating on the virtues of total abstinence from alcohol. Smell Lysol (too much cleaning)
D Q	112	titan	Picture an enormous robot with rivets all over its shell. Smell machine oil and steel.
D K	113	totem	Picture a Native-American totem pole with effigies of bears, salmon, and a huge, winged eagle on top. Smell a campfire and the wood of the totem pole.

That's it. Take your suit of diamonds along with you, pull it out and practice the visualization and imaginary smelling exercise whenever you find yourself having to kill time. Just a few times over this and you won't need the cards any more. You will INSTANTLY smell and see taffy when you think of the Diamond-8. The Diamond Queen will INSTANTLY bring to mind the smell of machine oil and steel. And vice versa, and through the entire suit.

15.8.2 Shuffle the Diamonds

... and go through the 13 cards, in order, forming a linked sequence of vivid mental pictures. Here's what I came up with yesterday

Alice in Wonderland (tot = 11 = DAce) is wiping a tears from her eyes with a

tissue (16 = D6). She is standing by a sizzling hot highway and her feet are sinking into the

tar (14 = D4). Across the highway, a six-foot tall beer

tap (19 = D9) is gushing beer onto the tarmac. The beer boils on contact, releasing a strong beer smell, and runs gently downhill into a grid of huge, multicolored

tacks (17 = D7) pushed into the tarmac. These tacks are also piercing and holding down a tiger's

tail (15 = D5), and the tiger is roaring in pain and licking a huge lump of pink-and-green alcoholic

taffy (18 = D8) by the other side of the road from Alice to soothe his pain. In front of the tiger, a

teetotaler (111 = DJ), specially summoned from AA for the situation, is wagging his finger at the tiger and delivering a stern warning about the evils of alcoholic taffy. The other hand of the teetotaler is exposed to the hot sun and is getting quite a

tan (12 = D2). With that overtanning hand, he is holding the right hand of a steel robot

titan (112 = Q2) who is wagging his head, clicking and beeping, emitting the odor of machine oil, and, with his other hand, painting his

toenails (toes = 10 = D10) and also the toenails of the

totem (113 = DK) standing right next to him (smell the toenail polish). The top of the totem is obscured by a huge

tam-o-shanter (13 = D3), which is baking in the hot sun and emitting a strong smell of roasting wool

Now, I came up with this phantasmagoria yesterday morning in less than one minute, I did not think about it all day, the day was very, very busy with other activities, I went out for a couple of drinks after work, slept through the night, and didn't think of this for a whole 24 hours until just now. I remembered it almost perfectly (I forgot the tacks, and all that means is that I didn't form the mental picture vividly enough – I had the tiger's tail threaded around the tacks, but I

recommended to you to imagine the pins pushed THROUGH the tail, adding the roaring and pain to the phantasmagoria). If I had been more careful, I would have remembered it perfectly.

Think about this: by exploiting the visual imagination and dragging in all the other senses – feeling the heat, smelling the smells, hearing the sounds, I was able to remember 24 hours later, after a minute of looking, a shuffled suit of cards (and, if I had been more careful, I would have remembered it perfectly!). When I am in practice, I can memorize an entire shuffled deck in 2 minutes, while you deal them out (I'll teach you the other suits later, while we go through the second layer of the memory system). When I am out of practice, 4 minutes is plenty. Even better, I can memorize 39 of the cards and then tell you the remaining 13 cards just by running down the suits in my mind and naming the cards that aren't in my phantasmagoria. I can even do it tomorrow from yesterday's shuffle.

This is a great party trick. The pros can do this really fast. The athletes in the world memory championships (`http://en.wikipedia.org/wiki/World_Memory_Championships`) can do it in less than 30 seconds, and they can memorize the order of 8 shuffled decks in 15 minutes.

Try this. You will amaze yourself. Then, think of practical applications.

15.8.3 The 40s and Hearts

Back with the 10's we learned pegs to memorize a shuffling of the diamond suit, in any order. This time, we'll learn pegs for the 40's and for the hearts suit. The reason to associate hearts with the 40's is that the first important consonant in 'hearts' is 'r,' a digit 4, which we promote to the number 40 for the application of memorizing cards. Here are my permanent pegs for the 40's. As always, feel free to tailor this set to your own memory style

Num	Sound	Peg	Description	Others
40	RS	race	A NASCAR race; colorful cars going around; oil, gasoline, smoke, loud engines	rose rice horse
41	RT	rat	huge, greasy rat scurrying through garbage in an alley; garbage smell, long teeth	road word radio
42	RN	rain	driving rainstorm; soaked clothes; dark street trying to hail a taxi; no visibility	huron horn roan
43	RM	ram	giant ram with huge, curled horns; it's saddled and you're riding it; leather & wool	arm worm room
44	RR	rear	distinctive rear end of Corvette automobile; driving it with the top down	rare roar horror
45	RL	rail	brass rail at a wood-paneled bar; can't reach it with your feet; beer and darts	whorl whirl reel
46	RJ	rash	painful, red rash on your forearm	ridge Raj rush
47	RK	rack	rack of antlers on caribou head mounted over the fireplace	wreck rag rug
48	RF	reef	coral reef; brightly colored tropical fish; SCUBA diving gear; rubber, seawater	roof rave wharf
49	RP	rope	heavy burlap or hemp rope in a coil or attached to the thing-to-remember	rib robe harp

Those pegs cover the playing cards Heart Ace (41) through Heart Ten (40). We need three more for the face cards

Heart Jack: 411, rat-a-tat-tat; picture a machine gun; you're carrying it and shooting it at things

Heart Queen: 412, rattan; picture white wicker furniture

Heart King: 413, radium; picture the item-to-remember pulsing with the (imaginary) green glow of radioactive radium.

Pick the diamonds and the hearts out of a pack of cards, shuffle them, and prepare to memorize the order. Take your time: don't rush it. Go for accuracy. The speed will come naturally with practice. Here's an example of a run I did just now, building this story in my mind in real time. If you follow along and build the miniature movie in your mind, you will no trouble recalling the order of the cards. Take special effort to chain the pegs one to another (yes, this is a combined application of chain-memory and peg-memory):

(D2) I'm outside in the hot sun getting a ***tan***, but I'm really burning and it hurts

(H5) There's a tropical drink station on the beach and there is a very prominent brass ***rail*** where I want to rest my feet while I order a cool drink, but I can't quite reach it with my feet

(HQ) I sit in a stool made of white ***rattan***; hear the distinctive creaking sound as you sit and try to reach the rail with your feet

(H3) Sitting next to me is a guy holding on to the reins of his riding ***ram*** and the ram is chewing the rattan (this is an important connection as it's easy to forget the rattan without making the ram do something with it)

(H10) he pulls out of his pocket a scale model of a NASCAR ***race*** track and places it on the counter, and the cars start going around

(DA) Alice from Wonderland (a ***tot***) walks up to take a look at the model race track

(DK) she puts a pint-sized ***totem*** pole right in the middle of the oval racetrack

(HJ) one of the effigies in the totem has a machine gun in its hand and starts shooting at the cars (rat-a-tat-tat)

(H4) many of the cars are Corvettes with their distinctive ***rear*** ends (four circular red lights, 2-by-2). The bullets are shooting out the red lights

(DJ) a teetotaler comes up to us at the tropical bar and lectures on the violent side-effects of drinking alcohol, using the machine-gunning totem as evidence

(H7) he points to a ***rack*** of antlers mounted on the wall of the tropical bar as further evidence: the hunter must have been drinking to go after such a huge caribou

(D9) we all laugh and him and order the bartender to pull us more beers from the ***tap***

(H8) some divers out in the ***reef*** hear the sound of the enormous beer tap pouring beer and pop their heads out of the waves and shout at us to pour some for them, too

(D3) we notice that some of the divers are wearing ***tam***-o-shanters over their dive helmets

(D5) tigers (with orange-and-black striped ***tails***) run along the beach into the waves and snatch the tam-o-shanters in their jaws.

(D6) they run a big circle over toward us, stop and wipe tears from their eyes (with ***tissues***) out of regret for stealing the tam-o-shanters

(H2) all of a sudden, it clouds over and a driving, tropical rain soaks the weeping tigers with tams in their teeth and pounds on the roof of the tropical drink station

(HK) the weather radio in the back of the bar starts to blare news at a high volume about the storm and pulse very brightly with green ***radium*** glow

(HA) huge black ***rats*** scurry out from behind the bar, scared out by the sudden noise of the radio – see the radium glistening on their greasy, rain-soaked fur

(D4) the rats run all the way out to the */tar/*mac in the parking lots near the beach, and the hot tar is steaming from the sudden rain; the rats are jumping on the hot tar

(D8) they run over to piles of hot green and white ***taffy*** piled up in one corner of the tarmac and start eating it voraciously

(D10) on top of the piles of taffy are young women painting their */toe/*nails – yes, in the driving rain – and dripping the toenail polish on the taffy

(D7) the toenail polish is running down the taffy and flowing around the shafts of enormous thumb*/tack/*s embedded in the base of the piles of taffy

(H6) under each thumbtack is a doctor's prescription for ***rash*** medication; it's going to help with all the sunburns we got

(H9) tied to each prescription is a burlap ***rope*** (don't know where this is going, yet, but just tie them up

(DQ) at the other end of each rope is a ***titan***, a robot, pulling on the rope

That's it – now close your eyes and run through the movie in your mind, and the cards will play out in order. Try not to go back and peek. If you have to do that, it means that you didn't form a sufficiently strong association linking one mental picture with the next. Practice will make perfect!

tan, rail, rattan, ram, race, tot (Alice), totem, ratatattat, rear (shooting out the Corvette brake lights) teetotaler, rack, tap, reef, tam, tail, tissue, rain, radium, rat, tar, taffy, toes, tack, rash, rope, titan

D2, H5, HQ, H3, H10, DA, DK, HJ, H4, DJ, H7, D9, H8, D3, D5, D6, H2, HK, HA, D4, D8, D10, D7, H6, H9, DQ

I had to really work hard to recall rattan and rear, which means I didn't form the connected images strongLY enough first time around. practice practice practice.

15.8.4 The 70s and Clubs

By now, you know the drill. The 70's are another decade in the regular peg list of the second layer of the system. Just as the 10's stand for the diamonds (because the 'D' of 'diamond' is a sound for 1, and 1 is the first digit of each visual memory word for the diamond suit), and the 40's stand for hearts (because the 'R' of 'heaRts' is a 4-sound), the 70's stand for the club suit. You will get three bonus visual-words from the third layer for the Jack, Queen, and King of clubs. Practice with the club suit alone, locking in the visual words, then mix in the hearts and diamonds. You will be able to memorize a shuffled pack of these 39 cards and impress yourself. Without further ado, here is my list:

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Num	Sounds	Card	Peg	Description	More
70	KS	C10	case	leather attache case with handcuffs	gas goose wax
71	KT	CA	cat	black cat, green eyes	coat kite goat
72	KN	C2	can	empty coffee can, label torn off	cane canoe gun
73	KM	C3	cam	camshaft from a big detroit V8 engine	cameo comb wigwam
74	KR	C4	car	colorful NASCAR race car	choir crew curry
75	KL	C5	coal	black messy coal or hot one	claw collie eagle
76	KJ	C6	cage	cage for an animal (rabid dog?)	cash gaucho gauge
77	KK	C7	cake	over-tall, toppling wedding cake	gong kayak keg
78	KF	C8	cuff	dress-shirt cuff hanging out of ill-fitting suit	calf coffee guava
79	KP	C9	cap	baseball cap	cape cowboy guppy
711	KTT	CJ	cadet	Navy cadet in dress uniform	cathode cathedral quietude
712	KTN	CQ	katana	samurai chopping with katana sword	cotton catnip kidney
713	KTM	CM	catamaran	absurd inflated raft catamaran	academy godmother Gotham

15.8.5 The 90s and Spades

This is the last set of ten, permanent peg words in the second layer of the system, with three bonus words from the third layer so we can memorize the last suit of a card deck, the spades. Every number in the 90's begins with the digit 9, and the consonant cluster for 9 is B and P. The word "Spade" encodes to 091; ignore the leading zero and pick the 9=P to stand for the spades, so the spades will come from the 90's. In later blogs, we'll summarize the entire second layer, add a couple of odds and ends, and work on applications like card tricks.

Num	Sounds	Card	Peg	Description	Others
90	PS	S10	Base	Home base in a ball field	abess boss pizza
91	PT	SA	Bat	baseball bat in the hands of a slugger	pot bath pod
92	PN	S2	bean	Steak-size kidney bean steaming on a bed of lettuce	bun piano ebony
93	PM	S3	beam	Huge timber in the low ceiling of a pub; hit your head on it	bum opium puma
94	PR	S4	bar	The pub itself: crowded, loud, stuffy	pier bear opera
95	PL	S5	ball	Baseball: stitches, tacky leather, autographs	pool belly bowl
96	PJ	S6	badge	Heavy, bronze sheriff's badge pulling your clothes	bush pasha peach
97	PK	S7	back	exposed back of ballerina	beak book bug
98	PF	S8	beef	Steak dinner (don't confuse with bean)	beehive buffet puff
99	PP	S9	pipe	Rene Magritte's famous painting; or Einstein	babe bobby pope
911	PTT	SJ	potato	baked potato with all the fixin's; see 92, 98	habitat beatitude pas de deux
912	PTN	SQ	baton	relay race; pass the baton	beatnik bedouin petunia
913	PTM	SK	podium	lecturer or preacher at the podium	badminton speedometer epidemic

15.9 Calculating Day of the Week

I blogged this out here:

`https://rebcabin.wordpress.com/2015/07/03/590/`

Here is an edited scrape:

Remember Dustin Hoffman's character in the movie "Rainman," who can instantly say the day of the week for a given date? Of course, he was an autistic savant and probably didn't know and certainly couldn't teach how he does it, but here is a method I learned from my Dad [Furst has the same method].

Preliminaries:

Do all math mod 7, that is, you only need to keep the remainder of the results of any calculation after dividing by 7. Thus, 15 is 1, 33 is 5, 49 is 0, etc.

Every day of the week has a number, starting with 1 for Sunday and ending with 0 for Saturday. Your goal is to convert any Gregorian calendar date into such a number.

Jan and Feb belong to the year BEFORE. The reason is that leap years add a day at the end of Feb and it makes all the math easier to think that Jan and Feb 1963 are really part of 1962, for example. Also remember that the Romans started their year with the month of March. Maybe it was for a similar reason.

Example:

29 Nov 1872.

First convert the century. 1900 happens to be a 0, so that's really really easy and convenient. The numbers for all other centuries go in a cycle 6 4 2 0, which is also really easy to remember: "AGES RUN IN SEQUENCE." Because 1900 is 0, 2000 is 6, 2100 will be 4, 2200 will be 2, etc. 1800 is 2.

Keep 2 in your head and move forward.

Now convert the year, and there are two steps. First, the year mod 7 because each year is one day longer than a multiple of 7, namely 364. 72 mod 7 is 2. Add 2 to the 2 we're carrying to get 4 so far.

Now get the number of leap years in the century. That's 72 / 4, or 18 discarding the remainder; that's 4. Add it to the 4 we're keeping, which is 8, which is 1.

Now get the month. Here are mnemonics for months:

Mar	Irish	4	St. Patrick's Day
Apr	Easter	0	
May	National	2	Memorial Day
Jun	Altar	5	Lots of weddings in June
Jul	Sun or Summer	0	First full sunny month of summer
Aug	Moon	3	Start seeing harvest moons in Aug
Sep	Job	6	Labor Day
Oct	Travel	1	Columbus Day
Nov	Restaurant	4	Eat turkey in a restaurant
Dec	Church	6	Christmas in a church
Jan	New	2	New-Year's Day
Feb	Love	5	St. Valentine's Day

If you forget one, you can recalculate a month number by just adding up the 30 = 2 and 31 = 3 days from the beginning of the year in March (mod 7, of course). The day counts for the months go like this: 32323/32323/30, nice easy pattern, or you

can sing the "30 days hath September, April, June, and November" ditty if you prefer. Whatever works.

We're carrying a 1 in head. Add a 4 for Nov, giving 5. Almost done.

29 mod 7 is 1; add our 5, we get 6. 29 Nov 1872 was a Friday.

Do this a few times and check yourself against Google or timeanddate.com. You'll make a few mistakes when getting started, but after a little while you'll be able to do it quickly and reliably.

Let's try today, 21 May 2022:

2000 = 6
22 = + 1 = 0
ly = + 22/4 = 5 discarding the remainder
May = + 2 = 7 = 0
21 = + 0 (Saturday)
Today is Saturday.

15.10 Person-Action-Object (PAO)

I recently learned the PAO—Person, Action, Object—system and it's quickly becoming my favorite. Almost magically, it helps you string long phantasmagorias together, 6 digits—three pegs—at a time. It's built right on top of the peg system and it is a 3X amplifier.

The PAO system associates 3 phantasmagorias in a little mini-movie to each number. You build up your PAO system by enhancing and stringing together the pegs you already know.

For example, consider these enhanced pegs:

	peg	Person	Action	Object
19	tap	CHIMP	TAPS	BEER
...				
29	nap	PUSSYCAT	NAPS	on the COUCH
30	mass	CONGREGATION	LEAVES	CHURCH
...				
58	leaf	BREEZE	RUSTLES	CANADIAN-MAPLE

Now, just from these, There are 64 different mini-movies you can make up:

192929 CHIMP NAPS on the COUCH 192930 CHIMP NAPS at CHURCH ... 293019 PUSSYCAT LEAVES the BEER ... 301958 CONGREGATION TAPS CANADIAN MAPLE syrup

and so on.

I had a document numbered 30291958 the other day. I needed to remember this number because I needed to refer to the document, by number, in some other places. Without memory training, I would have had to write it down, or copy-paste it to Notepad, or some such annoying, time-consuming, distracting paper-cut of an activity. Instead, I just created a mental movie.

If you had to memorize 30291958, you can do it the old way with four pegs as described earlier in this book, or you can just pick the Person field for 30, the Action field for 29, the Object field for 19, the Person field for 58, and so on if there were more pairs. I quickly get

The CONGREGATION NAPS after drinking BEER and a BREEZE rustles leaves.

I might end up with the same mental movie the old way, but this way is faster and more robust. How about 40331626? DALE EARNHARDT ROLLS EYEBALLS into a NICHE. How do I get DALE EARNHARDT? 40 is RACE, and my PAO for RACE is DALE EARNHARDT CLIMBS into the window of his famous #3 RACE CAR. 33 is MAMA, and my PAO is MAMA ROLLS DOUGH (with a rolling pin). 16 is TISSUE, and my PAO is ALL GORE DABS TEARS from his eyes with a tissue, as he weeps about global warming. 26 is just NICHE, so I didn't invoke PAO for the last two digits.

Knowing my pegs, it was actually almost too easy to come up with my PAO system. The downside for you, the reader, is that my PAO is even more personalized

than my pegs. That fact makes it easy for me to remember, but more difficult to teach. But I'll give it a go anyway.

The key to building your own PAO is to make make sure the persons are unique amongst themselves, ditto the actions and the objects.

My table is just an example. Fill out your own table to suit yourself. You'll find it very quick to do and you will love memorizing it because the pegs are a huge help.

	peg	Person	Action	Object
00	sauce	AMELIA	LADLES	SPAGHETTI SAUCE
01	sot	AQUALUNG	PICKS	a DOG END
02	sun	SUN CHILD (tarot)	REVELS	in SUNSHINE
03	seam	CRIMINAL	RUBS	FINGERPRINTS
04	soar	ALBATROSS	TORMENTS	the ANCIENT MARINER
05	sail	LEO	TRIMS	SAILS
06	sash	SHAH OF IRAN	ASCENDS	the DAIS
07	sock	GRINCH	DONS	HEAD SOCK
08	safe	SAFE	CRUSHES	WILE-E COYOTE
09	soup	CODGER	SPILLS	SOUP

	peg	Person	Action	Object
0	saw	LUMBERJACK	SAWS	LOG
1	tea	MISS MARPLE	POURS	TEA
2	Noah	NOAH	COUNTS	the ANIMALS
3	me	I	TIE	a BOW TIE in the mirror
4	ray	LUKE SKYWALKER	BRANDISHES	his LIGHT SABER
5	law	COP	STOPS	a LAND CRUISER
6	jaw	LURCH	CHEWS	ROSE STEMS
7	key	ALLADIN	TWISTS	KEY in rusty lock
8	fee	LAWYER	HORDES	GOLD
9	pea	PEA	BRUISES	the PRINCESS

Why do you need these single-digit entries? Here's one application: phone numbers, at least in the US, come in 3-3-4 patterns. To remember 258-876-5309 (this is a complete fake: an unassigned area code and the number from the old pop song that everyone knows), I might do

NOAH STOPS a shower of GOLD, his LAWYER TWISTS ROSE STEMS, a BARTENDER (53) SPILLS (09) a drink.

The only thing you didn't see from above is

53, BARTENDER SQUEEZES **LIME** into a bottle of Corona beer.

	peg	Person	Action	Object
10	toes	DANICA	PAINTS	her TOENAILS
11	tot	ALICE	DESCENDS	her THRONE
12	tan	LIZ	TANS	her CHIN
13	tam	ORCAS	FLIP	a TAM o'shanter
14	tar	TARMAC	SIZZLES	a SNAKE
15	tail	GAS-CAP	SNAGS	tiger's TAIL
16	tissue	AL GORE	DABS	his TEARS
17	tack	JUSTIN TRUDEAU	TACKS up	flag of QUEBEC
18	taffy	ROADRUNNER	HIGH-STEPS	in hot TAFFY
19	tap	CHIMP	TAPS	BEER

	peg	Person	Action	Object
20	nose	KING DAVID	BEHEADS	GOLIATH
21	net	FISHING NET	TRIPS	CADET
22	nun	NUN	WALKS	her pet CROCODILE
23	name	TEDDY BEARS	LEANS	a NAME CARD
24	Nero	NERO	PLAYS	his FIDDLE
25	nail	MAILMAN	FLINGS	MAIL
26	niche	ATHENIAN BUST	FILLS	a NICHE
27	neck	NEFERTIRI	POSES	for SCULPTURE
28	navy	NAVY	STEAMS	in the OCEAN
29	nap	PUSSYCAT	NAPS	on the COUCH

It would be a lengthy and uninstructive process to explain how I came up with all these. Suffice it to say that *they work for me* and you can make some that work for you!

16 All Pegs and Sound-Alikes

Letters for hex, ASCII, and R36, R64 alternative methods are included in the headers.

16.1 Caps and Rows

Items in ALL CAPS are recommended pegs: visually strong, easy to reconstruct by enumerating vowels, low collision potential. The first row in each block contains recommended alternative pegs in preference order. The rest of the rows are in alphabetical order.

Blocks without items in ALL CAPS do not have recommended pegs as of this version. Rows in blocks without items in ALL CAPS may not be ordered.

We work on this list in our hobby time. It will be several versions before it's finished.

It takes time to come up with good pegs. It's a quadratic process because one must pick pegs that don't collide visually with the ones already memorized. Picking a new peg means thinking about all the others picked so-far. I already have CHAIR

for 64 (used all the time), so SEAT for 01 and THRONE for 142 are not good ideas for me because they might incorrectly recall 64 at speed.

Most of this this list (except for the double leading zeros) is automatically constructed from a file named `core.clj` deeper in the directory tree of the source repository.

Pick your own pegs from this list, or mark your pegs in ALL CAPS in the `core.clj` file, pick four likely ones for the first line, sort the rest, and regenerate the PDF

```
visiting file core.clj

cider-quit
cider-jack-in
org-babel-execute-buffer

visiting the .org file

remove :results: drawers from any tables you changed
and Ctrl-C Ctrl-C in each babomorph block

org-latex-export-to-pdf
```

you must either manually evaluate `core.clj` via `Ctrl-C Ctrl-C` or start with a fresh cider-repl every time, as Clojure caches the results). You may have to generate the PDF file manually:

```
pdflatex -shell-escape how-to-memorize*.tex
```

at least twice to regenerate cross-references correctly.

000	SOY SAUCE	SASSY SAYING	assize	
001	CYST	SIESTA		
002	CESSNA	ASSASSIN	seasoning	season
003	SESAME seed bun	ASS'S MANE		
004	SCISSOR	CAESAR	ass's reins	
005	C S LEWIS	SIZZLING steak		
006	SAUSAGE	SIOUX SAGE		
007	SUZUKI scooter	SIZING	siezing	
008	SOUSAPHONE	ASS'S FOAL		
009	SWISS BANK	ASS'S BEARD		

00	SAUCE	OASIS	icehouse	sissy	sass	seesaw
	assess	assize	houses	sassy	say-so	seas
	seize	size				
01	SOT	SOD	Swede	acid	aside	asset
	city	east	haste	host	sad	seat
	seed	seedy	settee	side	sight	sit
	site	soot	sooth	south	stay	stew
	suede	suet	suit	suite	swat	sweet
	waste	west				
02	SUN	SWAN	swine	ozone	sinew	snow
	assign	hosanna	sane	scene	seen	sign
	sin	son	soon	zone		
03	SEAM	SWAMI	semi	psalm	asthma	sum
	assume	same	seem	some	somehow	swim
04	SOAR	SEWER	seer	soiree	sir	sour
	czar	eyesore	hawser	hussar	sore	sorrow
	sorry	wiser	zero			
05	SAIL	SEAL	swallow	slaw	sleigh	weasel
	assail	cell	easel	easily	hassle	hazel
	sale	sell	sill	silly	slay	slow
	sly	soil	sole	solo	soul	whistle
06	SASH	SWATCH	switch	sage	ice-show	siege
	assuage	ice-age	sashay	sedge	sewage	such
	usage					
07	SOCK	SACK	ski	sky	scow	sake
	ask	assegai	hoosegow	housing	husky	icing
	sago	sea-cow	seek	sick	soak	soggy
	squaw	suck	whiskey	wise-guy		
08	SAFE	SUFI	sofa	sieve	housewife	savoy
	ossified	save	savvy	suave		
09	SOAP	SPY	wasp	wisp	asp	soup
	sap	espy	sob	sop	spa	subway
	swab	swap	sweep	swipe	swoop	zebu
	zip					

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010	SAWDUST	SWEETS	seducer	hostess	citizen	cities
A	South-Sea	assets	assiduous	sideways	stays	
011	STATUE	STEED	studio	estate	iced-tea	wasted
B	astute	hesitate	sedate	sedateness	sedative	sit-down
	stadium	state	state-line	statesman	statistical	status
	steadiness	steady	stout	stoutness	stud	student
	studious	stutter				
012	SATAN	STEIN	sedan	stain	satin	stone
C	Estonia	Sweden	astound	hesitant	pseudonym	sadden
	sadness	satanic	satisfaction	sit-in	sodden	stainless
	stand	standard	stench	stenographer	stint	stoneblind
	stun	stunt	sudden	sweeten		
013	SEDIMENT	STEAM	sodium	stem	esteem	wisdom
D	abstemious	estimate	stamina	stammer	stamp	stampede
	steamer	steamfitter	steamship	stimulant	stomach	stump
014	SATYR	STAR	Saturn	stair	cider	cedar
E	Austria	Easter	Saturday	astir	astray	astronaut
	austere	historian	historical	history	hysteria	ouster
	oyster	satire	setter	staircase	star	starch
	stare	stark	starling	start	steer	sterile
	sterling	stir	stirrup	storage	store	storehouse
	storm	story	strafe	straight	strain	strait
	strap	stratagem	straw	strawberry	streak	stream
	streamer	street	strength	stress	stretch	strict
	stride	strife	strike	string	strip	stripe
	stroke	stroll	strong	structure	struggle	strut
	sturdy	sturgeon	suitor	swatter	sweater	yesterday
015	SADDLE	STOOL	still	stall	steel	style
F	Stalin	Steeler	hostile	sadly	settle	settlement
	settler	stale	stalemate	steal	stealth	stellar
	still-life	stillness	styleless	stylish		
016	STAGE	STOOGE	stitch	station	hostage	wastage
G	hesitation	stage-fright	stage-manager	stagecoach	stagehand	staging
	station	stodgy				
017	STOCKING	SIDECAR	stake	stick	stucco	stock
H	ascetic	sciatica	setting	sidekick	stack	stacker
	stag	stagger	stagnant	stakeholder	stalk	steak
	sticking	stickup	sticky	stigma	stigmata	sting
	stock-exchange	stockbroker	stockholder	stocking	stoic	stoke
	stoker	stuck	suitcase			
018	STIFF	STOVE	staff	stave	stevedore	sea-devil
I	set-off	staffing	stave-off	stifle	stove-pot	stovepipe
	stuff	stuffing	wasteful			
019	STAB	STEEPLE	stub	stoop	staple	steppe
J	eastbound	estop	seat-belt	sideboard	sideburns	southbound
	stable	steep	steeple	step	step-off	stepfather
	stepmother	stipend	stipulate	stop	stopover	stoppage
	stopping	stubborn	stupefy	stupid	sweetbread	

020	SENSE	SUNS	sneeze	sinus	seance	sunset
K	census	easiness	essence	iciness	science	sensation
	senseless	sensible	since	sincere	sinister	sinus
	snooze					
021	SAINT	SAND	cent	senate	synod	snot
L	Sunday	ascent	assent	sandal	sandbox	sandwich
	sanity	scent	scientific	sent	sentence	sentiment
	sentinel	snit	snoot	snout	sonata	sonnet
	sound	sound-box	sundae	swindle	syndicate	zenith
022	SIGN-ON	SAN JUAN	assonance	sea-onion	snow-knife	synonym
M	son-in-law	asinine	synonymous			
023	CINNAMON	SNOWMOBILE	snowman	cinema	sea-anemone	cyanamide
N	wisenheimer					
024	SINNER	SONAR	senior	snore	scenery	sunrise
O	snare	sneer	snort	snarl	sonorous	
025	SEE	SUNLAMP	snell	snail	sunlight	
P						
026	SIGNAGE	SNOWSHOE	cinch	singe	snatch	censure
Q	ascension	century	sensual	snitch	snow-job	sunshine
027	SINK	SYNAGOGUE	snack	snake	snick	zinc
R	cynic	isinglass	scenic	sea-anchor	sneak	snug
	sunk	sunken				
028	SNUFF	SNOWFLAKE	sniff	snafu	sniffle	saw-knife
S	Seinfeld	sinful				
029	SNAP	snowball	snob	snip	snipe	snoop
T	sauna-bath	sign-up	snapshot	snub	snub-nosed	sunbeam
	sunup	synopsis				

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030 U	SAMSON osmosis	SIAMESE sea-mouse	seamstress semicircle	semester	semisweet	swimsuit
031 V	CEMETERY assumed symmetric	SMITH smite symmetry	Semite smooth	housemaid something	smut sometimes	summit summed
032 W	SALMON Seminole summon	CEMENT awesomeness wise-man	seaman iceman wise-woman	semen sameness	seminary semantics	yes-man somnolent
033 X	OSMIUM Summum Bonum semi-metal	SEMI-MONTHLY same man semi-mobile	semi-mundane same mood semi-moist	same make semi-material semimonthly	same model semi-member semi-motile	same mont semimemo semi-moto
034 Y	SAMURAI smart	SWIMMER smirk	summer	simmer	smear	summary
035 Z	SAWMILL sea-mile assimilate	SMILE sea-mule simulate	semolina seamless simultaneous	smell seemly smolder	small similar small-minded	simile smelt smallpox
036 a	SMASH smidgen	SMOOCH so-much	smudge	smuggle	smuggler	semigloss
037 b	SMACK	SMOKE	sumac	smuggler	seeming	swimming
038 c	SEMAPHORE	SAMOVAR	symphony	semifinal	semiformal	semiofficia
039 d	CYMBAL assembly	SAMBA sample	somber simple	symbol somebody	zombie sympathy	swamp

040 e	SERIES	SEA-HORSE	sewers	source	Circe	serious
041 f	SARDINE assert swarthy	WIZARD assort	sardonic certain	sard seaward	sort seaworthy	hazard sortie
042 g	SARIN Sauron	SIREN surrender	serene sworn	Syrian	serenade	surname
043 h	SWARM	SERMON	serum	ceremony	ceramic	
044 i	SORORITY	SRIRACHA	zero-hour	sour-herring	searer	swearer
045 j	ISRAEL	CEREAL	serial	sirloin	seraglio	sorrel
046 k	SARGE sergeant	SURGEON surcharge	search	serge	surge	sea-urchin
047 l	CIRCUIT circle	SAUERKRAUT circumstance	sarcasm sarong	sour-cream zircon	circus zirconium	sirocco
048 m	SURF servile	SERF sorrowful	surface surfeit	service survey	serve survive	servant
049 n	CERBERUS Serb	SYRUP sauerbraten	surplice serpent	surplus sourpuss	surprise surpass	usurper

050 o	SOLSTICE solace	SALSA useless	slice	solicitor	sluice	sales
051 p	SLOT MACHINE household sled solid	OCELOT isolate sleet solitaire	slate oscillate slide sultan	salt salute slight zealot	salad silhouette slit	assault slaughter slot
052 q	SEA-LION aslant slant	CYLINDER saline sullen	saloon sea-lane swollen	salon silent zillion	sealant slain	slender slander
053 r	SALAMANDER slam	SLIME slim	cell-mate slow-motion	asylum slumber	soul-mate solemn	Islam
054 s	CELLAR salary	SOLAR PANEL sealer	celery seller	soiler slayer	sailor whistler	celerity
055 t	CELLULAR	CELLULOID	soliloquy	sea-lily	slowly	slyly
056 u	SOLUTION isolation	SLEDGEHAMMER zoology	slouch	sludge	slush	slash
057 v	SILK sea-legs	silica select	ceiling slake	slack sleek	sling sulky	slang
058 w	SULFUR salve sliver	XYLOPHONE salver solve	saliva salvo soulfood	salvage self soulful	salvation silver sylph	slave sleeve
059 x	SLOOP celebrate	SLIPPER celibacy	sailboat slip	slab slop	slap slope	sleep slowpoke

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060 y	SAGES suggest	SWATCHES	sieges	switches	sea-chest	sashes
061 z	SAGITTARIUS satiate	sagittal satched	siege-tower	sash-weight	switch-hitter	associate
062 +	SWITCH-KNIFE	SZECHUAN	sea-change	session		
063 ’/	SASHIMI	SWITCHMAN	sea-chum			
064	SUTURE switcheroo	SOJOURN	swisher	seizure	seashore	saw-sharpener
065	SEASHELL	SATCHEL	social			
066	ASSOCIATION	SUSHI CHAI	sash-chain	sage-judge	switch-shift	wise-judge
067	SAGE-GRASS	SWITCHGEAR	sash-cord	siege-coin	switch-cane	
068	SOUS CHEF	SWITCHOVER	swish fop	sash fob	switch-vein	
069	SWASHBUCKLER	SAGEBRUSH	s-shaped	sash-pocket	switchblade	switch-box

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070	SEX six	SOCKS squeeze	saxophone	Saxon	seacoast	success
071	SCOT scat skate	SKEET scatter skid	squid scout skit	socket scuttle waistcoat	ascot sea-god	squatter sect
072	ZUCCHINI askance sickness	scoundrel scan	scanty scandal	skin schooner	sequin second	signal sicken
073	ESKIMO	SYCAMORE	scheme	scum	skim	scumbag
074	CIGAR escarole scare screen squirt	SKIRT escort scarlet secret whisker	Oscar escrow score secretary	scorpion ice-cream scrap secure	screw scar scrawl singer	squire scarce scream square
075	SKULL Skoal sculptor skill	SCALE cycle sea-gull squall	scallop icicle sequel squill	school ossicle sickle suckle	scallion scalp single	scull scholar skeleton
076	SKETCH	SCOTCH TAPE	squash	sagacious	escutcheon	
077	PSYCHIC skeg	SUKIYAKI squawk	asking squeak	easygoing	seaquake	seeking
078	SCAFFOLD scoff	ICE COFFEE skiff	scofflaw	scuffle	scuff	scavenger
079	SKIP escapade squabble	SKIPPER escape squib	scab housekeeper	scoop icecap	scope skeptic	ski-boot ski-boat

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080	SAVES suffuse	CIVVIES	housewives	sophist	sufism	suffice
081	SEAFOOD sea-fight	SOVIET soft	swift	safe-deposit	sift	safety
082	SIPHON seven	SOUVENIR	Savannah	sea-fan	housephone	suaveness
083	SOPHOMORE	SEAFOAM	sea-vampire	suffumigate		
084	SAPPHIRE cipher severe	SUFFRAGETTE saffron suffer	safari savior whosoever	sphere savor	zephyr sever	sovereign several
085	ASPHALT civil	SOUFFLE safelight	seafowl safely	safflower suavely	ice-floe useful	swivel
086	SAVAGE	SAWFISH	icefish	housewifish	sufficient	safe-edge
087	ESOPHAGUS safe-conduct	SAFECRACKER savoy-cabbage	suffocate suffix	civic	safeguard	saving
088	SAFE-HAVEN	SAFE VAULT	save-face	sieve-field	save-virtue	
089	SOFA-BED	SAFE BOX	sieve-pit	sieva-bean		

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090	SPACE auspices	ASBESTOS specify	spice suppose	spouse	sea-biscuit	soupcon
091	SUBTERRANEAN September spite	SPIDERMAN scepter spot	Sabbath soup-tureen spout	hospital spate spud	houseboat speed subdue	spade spit
092	SPIN aspen subnormal	SPAIN espionage subpoena	spine has-been supine	soybean span	spoon spend	spun candy spun gold
093	SPAM subminiature	SPUMONI submit	submarine	submerge	submontane	spume
094	SPAR asparagus separate sport super	soprano aspire sober spray superb	spear aspirin sopor spread supper	spire cypress spark spree whisper	spare isobar sparrow spring zebra	saber osprey spirit spry
095	SEAPLANE esplanade split	ZEPPELIN sibilant spoil	spool sowbelly sublime	spell splash supply	spall spleen	spill splendid
096	SPEECH auspicious	seepage seabeach	species spatial	subject sub-age	special subagent	waspish
097	ICE-PACK aspect zip-code	SPOCK aspic	ice-pick ice-bag	spike sawbuck	spook speak	speck spectacle
098	SPOOF soupfin	soap-flake spiff	subway-fare	subway fire	subaverage	subfreezing
099	SOAPBOX house-paper	SOAP-BUBBLE	sea-poppy	soapbark	soapberry	subprincipal

0	SAW	HOSE	ice	house	haze	zoo
	a-sea	ace	as	ass	assay	ayes
	easy	essay	eyes	has	hawse	his
	is	ooze	sea	sew	so	sow
	us	use	was	wise	woozy	yes
1	TEA	HAT	hood	oat	yacht	widow
	ad	add	aid	at	auto	day
	die	do	dough	due	dye	eat
	eddy	etui	hate	head	heat	heed
	height	hide	hit	hoot	hot	hut
	idea	it	oath	odd	ode	tee
	thaw	thee	they	thigh	though	tie
	toe	tow	toy	wad	wade	wait
	watt	weed	weight	wet	what	wheat
	white	wide	wit	with	youth	
2	NOAH	WINE	honey	knee	ion	win
	an	ana	anew	ani	annoy	any
	ennui	gnaw	gnu	hen	hyena	in
	inn	know	neigh	new	nigh	no
	now	one	own	wan	wean	when
	wino					
3	ME	HOME	ammo	ham	hem	womb
	Emmy	May	may	mow	maw	aim
	him	hum	hymn	whim	whom	
4	RAY	ARROW	roe	hair	oar	wire
	air	are	area	aria	array	aura
	aware	awry	ear	era	err	hare
	harrow	hear	heir	her	here	hero
	hour	hurrah	hurry	ire	or	ore
	our	raw	rye	war	ware	wear
	weary	where	worry	year	yore	your

5	LAW	WHALE	ale	awl	aisle	eel
	Yale	Yule	ail	all	alley	allow
	alloy	ally	aloha	ell	hail	hall
	halo	haul	heal	heel	hell	hello
	hill	hole	hollow	holy	howl	hull
	ill	isle	lay	lei	lie	low
	lye	oil	ole	olla	owl	wail
	wall	wallow	waylay	wheel	while	whole
	will	willow	wool	yawl	yellow	
6	JAW	SHOE	witch	ash	edge	jay
	Asia	Jew	Joe	age	chew	chow
	each	etch	hash	hatch	hedge	hitch
	huge	hush	hutch	issue	joy	jus
	shah	she	show	shy	wage	wash
	watch	wedge	which	wish	woosh	
7	KEY	COW	oak	yak	hawk	hockey
	Hawkeye	IQ	Ike	ache	aggie	ago
	ague	awake	caw	coo	coy	cue
	echo	egg	ego	gay	go	guy
	hack	hag	hang	hick	hike	hock
	hog	hook	hookah	hooky	hug	quay
	queue	wake	walk	weak	week	whack
	wick	wig	yoga	yoke	young	
8	FEE	WAVE	view	hi-fi	hive	ivy
	ave	few	foe	half	halfway	have
	heave	heavy	hoof	huff	oaf	of
	off	via	vie	vow	waif	waive
	weave	wife				
9	PEA	PIE	hippie	buoy	bow	bay
	abbey	ape	be	beau	bee	boa
	bough	boy	buy	by	bye	ebb
	epee	happy	heap	hip	hobby	hobo
	hoop	hop	hope	hubby	obey	oboe
	paw	pay	pew	web	weep	whip
	whoopee	wipe				

10	TOES	daisy	odds	hothouse	idiots	deuce
	White-House	adds	ads	dais	days	daze
	dice	dizzy	does	dose	douse	doze
	eats	hideous	ideas	ides	idiocy	oats
	odious	tease	these	this	those	thus
	toss	twice	twos	wits		
11	TOT	TEDDY	date	tattoo	tidy	toad
	adit	audit	dad	data	dead	death
	debt	deed	deity	diet	dot	dote
	doubt	duty	edit	idiot	oddity	outwit
	thought	tide	tight	tithe	today	tooth
	tutu	without				
12	TAN	TUNA	Dane	dune	dawn	Eden
	Athena	atone	attain	attune	dean	deign
	den	deny	din	dine	done	down
	heathen	heighten	hidden	ten	than	then
	thin	thine	tin	ton	tone	tony
	town	tun	tune	twin	twine	
13	TAM	Adam	dime	dome	tomb	tome
	Edam	atom	autumn	dam	dame	deem
	demi	dim	doom	dumb	dummy	edema
	idiom	item	odeum	odium	tame	team
	teem	them	theme	thumb	thyme	time
14	TAR	tree	tire	tower	tray	tiara
	TRUE	adder	adhere	adore	attire	author
	dairy	dare	dear	deer	diary	door
	dower	dowry	draw	dry	dyer	either
	ether	hatter	heater	hither	hydra	odor
	other	otter	tare	tear	their	theory
	there	thorough	three	throw	tier	trio
	try	udder	utter	waiter	water	weather
	whiter	widower	wither			

15	TAIL	atoll	doll	doily	idol	tally
	Italy	daily	deal	delay	dial	dole
	dolly	dual	duel	dull	dwell	hotel
	huddle	ideal	idle	idyll	outlaw	outlay
	tale	tall	tallow	tell	tile	till
	toil	toll	tool	towel	whittle	
16	TISSUE	ditch	dish	adagio	Dutch	dash
	teach	touch	touche	thatch	duchy	dodge
	attache	attach	adage	whitewash		
17	TACK	DUCK	duke	thong	toga	thug
	Tokay	attack	attic	decay	deck	decoy
	dig	dike	ding	dock	dog	doing
	dong	dung	eating	ethic	haddock	headache
	heading	outing	tag	take	talk	thick
	thing	tick	tuck	twig	tyke	wedding
18	TAFFY	DOVE	toffee	diva	dive	thief
	TV	daffy	deaf	defy	edify	tiff
	tough	tuff				
19	TAP	TUBA	tabby	toupee	tip	tub
	adobe	dab	daub	debut	deep	depot
	dip	dope	doughboy	dupe	tab	taboo
	tape	tepee	top	tube	tup	type

20	NOSE	anise	noose	niece	ounce	news
	nice	noise	ensue	hence	hens	highness
	once	ones	onus	uneasy	wins	wines
21	NET	knight	knot	aunt	nude	nut
	Hindu	India	and	annoyed	ant	ante
	anti	end	endow	gnat	hand	haunt
	hind	hint	hound	hunt	into	knead
	knit	naught	neat	need	night	not
	note	undo	undue	unit	unite	untie
	wand	want	went	wind	window	wound
22	NUN	ONION	neon	union	noon	nanny
	none	nine	ninny	noun	no-one	anon
	anyone	inane				
23	NAME	GNOME	enemy	enema	enum	anemia
	annum	numb				
24	NERO	WIENER	honor	winery	whiner	winner
	anywhere	inhere	inure	narrow	near	newer
	nor	noria	nowhere	owner	unaware	
25	NAIL	NARWHAL	Nile	inlay	in-law	unholy
	anal	annual	annul	inhale	kneel	knell
	knoll	newel	newly	nil	null	only
26	NICHE	NACHO	notch	wench	haunch	hinge
	enjoy	gnash	hunch	inch	no-show	nudge
	unhitch	winch				
27	NECK	INK	Inca	nock	hunk	nook
	Yankee	awning	inning	knack	knock	nag
	nick	nog	unhook	unique	wink	
28	NAVY	KNIFE	nova	knave	nephew	naive
	enough	envoy	envy	inveigh	nave	unify
29	NAP	KNOB	nib	nub	nape	nappy
	Niobe	honeybee	nab	neap	nip	nob
	unhappy					

30	MASS	MOUSE	mace	maze	hummus	mice
	Messiah	aims	amass	amaze	amiss	amuse
	humus	mess	mews	miss	moose	moss
	muse	muss				
31	MAT	maid	moth	mud	mitt	midi
	amid	amity	emit	humid	mad	made
	mate	meadow	meat	meet	midday	midway
	might	mite	moat	mod	mode	mood
	motto	mouth	mute	myth	omit	
32	MAN	MOON	mane	hymen	mine	minnow
	amen	ammonia	human	main	mana	mania
	manna	many	mean	mien	moan	money
	omen	woman	yeoman			
33	MAMA	MUMMY	mime	imam	maim	Mau-Mau
	Miami	mayhem	memo	whim-wham		
34	MARE	HAMMER	Mary	mayor	merry	homer
	amour	emir	humor	mar	marrow	marry
	mere	mire	moor	more	myrrh	omer
35	MAIL	mule	mile	mole	mill	meal
	homely	homily	male	mall	melee	mellow
	moll					
36	MATCH	MASH	mesh	midge	mush	homage
	Amish	image	mashie	mooch	much	
37	MIKE	MUG	mica	hammock	mocha	Mecca
	Mohawk	amigo	amok	among	make	meek
	mock	muck	omega			
38	MUFF	MOVIE	mauve	Mafia	miff	move
	oomph					
39	MAP	AMP	hemp	hump	ump	amoeba
	imbue	imp	maybe	mob	mop	mope
	myopia					

40	RACE	ROSE	racy	raise	rase	raze
	Warsaw	arouse	erase	eros	erose	harass
	hearsay	hearse	heiress	heresy	hers	hoarse
	horse	iris	ours	rice	rise	rouse
	ruse	warehouse	worse			
41	RAT	ROAD	rod	radio	wraith	Earth
	aorta	arid	aright	art	award	erode
	errata	hard	hart	heart	hearth	herd
	hereto	hoard	horde	horrid	hurried	hurt
	raid	rate	read	ready	red	reed
	rid	ride	right	riot	rite	root
	rot	route	rut	ward	wart	weird
	word	worried	worth	wrath	wreath	wreathe
	write	yard				
42	RAIN	REIN	reign	renew	run	ruin
	Aryan	Irene	Orion	Reno	Rhine	arena
	arraign	earn	hernia	heroin	heroine	horn
	iron	irony	rhino	rune	urn	warn
	worn					
43	RAM	RIM	rhyme	ream	room	rime
	Rome	Romeo	arm	army	aroma	harem
	harm	roam	rum	warm	worm	
44	REAR	AURORA	rare	roar	rower	arrear
	error	harrier	harrower	hearer	horror	orrery
	warrior	wearer	worrier			
45	RAIL	RALLY	railway	real	reel	relay
	aerial	earl	early	hourly	hurl	oral
	oriole	rely	rill	roil	role	roll
	royal	rule	whirl	whorl	yearly	
46	RASH	ROACH	rage	ratio	Rajah	reach
	Irish	Russia	arch	harsh	orgy	rich
	ridge	rouge	ruche	rush	rushee	urge
	wretch					
47	RACK	ROCK	rake	rag	reek	ring
	airing	arc	argue	ark	ergo	hark
	hearing	heroic	herring	ragout	rig	rogue
	rook	rug	work	wreck		
48	REEF	rave	wharf	roof	revue	rife
	aerify	arrive	horrify	review	rough	rove
	ruff					
49	ROPE	RAP	rape	reap	repay	rip
	Arabia	Europe	harp	herb	hereby	orb
	rabbi	rib	ripe	rob	robe	rub
	ruby	rupee	warp			

50	LACE	Lassie	lass	lazy	lasso	lease
	Eliza	Lucy	alas	alehouse	alias	allies
	also	always	else	lees	less	lessee
	loose	lose	loss	louse	luce	
51	LOT	EYELID	lad	lathe	late	lady
	Hollywood	alight	alit	allied	allot	allowed
	allude	aloud	although	elate	elite	halt
	health	helot	hilt	hold	holiday	islet
	lath	lead	let	lid	lido	light
	load	loath	loot	loud	lout	lute
	old	wallet	wealth	wild		
52	LANE	LION	lawn	lean	lien	line
	Halloween	alien	alone	hellion	loan	loin
	loon	woolen				
53	LIME	LLAMA	lame	lamb	lima	lime
	Limey	alum	elm	haulm	helium	helm
	limb	loam	loom			
54	LAIR	LYRE	layer	lawyer	leer	liar
	Holy-Year	allure	healer	holler	lorry	lower
	lure	oiler	whaler			
55	LILY	OIL-WELL	holy-oil	Loyola	lowly	whale-oil
	Walhalla	hallelujah	hula-hula	ill-will	lolly	loyal
	lull	lulu				
56	LASH	LEECH	latch	leash	liege	ledge
	Welsh	allege	hellish	lodge	loge	lush
	ullage					
57	LAKE	hollyhock	elk	league	lug	lung
	ailing	alike	along	hillock	hulk	ilk
	lac	lack	lackey	lag	leak	leek
	leg	lick	like	lock	loco	log
	long	look	low-key	luck	whaling	
58	LEAF	WOLF	lava	olive	elf	loaf
	alive	aloof	laugh	leave	levee	levy
	life	live	love	luff	welloff	
59	LAP	WALLABY	elbow	lobe	lab	loop
	leap	alibi	alp	elope	help	lip
	lob	lobby	lop	lowboy	wallop	whelp

60	CHESS	CHEESE	chassis	chase	chaise	choice
	Jaycee	Jewess	ages	ashes	choose	joyous
	juice	schuss	wages			
61	JET	JUDO	cheetah	chateau	jade	jetty
	aged	chat	cheat	chit	chute	jot
	jute	shad	shade	shadow	sheath	shed
	sheet	shoot	shot	shout	shut	washed
	widget					
62	CHAIN	chin	gin	ocean	china	hygiene
	Jane	John	June	Shawnee	chine	join
	shine	shown	shun	widgeon		
63	CHIME	CHAMOIS	chum	gem	gym	jam
	Jim	jamb	sham	shame	shammy	shim
	shimmy					
64	CHAIR	JAR	sherry	washer	usher	cherry
	Hoosier	assure	azure	char	cheer	chore
	edger	hosiery	jeer	jury	osier	share
	shear	sheer	shire	shore	shower	sure
	wager					
65	JAIL	CHILI	jelly	jewel	shell	shawl
A	Chile	July	agile	chill	jolly	jowl
	shall	shallow	shill	shoal	shyly	usual
66	JUDGE	HASHISH	juju	Jewish	cha-cha	choochoo
B	chichi	chowchow	jaygee	josh	shoji	shush
	wishy-washy					
67	CHALK	JUG	jack	sheikh	chick	cheek
C	Django	aging	check	chic	choke	chuck
	chug	etching	itching	jingo	jock	jockey
	joke	shack	shag	shake	shako	shock
	shuck	washing	watching	wishing		
68	CHEF	CHEVY	chief	sheaf	chaff	chive
D	Java	Jehovah	achieve	chafe	jayvee	jiffy
	jive	shave	shove	wash-off		
69	CHIP	JEEP	jib	sheep	ship	Job
E	GP	chap	chapeau	cheap	chop	chubby
	gyp	jab	jibe	job	shabby	shape
	ship	shop				

70	CASE	GOOSE	gas	kiss	ox	eggs
F	accuse	acquiesce	ax	cause	causeway	chaos
	cozy	cuss	gauze	gaze	guess	guise
	guys	hex	hoax	quasi	wax	
71	CAT	GOAT	kit	kid	co-ed	caddie
G	God	Kate	acquit	act	acute	coat
	code	cot	cut	cute	gaiety	gait
	gat	gate	gaudy	get	ghetto	goad
	good	gout	guide	kite	kitty	quiet
	quit	quite	quota	quote	weekday	wicked
	wicket					
72	CAN	GUN	coin	queen	cone	cane
H	acne	again	agony	awaken	canoe	coon
	gain	gown	guinea	icon	keen	kin
	oaken	wagon				
73	CAM	COMB	acme	calm	cameo	coma
I	come	comma	gam	game	gum	hokum
	oakum	wigwam				
74	CAR	CHOIR	gear	curry	hiker	hooker
J	accrue	acquire	acre	agar	agree	anger
	care	carry	coheir	cohere	core	cower
	crew	crow	cry	cur	cure	eager
	eagre	ecru	euchre	gory	grey	grow
	guru	hugger	occur	ogre	okra	quarry
	queer	quire				
75	COAL	GLUE	coil	eagle	gull	quill
K	Gaul	call	claw	clay	clew	clue
	collie	cool	coolie	equal	gal	gale
	gall	galley	glee	glow	goal	gully
	hackle	haggle	heckle	kale	keel	keyhole
	kill	ogle	quail	ugly	weekly	
76	CAGE	GAUGE	cashew	gaucho	coach	couch
L	cash	catch	cache	cachet	gash	gage
	gush	gauche	ketch	kedge	hoggish	hogwash
	waggish					
77	CAKE	CAULK	cook	gong	king	cookie
M	cock	coke	cacao	cuckoo	gag	gangway
	gang	going	goo-goo	gook	kaka	kayak
	khaki	kick	keg	quack	quake	quick
	aching	agog				
78	CUFF	CAVE	calf	cafe	coffee	agave
N	cough	cove	covey	gaff	give	goof
	guava	guff	guffaw	quaff		
79	CAP	COWBOY	cob	cab	guppy	cube
O	cabby	cape	coop	cop	cope	copy
	cub	cup	gab	gap	gape	hang-up
	hiccup	keep	occupy	okapi	quip	wickiup
	wicopy					

80	FACE	VASE	fuse	vise	visa	vice
P	efface	effuse	fizz	fusee	fuss	fuzz
	office	phase	voice			
81	FAT	PHOTO	foot	fiat	voodoo	aphid
Q	afoot	aft	aviate	avid	avoid	effete
	evade	fad	fade	faith	fate	feat
	feed	fete	feud	fight	fit	food
	off-white	ovate	veto	void	vote	waft
82	FAN	PHONE	vine	fawn	van	fanny
R	Vienna	avenue	euphony	even	fain	feign
	fen	fin	fine	fun	haven	heaven
	hyphen	often	oven	ovine	phony	vain
	vane	vein	woven			
83	FAME	FOAM	ovum	fume	vim	
S						
84	FARE	FIRE	Pharaoh	fairway	ferry	ivory
T	afar	affair	affray	aviary	euphoria	ever
	fair	fairy	far	faro	fear	fiery
	fir	for	foray	four	foyer	fray
	free	fry	fur	furrow	fury	half-year
	heifer	hoofer	off-year	offer	over	vary
	veer	very	viewer	waiver	waver	weaver
	woofer					
85	FALL	EIFFEL	flea	veil	viola	waffle
U	avail	avowal	awful	fail	feel	fell
	fellow	file	fill	filly	flaw	flee
	flow	fly	foal	foil	follow	folly
	fool	foul	fowl	fuel	full	hovel
	offal	oval	phial	vail	valley	value
	veal	vial	vile	volley	vowel	
86	FISH	FUCHSIA	fudge	effigy	phage	voyage
V	affiche	fetch	fichu	offish	vetch	vouch
87	FIG	FANG	fog	fake	Vogue	havoc
W	evoke	fag	foggy	fogie	having	heaving
	offing	vague	vic	viewing	waving	weaving
88	FIFE	FIEF	five	viva	half-off	
X						
89	FOB	FOP	veep	vip	Phoebe	phobia
Y	faux-pas	fib	half-pay			

90	BASE	BUS	pizza	bass	booze	boss
Z	PS	abuse	abyss	apiece	apis	appease
	basso	bias	boys	busy	buzz	epos
	obese	oppose	opus	pace	pass	pause
	peace	piazza	piece	pious	poise	pose
	pussy					
91	BAT	BOAT	bath	opiate	poet	bait
	abate	abet	abide	abode	about	apathy
	apt	bad	bead	beat	beauty	bed
	bet	bid	bight	bit	bite	body
	boot	booth	both	bout	boyhood	bud
	buddy	but	butt	habit	opt	pad
	pat	path	peat	pet	petty	piety
	pit	pity	pot	pout	put	putty
92	BEAN	BUNNY	pan	pony	piano	pine
	ban	bane	bin	bine	bone	boon
	bun	ebony	happen	pain	pane	pawn
	pen	penny	pin	pone	pun	weapon
93	BEAM	bomb	puma	palm	bum	balm
	Bohemia	boom	opium	poem		
94	BAR	bear	berry	opera	pear	prey
	Boer	Hebrew	abhor	apiary	appear	bare
	beer	beware	bier	boar	bore	borough
	borrow	brew	brow	bureau	bury	buyer
	hopper	pair	par	pare	parry	payer
	peer	pier	poor	pore	pour	power
	pray	prow	pry	purr	pyre	upper

95	BALL	BELL	opal	bowl	apple	pillow
	able	appall	appeal	apply	bail	bale
	belay	belly	below	bile	bill	billow
	blow	blue	boil	bull	bylaw	eyeball
	happily	highball	hobble	hoopla	pail	pal
	pale	payola	peal	peel	pile	pill
	play	plea	plough	plow	ploy	ply
	pole	poll	pool	pull	pulley	uphill
	wobble					
96	BADGE	apogee	bijou	hibachi	bush	pouch
	Apache	abash	apish	batch	beach	beech
	botch	boyish	budge	page	pasha	patch
	peach	pitch	poach	posh	pudgy	push
97	BACK	BIKE	book	bouquet	buck	bog
a	aback	bag	bake	bang	beak	beck
	beg	being	big	bogey	bug	buggy
	epic	epoch	opaque	pack	pang	peak
	peck	peek	peg	pick	pig	pike
	pique	poke	puck			
98	BEEF	BEEHIVE	buffet	puff	buff	payoff
b	ab-ovo	above	behalf	behave	bevy	biff
	buffo	pave	peeve			
99	PIPE	BABY	papa	pope	poppy	puppy
c	babe	bib	bob	bobby	boo-boo	boob
	booby	bop	hubbub	pap	peep	pip
	pop					

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100 d	THESIS disuse	decease dissuasive	decisive daisies	desist deuces	disaster doses	disease diocese
101 e	TOAST twist destruct distress atheist	tacit deceit distance distribute attest	taste decide distaste district audacity	test deist distinct distrust outside	testimony destiny distinguish disturb	Tuesday destitute distract dust
102 f	DOZEN disinfect Whitsunday	decent disinterest Docent	descent disown	design dissent	dishonest dissonant	disinclined dizziness
103 g	TWOSOME dismember atheism	theism December decimeter	deism dismiss decimate	decimal dismount	disembark dissemble	dismay dissimilar
104 h	DOWSER desire disregard douser	dicer disarm disreputable	thesaurus disarrange disroot	desert discern dissertation	dessert dishearten disrobe	deserve disorder disrupt
105 i	DIESEL dislocate	tassel dissolve	dazzle docile	desolate deciliter	disillusion disloyal	dislike white-sale
106 j	WOOD-sage	dosage	decision	discharge	dissuasion	autosuggestion
107 k	TUSK disclose discourse disguise	task disconnect discover disgust	desk discontent discrete disquiet	disk discord discriminate dusk	discolor discount discuss outskirts	discomfort discourage disgrace doeskin
108 l	ADHESIVE disavow	tea-sieve disfigure	theosophy disfavor	deceive	decipher	disaffect
109 m	TEASPOON decibel disappoint dispel disposition	thespian despite disapprove disperse dissipate	toss-up despond disciple display	despot disabled disobey displace	despair diaspora dispatch displease	deception disappear dispense disposal

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110 n	TIGHTS	tedious	toothsome	tootsy	deduce	detest
111 o	TEETOTALER doodad	tete-a-tete dotted	tidied doubted	dietetic attitude	day-duty audited	deadwood outwitted
112 p	TITAN tighten	Titanic deaden	tetanus detain	tea-dance detonate	two-tone dead-end	tidiness
113 q	TOTEM datum	diadem daytime	teatime	two-timer	tedium	TeDeum
114 r	TATAR tooter deteriorate auditor	tater totter determine editor	tea-tray tutor detract headwater	tatter dater detriment hot-water	theater daughter daydreamer	thither deter diatribe
115 s	TITLE total deadly	tattle dawdle detail	tittle doodle thoughtless	tidal dateline	tootle deadline	titillate dead-letter
116 t	DEATHWATCH	detach	death-wish	dotage	hot-dish	dead-issue
117 u	HOT-dog detect	tie-tack detective	toothache tidings	dating idiotic	dedicate	deduct
118 v	AUTO-da-fe deadfall	tutti-frutti additive	thoughtful auditive	dative	dutiful	dutyfree
119 w	TEDDY-bear deadbeat	tadpole deadpan	tea-table deathbed	toothpick ditty-bag	tidbit	toothbrush

120	DANCE	tinsel	tenacity	tennis	tense	thence
x	tonsil	dense	dunce	ad-nauseam	Adonis	Athens
	audience	haughtiness	dynasty	utensil	whiteness	witness
121	DOUGHNUT	tent	taint	taunt	tenth	tendon
y	tend	thunder	tint	tender	twenty	TNT
	twi-night	tantalize	dainty	dandy	denote	dent
	dentist	denude	dint	dwindle	attend	authentic
	addendum	identical	identify	whodunit	astound	
122	ATHENIAN	tenant	tenon	teeny-weeny	denounce	adonian
z	hootenanny					
123	TENEMENT	thin-man	tonometer	town-meeting	denim	Denmark
	denomination	dynamo	dynamite	atonement	attainment	autonomy
	outnumber					
124	TENOR	tanner	teener	tenure	thinner	tone-arm
	tuner	diner	dinner	donor	downright	wet-nurse
	whitener	itinerary				
125	TUNNEL	toenail	thinly	tonal	town-hall	denial
	downhill	ethanol				
126	TEENAGER	tonsure	tenacious	tannage	tension	tonnage
	township	Danish	dunnage	donation	dungeon	twinge
	astonish	attention	wooden-shoe			
127	TANK	thank	think	tincture	tinker	tonic
	tunic	twinkle	dank	dinky	dining	donkey
	Don-Quixote	ethnic	hoodwink			
128	TUNA-FISH	tinfoil	tenfold	ten-footer	tea-knife	tuneful
	twinflower	de-novo	downfall			
129	DUNE-BUGGY	tenpin	tenpounder	tinplate	tune-up	twin-bill
	down-payment	downbeat	downpour	donnybrook	Danube	

130	THAMES damask atomizer	Thomas damsel atmosphere	teamster domestic utmost	team-song domicile	Thames demise	theme-song admissible
131	TOMATO thumbtack dumbwaiter esteemed	tamed daymaid adamite	tom-tom deemed admit	timid demit automate	tumid diameter odometer	timeout dumdum estimate
132	DOMINO tameness dimension admonish	T-man demand diminish autumnal	Tammany demeanor dominate headman	thiamine demonstrate domain ottoman	time-honored demon dominion	thumbnail diamond administer
133	DEMIMONDE	teammate	time-money	time-immemorial	dime-museum	demi-magic
134	TAMER tumor	tamarind demure	timer demarche	tomorrow demarcation	tommyrot admirer	Tia-Maria admiral
135	TAMALE timely oatmeal	time-lag time-lock	time-loan demolish	tumult dimly	timeless dome-light	time-limit etymologist
136	time-sheet admission	time-chart automation	tam-o-shanter atom-age	damage Edam-cheese	demijohn estimation	dumb-show
137	TOMAHAWK tummyache democrat	tomcat taming atomic	team-captain timing	time-card teeming	timekeeper demagogue	tommy-gun demigod
138	ATOM-FORCE dumbfounded	tomfool duumvirate	time-fuze hit-movie	time-value	team-foul	time-off
139	TOMBOY template tombolo dumbbell	temper team-play thimble dump	temperature team-boat thump automobile	temperance timber tumble	tempest timpani damp	temple tympan dimple

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140	TERRACE	Thursday	address	atrocity	authoress	dress
	dresser	dressing	drizzle	drowsy	duress	otherwise
	tea-rose	thirst	thrice	throes	torso	trace
	tress	trews	tricycle	truce	waitress	
141	TURRET	adroit	authority	dart	dearth	diehard
	dirt	dread	drought	druid	iterate	outright
	tardy	tart	territory	third	thirteen	thirty
	thread	threat	throat	thyroid	trade	trait
	tread	treadle	treat	treaty	trident	triton
	trout	truth	turtle			
142	TERRAIN	adherent	adorn	astronaut	attorney	darn
	dirndl	drain	drench	drink	drown	drunk
	dryness	eternal	outrun	thorn	throne	torrent
	tourney	train	trainee	trance	translation	transport
	trench	turn	turnip	tyrant	utterance	
143	DREAM	tearoom	term	terminal	thermometer	torment
	tram	tramp	trampoline	tremble	trim	trombone
	trumpet	theorem	turmoil	diorama	drama	dermatology
	dram	drome	dormer	dormitory	drum	dramamine
	uttermost	watermelon	watermark			
144	TERROR	terrier	trier	darer	dearer	drawer
	dreary	dryer	dry-run	adorer	outerwear	waterer
	water-rat	water-right				

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145	TRAIL	trial	trawl	thrill	trellis	trowel
	trolley	twirl	truly	truelove	tarheel	trilogy
	darling	derail	drill	droll	door-lock	water-hole
	water-lily	astrologer	Waterloo			
146	TRASH	tragedy	torch	thrash	thrush	traject
	treachery	treasure	tear-sheet	dredge	derision	dirigible
	drudge	duration	adoration	atrocious	authorship	hydrogen
	outrage	ostrich	attrition			
147	TREK	truck	track	tractor	tract	trick
	tricolor	turkey	target	turk	throng	dark
	daring	direct	doorkeeper	drag	dragon	drawing
	drug	during	director	dreg	dragnet	drake
	dirk	attract				
148	TREIF	turf	trophy	tariff	terrific	terrify
	thrift	thrive	traffic	travel	trifle	truffle
	tearful	trove	trough	drivel	draft	drift
	drive	drove	draught	derive	dwarf	atrophy
	hydrophobia	waterfall	weather-vane			
149	TRAP	tarpaulin	tripod	trapeze	treble	terrible
	throb	torpedo	trip	tribe	tribunal	triple
	troop	troupe	trouble	turbine	turbot	tribute
	drab	drape	drawback	drip	droop	drop
	durable	derby	adorable	attribute	water-boy	waterproof

150	TELESCOPE tollhouse idealize	tails delusive idolize	tales dollhouse headless	tools doilies hatless	tiles dailies outlast	tallies atlas utilize
151	TALIT too-late dilute etiolated	toilet toehold doldrums idolatry	telltale daylight dolt utility	tilt delight adult withhold	twilight delude adultery outlet	teletype dilate athlete
152	TALON DEW-Line idleness	tail-end delineate Italian	toluene delinquent outline	towline headline	talent hot-line	dalliance headland
153	THALLIUM Talmud hoodlum	tollman dalmatian	teleman dilemma	telemark dolomite	toolmaker head-lamp	tallyman hotelman
154	TAILOR dealer yodeler	tiller dialer	tolerate dollar	tailer dueler	teller hotel-room	toolroom idler
155	DAY-lily tallowless	telelens teleology	tall-oil Dalai-Lama	taillight	telial	tileless
156	TOOL-shed delicious etiology	tail-shaft deluge	tallish delusion	tillage diligence	tollage dilution	theologian adulation
157	TAILGATE dealings headlong	telegram delegate outlook	toll-call delicacy wedlock	telling dwelling	tollgate dialect	talc dialogue
158	TAIL-fin deliver	telephone doleful	television dolphin	twelve Delft	tell-off dollface	delve outlive
159	TULIP tallboy dilapidate	tailpiece tollbooth dollop	tholepin toolbox doll-up	tilbury talebearer ad-lib	tailpipe day-laborer	telepathy deliberate

160	DUCHESS audacious	tea-chest headcheese	touchstone	touch-system	digest	dishes
161	DIGIT dashed hotshot	touchwood dodged whitewashed	touchdown dish-towel	touch-type dishwater	ditched ditchdigger	dished attached
162	EDITION addition	tuition audition	teach-in outshine	tee-joint hesitation	dudgeon	Dutch-uncle
163	TAJ-MAHAL	touch-me-not	touchmark	Dutchman	attachment	
164	TEACHER autogiro	T-shirt etagere	dasher wheat-germ	dodger	dowager	de-jure
165	DASH-LIGHT	touchline	touchhole	Dutch-elm	Dutch-light	
166	DISHWASHER	touch-watch	dodge-chain	Dutch-cheese	head-judge	
167	thatch-grass hatcheck	teaching woodchuck	touching	dashing	dodging	deject
168	DUTCH-OVEN wheat-chaff	touch-off	Dutch-wife	dash-off	disheveled	dishful
169	DODGE-BALL dish-up	tissue-paper	toy-shop	touchback	touch-up	odd-job

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170	TAXI things doghouse	tax tea dogs	Texas cozy attics	text decks ethics	textile dachshund outcast	tongs dexterity
171	DOCTOR together docket edict	teakettle teakwood dictate etiquette	tact tag-day decathlon educator	thicket decadent ductile	ticktack decade dogwood	ticket decode adequate
172	TOUCAN technical	deacon thicken	decanter thickness	dignity token	taken tycoon	tea-wagon
173	DOCUMENT degum	Tecumseh dogma	autogamy thingumajig	day-camp	decamp	decompose
174	TIGER decorate digger ticker	attacker decrease digress	autocar decree doggerel	autograph decry outcry	dacron degrade outgrow	dagger degree tea-garden
175	TACKLE ducal toggle	dangle tag-line tongue-lash	decal tangle	declaim tequila	declare tickle	decline tingle
176	DICTIONARY eduction	dockage take-issue	dog-show tuck-shop	doggish	dogwatch	education
177	HOT-cake tacking	attacking tagging	decaying tea-cake	digging	dogging	outgoing
178	DOGFISH duckfooted	TAKEOFF hot-coffee	dog-food outgiving	dogface tick-fever	dogfight tick-off	dogfoot wedding-vow
179	teacup hiding-place	take-up	toe-cap	dig-up	decapitate	decoupage

180	DEVICE deface advise	two-faced devastate edifice	two-fisted deficit whiteface	two-phase diffuse	thieves divest	typhus dovehouse
181	DAVID deft diphthong taffeta	Tophet deviate divide theft	daffodil devoid eightfooter tuft	daft devote hotfoot typhoid	davit devout outfit	defeat diphtheria outvote
182	TIFFANY defiant advance	typhoon define advantage	toughness dauphin	tiffin divan	deafen divine	defense advent
183	DEAFMUTE	tea-family	tough-minded	TV-man	defame	
184	TAVERN defraud diverse	taffrail defray divorce	tovarich devour defrost	thievery diaphragm deaf-ear	defer differ adverse	deform diver advertisement
185	TEFILLIN duffel tofolli youthful	defalcation hateful tough-luck	defile hatful toyful	deflate outfield twofold	develop outflow white-elephant	devil tivoli wistful
186	DEVOTION diffusion	TV-show division	thievish audiovisual	deficiency whitefish	deviation hot-fudge	dovish
187	TOUGH-guy edification	de-facto	defect	difficult	diving	advocate
188	TV-fan	two-fifths	thief-vault	dive-off	eighty-five	
189	DIVE-BOMBER	taffy-pull	tough-boy	edifiable		

190	TAPAS	autobus	autopsy	debase	deposit	dubious
	headpiece	outpace	tapestry	tips	tipsy	top-secret
	top-sergeant	tops	topsail	topside	topsoil	topsy-turvy
	tubes	tubs	types			
191	DEBUTANT	adapt	adept	adopt	debate	debit
	depth	deputy	hotbed	outbid	output	teapot
	tepid	tiptoe	top-dog	top-hat	topdrawer	towpath
	two-bit					
192	HATPIN	tiepin	topknot	topnotch	thighbone	twopence
	debenture	deepen	debonair	depend	depone	Dubonnet
	Ethiopian	headpin	outbound	headband		
193	topman	topmast	topmost	top-milk	topminnow	
194	TYPEWRITER	dapper	daybreak	debar	debris	depart
	department	deportment	depreciate	depress	deprive	diaper
	dipper	outboard	outbreak	taper	tapir	taproom
	tea-party	tie-bar	top-rate	top-round	topper	topwork
195	TABLE	audible	dapple	deplore	diabolic	diploma
	diplomat	double	doublet	duplicate	edible	oddball
	tableau	tablet	tipple	topless	topple	tupelo
	two-ply					
196	DEBAUCH	adaption	adoption	tip-sheet	top-sheet	top-shelf
	two-page	type-wash				
197	TOBACCO	autobiography	debacle	debug	depict	dipping
	hatbox	tapping	tea-bag	toboggan	topcoat	topic
	topkick	topping	typical	typing	typography	
198	TYPEFACE	deep-felt	deepfreeze	tip-off	top-heavy	topflight
	two-by-four	typify				
199	TOP-BANANA	top-billing	top-boots	wet-pipe	white-poppy	white-paper
	depopulate	dope-peddler	dope-pusher	wastepaper	hot-pepper	

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200	GNOSIS insist census	necessity incessant	nieces unceasing	noses unseasoned	noises unsuspicious	incisor ounces
201	NOSTRIL honest instead unsuitable	nest incident instinct gnostic	nasty incite institute one-step	nicety inside instrument Wednesday	nose-dive insight unsatisfactory	anesthesia instant unsteady
202	ENSIGN ensnare uneasiness	newshen incinerator unseen	nice-nelly innocent unsound	niacin insane unison	nosiness insensitive unessential	nuisance insincere
203	INSEMINATION enzyme unassuming	newsman hansom unsimilar	noisemaker inasmuch	news-media insomnia	nizam in-sum	noisome unseemly
204	CENSOR insurmountable	answer insurrection	encircle newsreel	incertitude newsworthy	insert uncertain	insurgent unsworn
205	NACELLE insulin nozzle	ancillary insult nuzzle	enslave nasal oneself	insolence nestle unselfish	insolvent nicely wine-cellar	insular noiseless
206	NOSE-JOB	news-agency	incision	ionization	unsociable	
207	NOSE-cone insecure	nosegay insignificant	encyclopedia unicycle	honeysuckle unscrupulous	inscription wainscot	insect
208	IONOSPHERE uncivil	news-vendor unsafe	henceforth unicef	insufferable	insufficient	once-over
209	NEWSBOY inseparable unspent	nosepiece insipid	newsprint inspect	no-soap inspiration	newspaper insuperable	inhospitable insupportable

210	NUDIST entice industry undisturbed	Nazi handsaw no-dice	antacid hind-sight notice	anticipate indecent nowadays	antiseptic indistinct undeserved	enthusiasm induce undesirable
211	UNTIED handed night-duty united	annotate handout nitwit untidy	antidote haunted noted want-ad	antitrust indeed nudity	entitle knighthood undated	entity needed undoubted
212	ANTENNA centennial intend wanton	Anthony entwine intense	Indian handiness intonate	andante inattentive one-to-one	antiknock indent undeniable	antonym india-ink unthankful
213	ANTIMATTER anytime inadmissible ointment	anadem endemic indemnify on-time	anathema handmade intimacy onetime	anatomy handmaiden intimidate untimely	anthem handyman nightmare windmill	antimissile honeydew-melon nutmeg
214	NUTTER anthracite entrance interact interview nitrate unit-rule	Netherlands endear entry interest introduce nitric untiring	Unitarian endorse handrail interior neither nodder winter	another endure handwriting intern network notary wonder	anterior enter hinterland international neutral nutriment	anteroom entertain hundred interval nightrobe under
215	NOODLE indolent needle	antelope indulge needless	anthill intellect new-deal	endless intelligent night-owl	handle intolerable until	indelicate know-it-all windless
216	nightshade indigenous window-shopping	night-watch indigestion	notation intuition	nutshell untouchable	annotation windowshade	antechamber wantage
217	NIGHTGOWN antagonist handicraft indigo untangle	nothing antichrist hunting indignant winding	nightcap antics handcuff induction wanting	nautical antique handicap intact	nightingale anything inadequate integrity	nutcracker ending index intoxicated
218	NIGHTFALL antiphony undeveloped	native inadvisable undefeated	notification indifferent	endeavor individual	endive indefinite	handful undefined
219	NOTEBOOK antipathy handball windpipe	notable antipode inaudible	night-pay antipope independent	antibiotics handbag inedible	antibody handbill knee-deep	antipasto handbook knotty-pine

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220	NONCE annoyance	nonsense enhance	nonuse onions	nainsook onionskin	ananas unanswered	announce
221	UNKNIT nontechnical unwanted	noontime anoint union-dues	ninth innuendo	nineteen know-nothing	ninety unintelligible	nondenomina unintentional
222	NONUNION unknown	ninon unannounced	no-nonsense	nonnegotiable	nonentity	non-owner
223	NOON-MEAL nonmilitary	ninnyhammer nonemergency	nonemotional anonymous	nonmetal inanimate	nonmember unanimous	nonmedical unanimated
224	NUNNERY unhonored	nonracial	nonradical	onion-ring	union-organizer	unenrolled
225	nonliquid unenlarged	nonliterary unanalyzed	nonelect union-hall	nonallied union-labor	nonelastic union-leader	unenlightened
226	UNION-JACK uninsured	nonchalant union-shop	nonjuror uninitiated	nonage	knee-injury	uninjured
227	NONAGON nonacceptance unincorporated	nincompoop nonactive	nannygoat nonaggressive	nankeen nonequal	non-com nonexclusive	nonconformis unknowing
228	NINEVEH nonferrous uninformed	nine-footer nonavailable union-official	nonviolence unenviable	nonfat unenforcible	nonvoting uninventive	nonfeasance uninvited
229	NINEPINS nonplus nonpolitical	nonabsorbent nonpros union-pay	nonbeliever nonpartisan	nonbreakable nonoperative	nonpaying nonprofessional	nonpareil nonprofit

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230	GNOMES inmost	names enemies	namesake unmistakable	nemesis unmask	numismatist	animosity
231	NOMAD inmate	named pneumatic	new-math	name-dropper	animate	enmity
232	NEW-MOON unmanly	nominal unmannerly	nominee unmentionable	numbness unmindful	honeymoon unwomanly	inhuman pneumonia
233	ANEMOMETER	name-maker	new-member	unmaimed	unmemorized	unmomentous
234	NIMROD enumerate	namer unmarred	numeral unmarked	numerology unmerciful	newmarket unmarried	enamor un-American
235	ANIMAL	nameless	numbly	enamel	anomaly	pneumology
236	ANIMATION	unemotional	unimaginative	unmatched	unmeshed	
237	WINEMAKER	naming	anemic	inimical	unmake	
238	NYMPH	unemphatic	unmovable	nemophila	unmiffed	unmuffled
239	NAMEPLATE unimproved	nimble unimpressive	number unambiguous	unemployed unamiable	unimportant unhampered	unimpaired

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240	NURSE owners	nursery unrest	neurosis unhorse	nearsighted	honors	onerous
241	NARRATOR inheritance unredeemed	neurotic inroad unworthy	nard inward unwritten	north onward	inarticulate unearth	inert unhurt
242	NEURON unworn	no-iron unwarned	no-run unwarranted	nearness	narrowness	unearned
243	NORMAN narrow-minded	norm unarmed	normal innermost	near-miss	Norma	enormous
244	HONORARY	nearer	narrower	unrewarded		
245	NARWHAL knurl unroyal	narrowly one-reel	nearly unrelenting	New-World unreliable	enroll unruly	gnarl unreal
246	ENERGY inertia	narration onrush	nourishment ownership	Norwegian	enrage	enrich
247	NARCOTICS honoring	nark unrecorded	Newark unrequited	New-York unerring	anarchy unrecoverable	on-record
248	NIRVANA unrefined	nerve unruffled	no-refund	enervate	unravel	unrivaled
249	ENROBE unripe	nearby unrippled	enrapt unwrap	enwrap	honorable	unrepentant

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250	NELSON enlist	nails in-laws	nihilist unlace	annals unless	analyze unwholesome	awnless
251	NEWLYWED inlet unload	nailed inhaled unlit	annihilate unhealthy knelt	annulled unallowed	enlighten unallotted	inalterable unladylike
252	ENAMEL inalienable	nylon	nail	aniline	inland	on-line
253	ANNULMENT	newly-made	unlimited	unilluminated	unlamented	
254	INHALER	nailer	enlargement	kneeler	unlearned	unalarmed
255	HONOLULU	nailless	nail-hole	nilly-willy	unloyal	
256	KNOWLEDGE	analgesic	analogy	ineligible	unlatch	unleash
257	KNEELING unlock	nailing unlucky	no-luck unwelcome	new-look unwilling	onlooker unallocated	unlike unlooked-for
258	UNLOVED in-love	nail-file on-leave	New-Left unlawful	nullify unlive	analphabet	enliven
259	NAIL-POLISH unliable	nail-biting inhalable	nail-up unallowable	nol-pros unlabeled	analeptic unlabored	unhelpful

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260	NATCHEZ inches	nauseous injustice	niches unjust	notches	nudges	hinges
261	UNWASHED inched	nidget one-shot	notched unshod	nudged unshut	gnashed initiated	enjoyed
262	ENGINE enjoin ungenerous	nation ingenue unchanged	notion ingenious unchained	ancient injunction	enchant knee-jointed	engineer ungentle
263	NOTCHMARK	enjoyment	inchmeal	unashamed		
264	KNEE-JERK injury century	nature insurance	notcher onager	nudger on-shore	ensure unshrinkable	enshrine unsure
265	ANGEL unsheltered	ENCHILADA unusual	initial	unchallenged	unchilled	unshielded
266	HINGE-JOINT	unjudged	un-Jewish			
267	GNASHING unshackle	nudging	enjoying	inject	unchecked	unshaken
268	ANCHOVY unchivalrous	in-a-jiffy unchafable	unshaven	unshifted	unshuffled	unshoveled
269	HUNCHBACK wineshop	enjoyable	unshapely	unship	unchaperoned	unwashable

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270	inks	negus	next	nexus	knocks	annex
	encase	inaccessible	inexcusable	inexpensive	inexperience	innocuous
	inquest	unaccustomed	unexpected	unexplored	unquestioned	Yankees
271	necktie	naked	negative	nocturnal	nugget	nectar
	nicotine	anecdote	enact	inequitable	inquietude	inactive
	iniquity	honky-tonk	hunky-dory	knockout	uncut	ungodly
	uncaught					
272	nickname	knickknack	noggin	knock-knee	encounter	henequen
	inconceivable	incongruous	inconsiderate	inconsistent	inconsolable	inconvenience
	inconvertible	unacquainted	unconcerned	unkind		
273	newcomer	encamp	encomium	encumber	enigma	encompass
	honeycomb	income	incomparable	incompatible	incompetent	incomplete
	incumbent	uncomfortable	oncoming	unicum	unkempt	
274	nacre	nagger	neckerchief	Negro	neck-rein	necrology
	Niagara	nicker	niggard	anchor	encore	encourage
	engrave	hunker	inaccurate	inaugural	incarnate	incorporate
	incorrect	increase	incur	inquire	knackwurst	knickerbockers
	knocker	ungraceful	unguarded	unicorn	winker	winegrower
275	nickel	necklace	neckline	neglect	negligee	niggle
	no-cal	nucleus	ankle	enclose	incalculable	incline
	include	inequality	inkling	inoculate	knuckle	unclaimed
	uncle	unequal	wine-cooler	wineglass		
276	negation	negotiate	nicotiana	noxious	anguish	anxious
	encage	engage	inaction	incautious	unction	
277	nagging	necking	nogging	honey-cake	inking	knocking
	ongoing	ungag	winking			
278	no-cavity	no-caf	encage	knock-free	knock-off	uncover
	unequivocal					
279	hanky-panky	incapable	innkeeper	incubate	inkpot	kneecap
	knockabout	uncouple	unequipped	unoccupied		

280	naves infuse knaves	novice invest unfasten	nephews investigate	nevus invisible	envious invoice	infest knives
281	naphtha infidel unfadable	neophyte invade unfit	nifty innovate	no-fat invidious	Nevada invite	inevitable unavoidable
282	novena invent unfounded	newfound inoffensive unwoven	no-fun invincible	infant uneven	infantry uneventful	infinite unfinished
283	navy-man	November	infamous	one-family	unfamiliar	
284	nefarious enfranchise invariable unfortunate unversed	nephritis environ invert unfurnished	nevevvr infer knavery uniform	nouveau-riche infirm unfair universe	anniversary inform unforeseen university	enforce infringe unforgettable unvaried
285	navel inflame inflow unavailable	naval infield invalid unfilled	novel inflate invaluable unflattering	anvil inflict involuntary unfold	envelope influence involve univalve	infallible influenza invulnerable unveil
286	new-fashion infatuate unfashionable	navy-ship innovation	novation invasion	envision knavish	inefficient knife-edge	infusion unofficial
287	navigator invoke	novocaine unaffected	ineffective Univac	infect	invective	invigorate
288	navy-wife	any-five	in-view-of	knife-file	unfavorable	
289	navy-bean unifiable	navy-brass	enfeeble	enviable	unevaporated	unfabricated

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290	nabs knapsack	naps knobs	nibs unbiased	nips unobserved	NBC unobscured	nubs unopposed
291	nobody inept unbutton	no-bid inhabit unobtrusive	nepotism inhibit unhoped	Neptune input unpaid	anybody knapweed unpatriotic	inapt unapt
292	nepenthe one-by-one	Nippon unbeknownst	nubbin unbind	henbane unbend	inbound unpin	kneepan unopened
293	napalm	niobium	honey-palm	one-upmanship	wine-palm	
294	napper inebriated unprepared	neighbor inborn unpromising	napery unbearable unproved	newborn unbroken unprovoked	inappreciable unburden unpure	inopportune unparalleled
295	nibble Naples inapplicable	nipple Napoleon unable	nebulous Hannibal unbelief	niblick Nobel unbolt	nobleman enable unbalanced	nubile inability unpleasant
296	nuptials	honey-beige	one-page	inhibition	unabashed	
297	napkin Winnipeg	nipping unbuckle	henpecked	knobkerrie	unbecoming	unpack
298	napiform unbaffled	unpaved	unbefitting	unbeveled	unbehaved	unhopeful
299	nabob	nip-up	unbiblical	unpayable	unpopular	unpublished

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300	masseuse muses	mesas Messias	masses Mississippi	misses Moses	misuse	mosses
301	mast mister must homestead	master mistral moustache homicide	masticate mistrial mustard home-study	misdeed mistrust muster Mae-West	mist moist mystery	mistake most hamster
302	mason misnomer	messenger misunderstand	mezzanine mizzen	misanthrope muezzin	misinterpret Amazon	misinformed
303	museum amazement	messmate amusement	miasma	mesmerize	mismanagement	mismatch
304	masseur misericord	miser amuser	miserable emissary	misery Missouri	misrule	miserere
305	missal miscellaneous mussel	missile misology muzzle	measles Moselle himself	mislay Moslem myself	misalliance muslin	mausoleum muscle
306	message misogynist	massage moisture	mischance musician	mischief	misjudge	mischosen
307	mascot misconduct mosque mosaic	mask misgivings mosquito homesick	mascara misgovern muscular amazing	masculine misguided music	masquerade misogamist musk	massacre missing musket
308	misfit hemisphere	massive whomsoever	misfire amusive	missive	misfortune	misfeasance
309	mishap misprint	misplay homo-sapiens	misplace homespun	misapply mass-produced	misbehave	misbelief

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310	maids medicine moths	mats midsummer amidst	mates medusa mudslinger	madhouse midsemester immodest	matzo modes	meadows modest
311	method humidity	motet imitate	mediator immediate	Methodist	matador	Mediterranean
312	madonna methane emptiness	maiden median hometown	mitten matinee	madness mutiny	midnight midwinter	meatiness mutton
313	medium meatman	madam hematoma	madman home-team	mathematics	mad-money	mahatma
314	matron meteor mutter cemetery	material metropolis Madeira	maternal moderate Mata-Hari	matter modern matrimony	mattress mother humdrum	meter motor humidor
315	medal middling mutilate	meddle model mythology	medial motel	metal module	mettle mottled	middle muddled
316	mod-show mutation	mod-shop home-dish	mediation imitation	amidships	midshipman	modish
317	madcap meeting humdinger	mating meticulous omitting	matting mitigate	medical modicum	medicare mythical	mediocre mudguard
318	midwife modify	medieval humidifier	metaphysics	metaphor	midfield	motive
319	meatball	made-up	metabolic	mouthpiece	mutable	mudpie

320	manse minus monster ominous	mayonnaise miniskirt monsoon	manslaughter means municipal	mainsail minister amnesty	menace minstrel emancipate	mince monastery immense
321	amend maintain manhood mentor monitor	amenity man-eater mantle mind monotonous	amount man-trap maundy minidress month	emanate mandate meantime mint mountain	humanity mandolin mend minute womanhood	immunity Manhattan mental Monday
322	minion mononuclear	manana mononucleosis	manhandle amanuensis	meningitis eminent	Mennonite imminent	manhunt
323	monument monomania	mainmast moneymaker	man-made moneyman	manometer ammonium	manumit homonym	minimum omnium
324	manor minor Monroe	manner minority homeowner	manure monarch	miner monaural	man-hour money-order	mineral monorail
325	manhole moneylender hymnal	mainland meanwhile womanly	mainly monologue	manly moonlight	manual Manila	monolith Emmanuel
326	manager monition omniscience	manger moonshine ammunition	mansion moonshell	mention moonship	menagerie moonshot	miniature munch
327	maniac minicab monkey	mankind mining monogamist	manikin mink moneygrubber	manacle monk Munich	manicure monocle ammoniac	meaning monogram
328	manifest money-flow moon-voyage	manful monophonic munificent	manifold moonfaced omnnificent	maneuver moonfish omnivorous	manufacturer moon-flight	menfolk moon-flower
329	man- money-belt amenable	ape monoplane hymenopteron	manipulate monopoly omnibus	manpower moonbeam omnipotent	minable moon-blind	moneybags myna-bird

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330	mammas mimeses	maims mimosa	mayhems momus	memos mummies	memsahib	mimes
331	maimed Mohammedan ohmmeter	mammate Muhammad	mammatocumulus mummy-tomb	mammoth myomata	mayhemmed homemade	mimetic imamate
332	mammon	memento	Miamian	moment	momentum	
333	mimamsa	mamamu	mamma-mia			
334	memorial mummer	memory mammer	memoir mammary	memorandum homomorphy	maimer	mimer
335	mammal	my-mail	mamluk	mamelon	hamamelis	mammillary
336	home-mission	home-match	mumchance	mummichog		
337	mammock homemaker	mimic	mimeograph	maiming	mumming	mummy-case
338	Memphis	home-movie	humhum-fabric	mummify	mammiform	
339	mamba memo-pad	mambo mummy-bag	member mumbo-jumbo	mumps	mumble	membrane

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340	mercenary Mars	mercer Morse	mercy amerce	morass amorous	morose humorous	morsel immerse
341	mart martyr myrtle	maraud meridian myriad	maritime merit hammer-throw	married mirth homeward	marten mortal	martini murder
342	marine morning	mariner mourner	Marianne maharani	marihuana moron	maroon home-run	marionette
343	marmalade merrymakers	marmoset Mormon	mermaid murmur	marmot myriameter	marimba homeroom	merriment
344	mirror	merrier	moorier	marara	hammerer	
345	marl merrily immoral	moral amarillo	morale hammerlock	mural home-rule	marlin humorless	merely emerald
346	march margarine marshal	Martian marchioness merchandise	maharaja marriage emerge	marchpane meerschaum emergency	margin merger mirage	marsh merchant hemorrhage
347	mark mercantile homework	market miracle	marigold mercury	maracas morgue	marksman marque	marquess America
348	marrowfat Morpheus	marvel Moravia	moreover	MIRV	morphine	murphy-bed
349	marble moribund	marabou hammer-beam	mariposa hammer-blow	marplot	marrowbone	morbid

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350	MALICE aimless	homeless mails	maelstrom molest	mills	miles	millisecond
351	MALT hamlet melodrama muleteer	melt homiletics melody multiple	mold humility milady multitude	omelet malady mild umlaut	mulatto mallet military	emulate meld millet
352	MELON melancholy	Himalayans millennial	hemline millennium	milliner million	homeland	malign
353	mailman mule-mark	maim mealymouthed	melamine emolument	millimeter	millman	muleman
354	molar mailer	miler milord	malaria millwright	miller ameliorate	milliard	mealworm
355	may-lily	molehill	milliliter	mill-wheel	melilot	
356	militia mulish	malicious emulsion	mileage humiliation	maladjusted homology	millage	mulch
357	malcontent mollycoddle	mailing molecule	malignant Malaga	malkin amalgamate	milk hemlock	milligram homologous
358	mellophone mellifluous mollify	malevolent milee-feuille homelife	malefactor millefleurs	malfeasance millfeed	malformed millefiore	malfunction mule-foot
359	mailbag malapert	mail-boat malaprop	mail-pouch mulberry	mailbox muleback	malleable	melba

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360	magistrate	majesty	majuscule	matches	images	
361	machete	matchwood	mashed	midget	meshed	emaciate
362	machine imagination	mission homogenous	motion omission	magenta	mushiness	emotion
363	matchmaker	matchmark	mishmash	mushmelon	MGM	
364	major moocher	majority mushroom	mature meshwork	measure homager	majorette imagery	masher
365	majolica	Magellan	matchless	matchlock	matchholder	mutual
366	magician	muchacha				
367	magic	mashing	meshing	matching	Michigan	
368	midge-fly	home-chef				
369	matchable	matchbook	matchbox	matchboard	match-up	mushball

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370	macassar mix	maccus mucus	magazine mugs	maxim amongst	megacycle hammocks	mikes
371	maquette mocked	maggot mugged	macadam	megaton	makedo	mikado
372	magnet magnitude	magnate mahogany	mackintosh magnolia	magnesium mechanic	magnanimous mackinaw	magnificent
373	magma	mugwump	homecoming	home-game	homogamy	
374	macaroon microscope immigrant	mackerel mockery homecraft	maker mongrel homegrown	make-work muckraker	meager mugger	microbe emigrant
375	mackle immaculate	mangle	mingle	Mogul	Michael	meekly
376	makeshift	mug-shot	mawkish			
377	making	mocking	mugging	mucking		
378	megaphone	make-off	makefast	make-over	Machiavellian	mugful
379	magpie hummingbird	make-believe Yom-Kippur	makeup amicable	Maccabee	megapolis	mock-up

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380	mavis emphysema	movies ham-fisted	moves home	muffs office	Mephisto	emphasis
381	moviedom emphatic	moved	miffed	mufti	amphetamine	amphitheater
382	maven	muffin	homophony			
383	moviemaker	movement				
384	MAYFAIR myophore	AMPHORA	mover	maverick	hamfare	whomever
385	mayflower homefelt	mayfly	movieland	moveless	muffler	hemophiliac
386	mayfish	movie-show	home-voyage			
387	moviegoer	moving	movie-camera	home-viewing		
388	movie-fan	Mafia-family	move-off	move-over		
389	movable	movie-plot	move-up	amphibious	amphipod	amphibole

390	maps	mobster	ambassador	ambuscade	embassy	embezzle
	emboss	embosom	home-base	impasse	impassive	impose
	impossible	impostor	symposium			
391	moppet	ambidextrous	amputate	embitter	embody	embed
	empathy	impediment	impudent	impute	impetus	homebody
	ombudsman	wombat	symptom	sympathy	somebody	
392	ambient	embankment	impanel	impend	impenetrable	impenitent
	impinge	impound	impugn	impunity	homebound	sampan
393	map-maker	embalm	empyema	wampum		
394	mapper	mayberry	moper	amber	embark	embarrass
	embrace	ember	embroider	embroil	empire	empower
	emperor	empress	hamburger	hamper	homburg	impair
	impart	imperative	imperfect	imperial	impertinent	import
	impress	imprint	imprison	improper	improve	impure
	umbrella	sombrero				
395	maple	mobility	maypole	ambulance	amiable	ample
	amplify	ampul	embellish	emblem	employ	home-plate
	humble	impel	implement	implicate	implore	imply
	impolite	impala	impulse	umbilical	sample	simple
	semblance	symbol	assembly			
396	ambush	ambition	impatient	impassionate	impetuous	impish
	impeach					
397	mobcap	moping	mobbing	mopping	myopic	ambiguous
	humbug	hymnbook	impact			
398	ambivalent	ambivert	amebiform	impavid	impoverished	swamp-fever
399	maypop	mopboard	mop-up	humpback	imbibe	impubic

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400	racist irresistible	recess warehouses	resist	raises	roses	horses
401	receipt restrict airstrip worst	recital roast arrest wrest	residual roost erst	rest roster hirsute	restore rust wrest	restrain rustic wrist
402	raisin resign hoarseness	reason resound orison	resin arsenal war-zone	recent arson Arizona	resent arsenic	rescind horizon
403	reassemble horseman	raceme oarsman	razzmatazz warehouseman	resemble wearisome	resume	heroism
404	racehorse resort	racer resource	razor eraser	reassert horse-race	research horseradish	reserve
405	razzle roselike wrestle	resale rosella	resolute rustle	resolve herself	result ourselves	roseleaf yourself
406	recession	rescission	rose-cheeked	rose-show	horseshoe	
407	rascal rose-cut harassing	racecourse Rosicrucian	rescue rusk	racing rye-whiskey	rising warehousing	risk arousing
408	receiver horseflesh	rosefish	horsefece	horsefeathers	horsefish	horsefly
409	rasp reciprocal responsible	raspberry respect airspace	receptacle respiration horseback	reception respite horsepower	recipe resplendent horsewhip	recipient respond irascible

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410	ARTIST radius	ARTISAN raticide	yardstick reduce	roadster righteous	Rhodes ritzy	roads arduous
411	RAT-A-TAT airtight irritate rutted	REDHEAD aridity radiate	redwood certitude red-hot	hardtop hardhead redoubtable	erudite hardwood redtop	redoubt heredity rotate
412	RATTAN ardent readiness rudeness	RED-NECK earthen retain	red-nose harden roadnet	redness hearten rotten	retinue ordain routine	warden radiant ruddiness
413	RADIUM ready-made	RADIOMAN redeem	yardman rudiments	radome	roadman	rhythm
414	RADAR Rotarian retard retrieve hairdresser	RIOTER rather retire return orator	artery reader retort rider order	air-to-air raider retrace rotary wardrobe	aerodrome redress retract rudder writer	hoarder reiterate retreat ardor
415	RADIAL airedale heartless retell	HURDLE artless ordeal riddle	retail earthling ratline ruthless	redolent hardly readily whortleberry	rattle heartily redleg	redial heartland retaliate
416	RADISH retouch	RADIATION ruttish	artichoke erudition	yardage hardship	road-show heritage	irritation hard-shell
417	RADIOACTIVE articulate red-oak wording	RED-CARPET erratic redact writing	earthquake heartache redcap	radical radiocast ridicule	redcoat radiograph riding	article reading surtax
418	RADIO WAVE reedfish write-off	HORS-D-OEUVRE artful certify	radiophoto artifact	ratfish artificial	ratify heartfelt	redfin hurtful
419	HEARTBREAK hartebeest ratproof	REDBREAST air-tube redback	heartbeat hardpan redberry	rutabaga irritable redouble	yardbird ratable	wiretap rataplan

420	rains	rancid	ransack	ransom	reigns	reinsert
	reinstate	renews	rhinoceros	rinse	ruins	earnest
	harness	irons	ironsides	weariness	urns	Uranus
	soreness					
421	rand	rant	random	rainy-day	rain-out	raindrop
	reenter	reindeer	render	rent	reunite	rind
	rondeau	rondo	round	rundown	runt	aeronaut
	around	errand	errant	hairnet	hornet	orient
	ornate	warrant	worn-out	surrender	surround	
422	rayonnant	renounce	renown	reunion	run-in	ironhanded
423	rainmaker	reanimate	rename	renominate	renumerate	ironmonger
	ornament	uranium	surname			
424	rainwear	renewer	ruiner	runner	earner	ironer
	iron-ore	ironworks	ornery	warner		
425	rainless	renewal	run-well	one-reel	unreal	
426	ranch	range	rainwash	rain-check	reenjoy	reinsurance
	renege	arrange	orange	wrench		
427	rank	rancor	raining	rink	running	raincoat
	earnings	ironclad	ironic	ironing	warning	wrinkle
428	rainfall	rainfrog	rainful	aeronaval	reinforce	renovate
	reinvest	run-off	Herrenvolk			
429	rainbow	rainpool	rainproof	runabout	runup	hornbill
	hornblende	hornbook	hornpipe			

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430	rams armies	reams armistice	rooms airmass	remiss harems	rhymes surmise	arms
431	remedy armada	remit aromatic	remodel armed	remote eremite	rheumatism hermit	roomette warmth
432	raiment remunerate Armenian surmount	remain Roman ermine	remind romantic harmony	reminisce Romania hormone	remnant ruminant ceremony	remonstrate airman sermon
433	remember	rememorate	roommate	rum-mill	armament	
434	rammer roomer	ramrod rumor	remark rumrunner	remorse armor	rhymer armory	roamer warmer
435	remilitarize armhole armlock	rimland armlet	ramulous harmless	airmail ormolu	air-mile warmly	armless armload
436	ramjet	ramshackle	rematch	remission	rummage	armchair
437	remake arming	rhyming harming	roaming warmonger	rooming	Armageddon	armguard
438	ramify harmful	reemphasize	removal	roomful	armful	earmuff
439	ramp reimburse rambunctious	rampart rhombus rambutan	ramble rhumba armpit	rampage romp warm-up	reemploy rumble	Rembrandt rumpus

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440	rehearsal warriors	roars	arrears	errors	horrors	warhorse
441	rarity reward	reared reword	reread rewrite	reredos roared	reorder Ararat	reroute arrowroot
442	rear-window war-horn	rearrange warworn	rerun	rewarned	reworn	rareness
443	rearmament	reremouse	rearmost	rewam	earworm	wireworm
444	rare-earth rewirer	rarer roarer	rareripe	rere-arch	rearhorse	rearward
445	rarely	rear-alley	reroll	rerail	rural	war-rally
446	raree-show	rear-echelon	rerush	reration	heroworship	arrearage
447	reargue rework	rear-guard rehearing	reorganize earring	rearing hierarchy	roaring	rewiring
448	rara-avis	rear-fanged	rearview	rarefy	errorfree	
449	rarebit	re-orbit	roorback	wire-rope		

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450	rails hairless	realize orioles	release wireless	realest	realist	airless
451	reality roulette hurled	realty royalty world	reallot air-letter	relative air-light	rolled harlot	roulade herald
452	realign airland sirloin	relent airline	reliance earliness	relinquish Ireland	reloan orlon	roll-on whirlwind
453	railman heirloom	realm	relumine	rollman	rollmops	Harlem
454	railroad ruler	raillery	rallier	reeler	reelrall	roller
455	railwayless	roleless	royal-law	royal-yellow	ruleless	
456	railage earlship	reallege horology	reeligible	relation	religion	relish
457	railing relocate harlequin	reelect roll-call oarlock	relax rollicking warlike	relegate rolling whirligig	relic hair-lock yearling	reluctant hireling yearlong
458	railfan airlift	relevant werewolf	relief	relieve	relive	roll-off
459	railbird earlobe	reliable harelip	relapse hurly-burly	rollback orlop	roly-poly whirlpool	earlap

460	rages ridges	ratios righteous	regisseur rushes	register arches	rejoice	riches
461	ratchet archdiocese airsheet	recheat arched	reshod earshot	rigid urged	reached archduke	rushed wretched
462	reagent richness origin	ration Russian urchin	regent archenemy urgency	regenerate Argentina surgeon	region erosion assertion	rejoin Eurasian
463	regime	regiment	Rushmore	hair-shampoo	Irishman	
464	rasher archer searcher	richer harsher surgery	reassure erasure	Roger orchard	rusher urger	rush-hour surcharge
465	racial ear-shell	Richelieu harshly	richly searchlight	ritual	rushlight	archlute
466	rejudge	Rosh-Hashanah	rush-job	arch-joint		
467	raging air-check	reaching urging	recheck	reject	reshake	rushing
468	rechauffe arch-foe	reshave arch-wife	reshuffle archfiend	rejuvenate Irish-ivy	rush-off	rye-chaff
469	reshape worship	reship	airship	archbishop	hairsheep	warship

470	racks requisite orgasm	rags ruckus works	recast arcs wrecks	request heroics circus	raucous irksome sarcasm	rex orchestra
471	racket requite arcade circuit	ragged rickets arctic	ragtime rocket architect	react regatta erect	rectangle rigadoon irrigate	rector rugged haircut
472	raccoon rekindle hurricane	reckon air-gun organ	reconcile arcane Oregon	reconsider archangel	reconstruct arrogance	regain hearken
473	ragman requiem circumstance	ragamuffin rigmarole	recommend argument	recompense origami	recommit war-game	recumbent workman
474	raker regard aircrew	record regret hara-kiri	recourse require war-cry	recreation rigor warcraft	recruit rocker worker	recur aircraft
475	raglan recoil ringleader wriggler	rakehell recollect air-cooled oracle	recall regular argali circle	reckless regalia argyle	reclaim recluse hiring-hall	recline row-galley hourglass
476	reaction erection	rickshaw irrigation	ricochet workshop	roguish wreckage	air-coach	air-gauge
477	raking rococo	recognize archaic	recook arguing	rigging working	ringing	rocking
478	recover	rockfall	rock-fill	Roquefort	rockfish	archives
479	recap rock-bottom	recapitulation rugby	recuperate archipelago	rockabye ark-boat	rockberry herringbone	rockbird arquebus

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480	refuse orifice	reviews Orpheus	revise wharves	revisit war-office	airwaves service	harvest surface
481	raft roughdry	refit arrived	rift hereafter	rivet horrified	roofed war-veteran	rooftop servitude
482	raven roughness	ravine ruffian	refine earphone	revenge orphan	revenue wharfinger	roughneck servant
483	revamp	refumble	Orpheum	war-fame	wharfmaster	
484	reefer refrain river	refer refresh rover	reform refrigerator roughrider	referee refurbished wherever	referendum reverse surveyor	reverend revert
485	raffle reveal revulsion airfield	ravioli revel rifle airflow	refill revolt rival arrival	reflect revolve rivulet oriflamme	reflux revolution rueful servile	revalue revolver ruffle
486	ravage roughshod	ravish oarfish	refashion wharfage	refuge	revision	roughage
487	revocation	raving	roofing	horrifying	irrevocable	
488	revival	rough-fish	rough-footed	wharf-fish	survivor	
489	rev-up	reviewable	refabricate	roofable	rough-up	wharf-boat

490	rabies	repast	repose	repost	rebus	robust
	rhapsody	robes	ropes	airbase	airbus	Arabs
	arabesque	hairpiece	harps	herbs	surpass	
491	rabbit	rabid	rapid	rapt	rebate	repeat
	rebuttal	reptile	repute	robot	rowboat	rubdown
	rubytail	wrapped	arbiter	orbit	whereabouts	acerbity
492	rapine	rebound	reopen	repent	ribbon	ripen
	robin	ruby-wine	hairpin	harpoon	European	urban
	whereupon	warbonnet	serpent			
493	repayment	air-boom	robomb	europium	herbwoman	war-poem
494	rapier	rapper	rapport	rapparee	reappear	reaper
	repair	report	repartee	reprehend	represent	repress
	reprieve	reprimand	reproach	reprobate	reproof	ribroast
	rhubarb	robber	rob-Toy	roper	rubber	airport
	arbor	harbor	harebrained	wrapper	surprise	Cerberus
	cerebrum					
495	rabble	rape-oil	rebel	rebuild	repeal	repel
	replace	replant	replenish	replete	replica	reply
	repulse	ripple	rubble	ruble	rubella	airplane
	horrible					
496	rapacious	rapture	rubbish	rupture	repatch	arpeggio
	eruption	harp-shell	herbage			
497	rebuke	repugnance	repack	roebuck	rope-key	rubicon
	rubicund	Arabic	Areopagus	harbinger	wrapping	yearbook
498	rebuff	repave	riboflavin	rope-off	rubify	rub-off
	herbivore					
499	republic	republish	repeople	rip-up	air-pipe	earbob
	orb-web	war-baby				

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500	lasses alehouses	lassoes aliases	leases Ulysses	lazy-Susan Alsace	losses	Louis-Seize
501	last lust all-star whilst	lassitude lucid elastic celestial	lawsuit luster elucidate solicit	least lowest holster	lest LSD illustrate	list all-set ulster
502	laziness loosen	lessen lozenge	lesson hallucination	license halcyon	liaison Wilson	listen
503	lissome	lyceum	wholesome	Elysium		
504	lacerate elsewhere	laser illusory	lessor ulcer	lizard	loser	also-ran
505	lazily whole-souled	leasehold	low-slung	luzula	oil-seal	wholesale
506	loose-jointed	Alsatian	oil-switch	wall-switch	yellow-sage	yellow-sedge
507	lacing	leasing	losing	Alaska	oilskin	woolsack
508	lascivious	laissez-faire	Lucifer	elusive	illusive	
509	lazybones allspice	lesbian	lisp	low-spirited	wellspring	well-spoken

510	lattice loudspeaker waltz	lettuce Lao-Tse	ladies eldest	lads halitosis	lighthouse oldest	lotus holidays
511	latitude altitude wilted	lighted eluded yuletide	loaded halted solidity	la-di-da holdout solitude	low-tide old-time	allotted well-to-do
512	Latin lighten	latent lightning	laydown lowdown	lieutenant oldness	litany old-wine	leaden well-done
513	lithium altimeter seldom	leitmotif allotment	loudmouth illtimed	light-meter ultimate	light-minded yuletime	litmus well-timed
514	ladder letter looter alderman holy-water	lateral leotard lothario altar older	lather literal lottery alter ulterior	latter literature louder elder yielder	leader litter Luther halter	leather loiter altruism holder
515	ladle little	lately wall-to-wall	ladylike wildlife	ladylove wildly	lethal yellowtail	leadless
516	ladyship old-shoe	late-show salutation	latish	Low-Dutch	old-age	oldish
517	leading altogether sliding	latecomer Eolithic	litigant holding	lithograph welding	loading wildcat	ludicrous yielding
518	ladyfinger healthful well-advised	leadoff holdover wildfire	lightface illadvised wildfowl	lightfast oldfangled	lightfingered old-fashioned	light-foot oldwife
519	ladybird let-up	ladybug hilltop	laudable holdback	leadback hold-up	lead-up wildebeest	lead-pipe

520	lance lyonnaise silence	lens alliance	lioness aliens	linseed allowance	lonesome holiness	lunacy illness
521	land laundry alienate wall-unit	lend lentil island well-knit	lent lindy eland cylinder	lint linden Holland assailant	lintel lunatic oleander	lean-to London walnut
522	lenient all-in-one	Lenin hole-in-one	leonine illanun	linen Illinoian	linin wellknown	alanine
523	lineament illinium	lineman	liniment	Holy-Name	alignment	allonym
524	tanner loaner	lanyard loanword	leaner loner	liner lunar	linear Leonardo	lionhearted alienor
525	lanolin lonely	lienal lunule	lineal willy-nilly	line-haul	linoleate	linoleum
526	launch luncheon ill-natured	lounge lunge	lineage lynch	longitude alienation	lingerie elench	lunation elenge
527	lanky loaning	lining loincloth	link low-necked	links lunkhead	linocut lynx	leaning all-inclusive
528	lanuvium well-enough	lenify well-informed	line-firing	line-off	lion-fish	ill-informed
529	lenape loanable	unable loanblend	linebacker alienable	line-up	linebred	linebreed

530	lambs elms	limbs Hallowmas	limousine slums	looms	low-mass	almost
531	lamdan ill-mated	limit well-made	a-la-mode	almighty	elm-tea	helmet
532	lawman limnologist alimony illuminate	layman luminary almanac oilman	lameness low-minded aluminum well-mannered	lament low-man alumnus well-meaning	lemon ailment element whaleman	limen aliment eliminate
533	lamium	limeman	alma-mater	well-memorized		
534	lemur halomorphic	limerick oleomargarine	all-American	allomorph	Hail-Mary	hallmark
535	lamella elemol	lamellate elm-hill	limelight oil-mill	lime-oil	lowmelting	all-male
536	limation ill-matched	lime-ash well-matched	lime-juicer	Limoges	limewash	lomatia
537	lawmaker almicore	lambkill almucantar	lambkin whelming	limekiln salmagundi	lummox	looming
538	loomfixer	lymph	lymphad	well-emphasized		
539	lamp limbo Olympics	lamprey Limburger slumber	limber lumbago	limp lumber	limpid lump	limbus Alhambra

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540	larceny hilarious	liars	lawyers	lower-house	walrus	wheelhorse
541	laird lured hilarity	lard laureate well-read	lariat lurid whole-hearted	leeward alert celerity	low-rate all-right salaried	Lord already
542	laryngitis aileron	laurin all-or-none	layer-on hollow-horn	loran well-run	learned well-earned	Lorraine willow-wren
543	larmier lyreman	larryman alarm	lorimer allurement	lormery eelworm	lorum willowworm	lowermost
544	lowerer	lyrurus	allurer	hollerer		
545	laurel	lauryl	lawyerlike	Lorelei	lorilet	Holy-Roller
546	large allergy	largess alurgite	largeness oleraceous	laureation	layerage	lurch
547	lark lower-class	largo lurk	layering lyric	lorica alluring	lorikeet oil-ring	lowercase oil-rock
548	larva	larvule	low-roof	lyreflower	lyriform	
549	larb lyre-back	larboard lyrebird	larrup	layer-up	low-warp	lowerable

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550	lawless hula-hulas	lilies oil-wells	lowliest	lulls	lulus	Allhallows
551	lilt well-heeled	lilied well-willed	lily-white ululate	low-yield	loyalty	lulled
552	lillianite	lowland	lowliness	ululant	whale-line	Sally-Lunn
553	law-lamb	lilium	lolium	allelomorph		
554	lowlier	Lollard	well-larded	well-learned	cellular	
555	liliales	lilylike	lily-livered	lily-low	letwel	lowlily
556	lallation	liliaceous	ululation	well-latched		
557	lilac yellowlegs	lolling Yule-log	low-lying soliloquy	lulling	ill-luck	owllike
558	low-level	lowlife	low-lived	well-loved	yellow-livered	all-alive
559	lalapalooza	lilliputian	lollipop	lullaby		

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560	latches luscious	lodges elegies	loges eulogist	loggias eyelashes	legislator zoologist	logistics
561	latchet	elegit	legitimate	lodged	alleged	woolshed
562	legend elision	legion elysian	lesion halogen	lotion illusion	allusion solution	elation
563	lodgement	hall-chime	Welshman	eulogium		
564	lasher lushier	ledger Algeria	lecher alleger	leisure well-wisher	lodger wheelchair	lusher
565	lushly	hellishly	oil-shale	owlishly	wellchilled	slouchily
566	ill-judged	all-Jewish	logician	well-judged		
567	latchkey illogical	lodging	logic	lashing	alleging	yellow-jack
568	low-shafted	well-shaven	well-shuffled			
569	awl-shaped	hell-ship	algebra	oil-job	oil-ship	well-shaped

570	laxative locust elixir	laxity low-cost holocaust	legacy log-house	legs likewise	locks lungs	looks elks
571	languid locket elector	liquid lockout electricity	legate lookout hellcat	length low-cut select	locate alley-cat	locked alligator
572	laconic ill-conditioned	lagoon Allegheny	Lackawanna elegance	likeness eloquence	longhand oilcan	well-conditioned silken
573	lagomorph alchemy	legman algometer	legumes illcome	leukemia ill-kempt	ligament welcome	locomotive
574	lacquer logger lucre well-groomed	laggard locker lukewarm well-grown	languor longer lugger	legroom longhair allegorical	linger looker licker	liquor lucrative slugger
575	laical low-calorie oilcloth	legal low-class willingly	likely luckily	local alcohol	locule alkali	long-lived illegal
576	language lockage elocution	languish logjam election	legation luggage sluggish	leakage allegation	lockjaw allocution	loquacious allocation
577	lagging locking	leggings lugging	licking oil-cake	liking	longing	looking
578	lakefront well-covered	lawgiver	liquefy	locofoco	alcove	ill-given
579	legbar look-up	leg-break longboat	leg-pull lugubrious	leucoplast helicopter	likable allocable	lockup

580	laughs life-span olives	layoffs life-size slaves	leaves loaves	levies love-sick	lives lovesome	lifesaver elves
581	laughter livid elevator	lavatory loft ill-fated	lafayette loved slaved	left alleviate solved	lifetime aloft well-fed	lift elevate
582	lavender elephant	liven eleven	live-in elfin	live-one well-founded	love-in slovenly	alphanumeric solvent
583	lavement self-made	leaf-mining	lifeman	lovemaking	alluvium	ill-fame
584	laugher livery hellfire silver	layover loafer illfare slavery	lever lover welfare sulfur	lifer love-arrow well-over	liver louver wolfram	livewire alfresco wolverine
585	lavalava lifelong lovely	lavaliere lifeless alluvial	lawful livelihood alfalfa	leaflet lively olive-oil	level lovelock wallflower	lifeline lovelorn willful
586	lavage wolfish	lavish salvage	leafage salvation	alleviation selfish	elevation	olivaceous
587	layfolk loving	laughing ill-affected	leaving olfactory	living self-confidence	lifeguard well-affected	live-oak
588	laugh-off self-evident	leave-off well-favored	life-force	live-off	Love-wave	ill-favored
589	lavabo live-by	laughable live-up	leafhopper lovable	lifebuoy lovebird	lifeboat love-potion	livable alphabet

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590	labs lobbies elopes	laps lobster helps	lapse alabaster wall-base	lips alibis celibacy	loops alps	lopsided ellipse
591	lapidary law-abiding ill-boding	lappet albatross well-paid	leaped albeit whaleboat	libido eloped	lipid allopathy	looped halibut
592	lupine whalebone	albino well-appointed	Albion sleepiness	alpine	hell-bent	olibanum
593	low-beam oil-palm	album	Alabama	albumin	elopement	helpmate
594	labor liberal lubricant wellborn	laboratory liberty elaborate wellbred	lawbreaker Libra elbow-room sleeper	leopard librarian eloper slipper	leap-year looper helper celebrate	leprosy lowbrow ill-bred
595	label lobule hillbilly	lapel tube-oil hullabaloo	liable loophole well-beloved	libel lowbell soluble	lobelia allowable syllable	loblolly helpless
596	libation	lobation	low-budget	lowbush	lowpitched	slipshod
597	lawbook helping	leaping ill-being	lobbying oil-bag	looping well-being	alpaca sleeping	eloping
598	leapfrog	lop-off	all-beef	helpful	wellbehaved	lapful
599	lap-up	leap-up	ale-pub	Ali-Baba	oil-pipe	wallpaper

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600	chases joss-house	cheeses shoe-size	choices	jazzist	Jesus	juices
601	chastity geste just suggest	chest gesticulate Jesuit	chestnut gist shoestring	cheese-eater jest showstopper	cheese-head joust adjust	gest justice washstand
602	chasten adjacent	chosen show-scene	geocentric juiciness	jazziness	jacinth	Jason
603	chessman	jasmine	jawsmith	jazzman	chassemaree	
604	chaser	chasseur	chooser	cheesery	jowser	juicer
605	chisel shoe-sale	choiceless wash-sale	jazzily witch-hazel	jostle hashslinger	joyously	juiceless
606	cheese-shop	chess-champion	gestion	gesture	egestion	suggestion
607	chasing joskin	chasse-cafe	chasse-croise	cheesecake	cheesecloth	choosing
608	cheeseflower	Joseph	Josephine	jussive	shows-off	
609	chassepot jazzbow	cheeseparer juice-up	cheeseburger Jezebel	chess-pie shoespoon	chessboard Josup	jasper Shuswap

610	chiteaus shadows	jets sheets	jetsam shots	jettison shouts	Judas geodesy	shades
611	cheated shutdown	jaded shutout	Judith agitate	shaded geodetic	shadowed	shoot-out
612	chatoyant shodden	chutney showdown	jetness shoddiness	jetton shut-in	jitney	shadiness
613	shedman show-time	sheet-home	shittim	shootman	shotmaker	shotman
614	chatter shooter jittery	cheater shouter geotherm	cheddar shudder	chowder shutter	shatter ashtray	shoe-tree watchtower
615	chatelain shadily	chattel shadowland	chatelet shadowly	jetliner shot-hole	shuttle	shadeless
616	chitchat agitation	judiciary	judicious	schottische	shotshell	agiotage
617	chatting sheeting geoduck	cheating shooting watchdog	judicature shotgun Asiatic	shaddock shouting	shading shutting	shadowing jotting
618	shadflower sheetful	shadoof shutoff	sheatfish	she-devil	sheetfed	sheetflood
619	jetbead sheet-block shotbush washed-up	jetport sheathbill shot-put washtub	jodhpur shootable shut-up	shadowbox shootboard chateaubriand	shadbush shoot-up agitable	sheetpiling shotblast hushed-up

620	chance genius shyness	chains genocide	chancellor agency	chanson ginseng	Chinese issuance	genesis oceans
621	chandelier gentle jaundice Shinto	chandler gentry joint shunned	chant genetics Jane-Doe shunt	Chinatown giant John-Doe agenda	gendarme jaunt shanty agent	gender janitor shindig witch-hunt
622	chignon Shenandoah	genuine	Shannon	shenango	shenanigan	shoneen
623	chainman show	chain-method name	Chinaman	chinampa	chunam	genome
624	chinaware January	general joiner	generation junior	generic shiner	generosity	genre
625	chainless oceanology	chainlet shinleaf	channel	chenille	genealogy	genial
626	change gingivitis	chain-shaped agential	chinch jointure	chinchilla	ginger	gingelly
627	chink jonquil shunning	Chinook junk shank	chunk junkie adjunct	gynecologist junction adjoining	johnnycake schnook eugenic	joining shining hygienic
628	Geneva shoe-knife	genuflect	janiform	junefish	juneflower	oceanfront
629	china-boy joinable shinplaster	chinbone Juneberry	chin-up junebud	janapa juniper	johnapple schnapps	johnboat shinbone

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630	chemise shames	chimes shamus	chums	gems	James	jams
631	geometry age-mate	jammed	jamwood	jimdandy	jimmyweed	ashamed
632	chimney gymnast hush-money	chow-mein shaman wash-woman	G-man showman watchman	Gemini ashman	geminate hegemony	gymnasium hatchman
633	Jemima	gym-master	jam-maker	show-member		
634	chimer shamrock	chimere shimmer	gemmary shomer	jammer	shamir	shammer
635	chummily shameless	gemlike Ishmael	gemmologist	gemmule	gym-hall	schmaltz
636	gemmieeous	gemmation	gym-shoe	Jamshid	jimjams	shim-sham
637	chiming jamoke	gemming shoemaker	gimcrack watchmaker	gymkhana yashmak	Jamaica	jamming
638	SHAM-FRUIT shameful	CHAMFER sham-feed	gem-fruit chamfron	gemmiferous	shamefaced	shamefast
639	chamber jambeau jumbo	champagne jambalaya jumble	champion jambone jump	champignon jam-pack shambles	chimpanzee jam-up shampoo	chump jamboree

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640	chairs jersey	cherries jurist	chores Scheherazade	gyroscope shares	gyrus shears	jars
641	charity chariot shrewd	chard joyride shroud	chart shirt surety	charter short watchword	cheroot shard ushered	cheet shred
642	charnel journal sojourn	churn journey sureness	gerent shrine adjourn	Geronimo shrink assurance	gerund shorn usher-in	geranium shoehorn
643	chairman germinate showroom	charm gerrymander shrimp	charwoman jeremiad washerman	germ juryman washroom	German jurywoman	germane ashram
644	cheerer shirrer	Gerardia shorer	juror surer	sharer assurer	shearer usherer	shearier usurer
645	charlatan schorl usherless	charily sherlock	cheerleader shoreline	Charlie shareholder	cheerless shrill	churlish surely
646	charge shrewish	cherish ushership	church	Churchill	George	gyration
647	charcoal jargon shrug	Cherokee jerrican	gyrocompass shark	Jericho sharing	jerk shirk	jarring shriek
648	chervil shroff	cheerful shrove	sharif surefire	sheriff surefooted	shrift giraffe	shrive
649	charabanc cherub shrapnel	charpoy sharp shrub	cheer-up sharpy	cherry-pie sherbet	chirp shorebird	chirrup showerproof

650	chalice joyless	chills Julius	jalousie shells	jealousy shoelace	jailhouse ageless	jewels socialism
651	child shelter	gelatin shield	gelid shoulder	jilt age-old	jolt agility	shallot
652	challenge jolliness wash-line	chalone julienne	chilliness shalloon	chow-line shallowness	jawline show-line	jillion echelon
653	chalumeau shellman	chillum gelometer	schalmei	schlemiel	shalm	shalom
654	chiller sheller	jailer shalier	jelerang shellier	jewelry shellwork	jollier ashlar	shallower
655	jellyleaf shell-less	jellylike shell-like	jollity agilely	shale-oil shillelagh	shallowly	shell-hole
656	geology	shallowish	shellshake	shell-shock	shilly-shally	hagiology
657	Shalako shellycoat	shealing shilling	shellac Shylock	shellcracker age-long	shell-egg gelignite	shelling hatchling
658	jellify shellflower	jellyfish shell-vault	jollify shelver	shelf	shellfire	shellfish
659	chilblain julep	jailbait shallop	jailbird shellapple	jailbreak shellback	jalopy shellproof	jellybread shilpit

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660	cha-chas judges	chiches shojis	chi-chis	cicisbeo	joy-juice	joshes
661	chichituna	jo-jotte	jujitsu	joshed	judged	shushed
662	jejune wishy-washiness	jejunum	Jewishness	shoeshine	Shoshonean	watch-chain
663	judgmatic	judgment	chechem	Chichimec	shisham	hashish-hemp
664	Cheshire	chicharro	josher	judger	Jewish-rye	ash-shower
665	chachalaca	shashlik	wedge-shell	wishywashily		
666	cha-cha-show	judgeship	hashish-ash			
667	cheechako	joshing	judging	Shoshoko	shushing	
668	Jewish-faith	Jehoshaphat	Cheshvan			
669	Chachapoya shoe-shop	chichipate showshop	chichipe	jojoba	jujube	shoe-shaped

670	checks jockeys watchcase	checkstones jokes	checkstrap shakes	cheeks showcase	chicks Ajax	jackass hedgehogs
671	checked gigot sheikdom shoe-shop	checkout jackdaw shocked showshop	chickadee jacket adjective	Chiquito jagged eject	chickweed shakedown Washington	choked shake-out shoe-shaped
672	cheekiness shaken	checkin shake-hand	chicken ashcan	chicanery	gigantic	jackknife
673	chagoma shaggymane	checkmate	Chickamauga	geochemist	jackhammer	jigamaree
674	chagrin checkwork jaywalker witchcraft	chicory choker joker sugar	checker jaguar geography	checkrein jigger age-group	checkroom jogger hitchhiker	checkwriter jayhawker shaker
675	chagul chuckle jocular shekel	checkless gigolo joggle shingle	checkline jackal juggle washcloth	checklist Jekyll jugular wishing-well	chicle jiggle jungle	chocolate jingle shackle
676	checkage hootchy-kootchy	chokecherry	Chukchi	jackshaft	jigjog	ejection
677	checkhook joking	checking shaking	check-key shocking	Chicago ashcake	chugging jacking	jogging hijacking
678	checkoff jackfruit	check-over jugful	check-weave shakefork	chock-full shake-off	choke-full shake-wave	jackfish shockwave
679	cheekbone check-up shakeup	cheekpiece chick-pea	checkback chockablock	checkbite Jacob	checkbook Jacobin	checkpoint shakeproof

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680	chafes jives	chauffeuse shaves	chefs sheaves	chiefs shoves	chives watch-face	geophysics watch-officer
681	chafed shavetail	chiefdom shift	chieftain shoved	cheviot Java-tea	shaft achieved	shaved
682	chaffinch shaven	chiffon ash-oven	chauvinist washfountain	chavender	Javanese	juvenile
683	chief-mate	sheaveman	show-fame	achievement		
684	chafer shaver witch-fire	chaffer shivaree whichever	chaffron shiver Jefferson	chauffeur achiever	chevron ashiver	shofar watch-fire
685	chevalier joyful shuffle	chiefless shaveling ashfall	chiefly sheaflike age-fellow	chivalry shoeflower wishful	javelin shoofly	jovial shovel
686	chief-justice-ship geophagy	chiffchaff	chivage	jewfish	shavish	she-fish
687	chafing jiving	chaffing showfolk	chaffcutter achieving	shaving	shovegroat	shoving
688	chaff-flower	chief-of	shave-off	shove-off		
689	chaffable	chavibetol	achievable	shove-up	watch-fob	

690	chaps gypsy shops	chapstick gypsum showbiz	chips jabs showpiece	chops jobs	chopsuey gyps	chophouse ships
691	chapter ashpit	chopped hashpot	gibbet hotchpot	shipped Egypt	shopped Jupiter	showboat
692	cheapen Japan ashpan	chip-in jawbone hash-pan	Chippendale shabbiness watchband	Chopin sheepnose wishbone	chubbiness sheepnut	chupon shipowner
693	chapman shipment	chapwoman shipmaster	chipmunk shipmate	sheepman shopman	sheepmaster hatch-beam	sheepmint
694	chaparral gypper jeopardize shipyard	chaperon jabber shipper hatch-bar	cheaper jabberwocky shopper washboard	chipper jobber shepherd switchboard	chopper jawbreaker shipwreck	gibber jabiru shipwright
695	chapel shapely ash-blonde	chaplain shoeblack washable	Chablis shoebill sociable	giblets showbill	jobless shibboleth	jubilee shoplifter
696	chippage sheepshearer	chop-cherry sheepwash	chop-chop shipshape	jib-o-jib hodgepodge	she-beech Egyptian	sheepish
697	chapping shipping	chipping shoepack	chopping shopping	jabbing shopkeeper	shaping shebang	sheepgate hedgepig
698	chapfallen sheepfoot	chop-off shopful	shapeful	sheepfaced	shipfitter	sheepfold
699	chapbook sheep-pod	chip-blower shipboard	chip-proof shipbuilder	jibboom hushpuppy	shapable	shape-up

700	cases	causes	cusses	gases	gazes	guesses
	gashouse	kisses	axis	access	excess	excise
	exhaust	exist	ecosystem	success		
701	cassette	cast	caustic	cost	costume	coast
	custard	custom	custody	gastronome	gazette	ghost
	guest	gust	gusto	quest	accident	acoustics
	august	ecstasy	excitement	exit	extend	extra
	Occident	oxidize				
702	casino	cosine	coziness	cousin	caisson	cozen
	cuisine	gesundheit	goosander	gooseneck	keeshond	accent
	acquiescence	eccentric	oxen	waxen		
703	casement	chasm	cosmos	gasometer	gizmo	keysmith
	kismet	axiom	axman	exam	example	exempt
	egoism	quasi-medical				
704	causerie	casserole	gasser	gasworks	gazer	geyser
	gizzard	guesswork	Kaiser	accuser	exercise	exert
	exhort	exorbitant	waxer			
705	cas-elaw	caseless	castle	causal	gaslight	gasoline
	gazelle	guzzle	quasi-legislative	quisling	axle	auxiliary
	exalt	exhilarate	exile	excel	exhale	
706	caseation	gasogene	quasi-judicial	quiz-show	accusation	acquisition
	exchange	exchequer	exigent	oxygen		
707	cascade	casing	cask	cassock	cossack	casket
	gasket	goose-egg	kiosk	kissing	exact	excavate
	exclaim	exclude	excuse	execute	excursion	exquisite
708	cohesive	case-phrase	gas-fired	gasiform	gasify	goosefoot
	gooseflesh	kiss-off	oxfly	oxford	quasi-official	
709	casaba	Casbah	cusp	cuspid	cuspidor	gasp
	gazebo	gooseberry	gospel	gossip	hocuspocus	accept
	except	exhibit	expand	expect	expend	experience
	expire	explain	explore	export	express	exuberant

710	cats co-eds quits	catsup codicil	catastrophe egotism	coats goddess	caddies godson	codes kites
711	cadet godhead acquitted	cathead gutted hog-tied	cathode quietude	cathedral quotidian	cutout quoted	coded acted
712	cadence giddiness kidney	catnap good-natured kitten	catnip goodness aquatint	codeine guidance acquittance	cotton guidon quietness	Gideon kidnap wickedness
713	catamaran Gotham	catamenia academy	catamount hecatomb	gateman	godmother	good-humored
714	cataract cutter quadrant squadron	catarrh gather quadruped skater	catering guitar actor	caterpillar gutter actress	coder kidder equator	cotter quitter scatter
715	catalog cuddle kettle	catholic cutlery acquittal	cattle cutlet schedule	caudle godly Scotland	coddle goodly scuttle	cotillion goodwill
716	cattish kiddy-show	cottage quotation	godchild	good-show	go-dutch	kaddish
717	catcall cuticle kittiwake skating	catacomb cutting acting	catgut gatecrasher aquatic	catechism goody-goody hectic	coeducational Gothic octagon	coating kidding walkie-talkie
718	cadaver get-off scot-free	catafalque go-devil	catfish godfather	codfish khedive	cutoff active	gadfly octave
719	catapult cutup equitable	catbird getup October	catboat giddap octopus	catbrier guidebook	cutback good-bye	cutiepie Good-Hope

720	cancel	cancer	coincidence	conceal	concede	conceit
	concentration	consent	consequence	conserve	consider	consign
	consist	console	conspire	constable	constitution	consul
	consume	counsel	gunhouse	gainsay	gunnysack	queens
	agonize	Kiwanis				
721	candid	candle	candy	cant	canter	condense
	condition	conductor	contact	content	contest	context
	continue	contract	contribute	count	counter	county
	country	gander	gauntlet	keynoter	kind	quandary
	quantity	quaint	account	acquaint		
722	canine	cannon	canon	canyon	conundrum	ganoin
	gunhand	Queen-Anne	quinine			
723	gunman	gunmetal	agnomen	economy	equanimity	ignominy
	kinema	konimeter				
724	canary	canard	caner	cannery	coiner	coowner
	goner	gunner	gainer	ignoramus	skinner	
725	canal	gunlayer	gunline	gunlock	gunwale	kennel
	keenly	queenly	acknowledge	hangnail	signal	
726	coinage	coinsurer	concha	conjecture	conjugal	conjunctive
	conjurer	conscience	conscientious	conscious	gunshot	gun-shy
	kinship	quench	signature			
727	cannikin	conclave	conclude	concoct	concussion	concord
	conk	concrete	concur	congress	connect	conquer
	conquest	congratulate	cunning	eggnog	hakenkreuz	iconoclast
	kink					
728	canvas	confection	confederate	confer	confine	confirm
	confiscate	conflict	conform	confuse	convalesce	convenient
	convention	convert	convey	convict	convince	convoy
	connive	gainful	gonfalon	gunfight	gunfire	queenfish
729	canape	cannibal	canopy	connubial	gunboat	gunplay
	gunpoint	gunpowder	queenbee	ignoble		

730	cameos games	chemist gums	comestible	combs	commas	commissary
731	comedian commodity accommodation	comet commodore	commit commuter	comity gamete	committee gamut	commode hook-method
732	commander communist keyman	commence communicate kimono	comment community acumen	common cowman augment	commons cumin	commune gamin
733	Camembert	commemorate	co-member	chamomile	Agamemnon	echomimia
734	camera gamma-ray	comer aquamarine	commerce	commercial	comrade	cummerbund
735	camel comely accumulate	camellia cumulative	Camelot cumulus	camlet gamely	calmly gimlet	chamelion kiimmel
736	camshaft gumshoe	commeasure	commission	commissure	commotion	cumshaw
737	chemical gimmick	coamings gumming	combing kamikaze	comic kumquat	coming kymograph	gaming
738	camphor comfort	camphene gumfield	camphire gumflower	camouflage keymove	come-off	comfit
739	camp company competence comply gamble	campus compare compete component gambit	combat compass complain composer gambol	combine compassion complete compress gumbo	combo compel complex compromise	compact compensate compliment computer

740	caress	coarse	correspond	corsage	corset	course
	craze	crease	crest	crass	crisis	cross
	crucifix	cruise	crust	crystal	curious	curse
	Christian	Christmas	chrysanthemum	charisma	chorus	garcon
	garrison	grace	grasp	grass	graze	grease
	Greece	grocer	gross	accuracy	across	Aquarius
741	carat	card	cardinal	carrot	cart	cartoon
	cord	cordial	corridor	courtesy	court	coward
	cradle	crate	crater	create	credit	critic
	crowd	crude	curate	curd	curt	curtail
	curtain	curtsy	chord	garden	garter	gird
	grade	grate	gratify	gratis	great	greet
	growth	grid	grit	guard	karate	krater
	keyword	quart	quarter	accord	accurate	egret
	yogurt					
742	careen	carnage	carnation	carnival	chronic	corn
	corner	cornet	cornice	corny	coronation	crane
	crank	crayon	crony	crown	currant	current
	crooner	cranberry	garnet	garnish	grain	granary
	grand	granite	grant	granny	greenhorn	greyhound
	grin	grind	ground	grown	guarantee	groan
	green	kernel	ocarina	quarantine	hanger-on	
743	Carmen	chromium	cram	cramp	cream	cremate
	crime	crimson	crumb	crumble	garment	gram
	grammar	grampus	grim	grime	groom	grumble
	gourmet	gourmand	gremlin	Karma	kermis	Kremlin
	acrimony	aquarium	agreement	sacrament	scream	skirmish
744	carara	career	careworn	cariri	carrier	courier
	coherer	corer	crier	curare	grower	Gruyere
	agrarian					

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745	carryall	careless	carol	coral	corral	crawl
	cruel	curl	cure-all	chorale	garland	garlic
	girl	gorilla	grill	growl	gruel	quarrel
	aquarelle	eggroll	squirrel	scarlet		
746	carriage	carwash	courage	crash	cache	creature
	crouch	crush	crutch	garage	gorge	gorgeous
	gracious	gurge	grudge	kirsch	aggression	accretion
747	carcass	cargo	caricature	character	cork	correct
	crack	crackle	crag	creak	creek	crick
	cricket	croak	crock	crocket	crocodile	crocus
	crook	croquet	co-worker	curriculum	gargle	Greek
	grog	gherkin	aggregate	segregate		
748	carafe	caravan	careful	carryover	carve	corvette
	craft	cravat	crave	crevice	curfew	curve
	graft	graffiti	graph	grave	gravel	gravy
	grief	groove	grove	gruff	aggravate	sacrifice
	scarf					
749	carabao	carbide	carbine	carbon	carboy	carp
	carpet	carry-back	corporal	corporation	corpse	corpulent
	corroborate	crab	crepe	creep	crib	cripple
	crop	croup	curable	curb	chiropodist	chiropractor
	garbage	grab	grape	grapple	grip	group
	grub					

750	calls	calcium	callous	clasp	class	clause
	close	closet	cluster	colossus	cholesterol	gallows
	glass	glaze	glissade	gloss	equalize	oculist
751	cauldron	clad	cleat	clod	cloth	cloud
	cold	collide	colitis	colt	cult	geld
	gilt	glad	glade	glide	glitter	Goliath
	gloat	glut	gold	guild	guilt	kilt
	killed	kilowatt	quilt	equality	occult	wiglet
	escalator					
752	calendar	call-in	clandestine	clean	clench	client
	clinch	clinic	clink	clown	colin	colleen
	colon	cologne	colony	coolness	gallant	gallon
	glance	gland	glean	glen	kiln	ugliness
	esculent					
753	calamity	calumniate	claim	clammy	clamor	clamp
	clemency	climate	climb	clump	clumsy	column
	culminate	glamour	gleam	glimmer	glimpse	gloam
	gloom	glum	acclaim	acclimate	kilometer	schoolmaster
754	caller	calorie	chlorite	claret	clarify	clarinet
	clarion	clear	clergy	clerk	collar	color
	cooler	galore	gallery	glare	glory	glowworm
	killer	Kilroy	eclair	heckler	ocular	ogler
	scholar					
755	Galilean	Galilee	clawless	coalhole	coaloil	glow-lamp
	guileless	keelhaul	echolalia	ukulele	uglily	
756	clash	clutch	collage	collation	college	collision
	coolish	galosh	Gaulish	glacier	goulash	keelage
	killjoy	klatch	ecology	English		
757	calculator	calico	calling	clang	claque	click
	cling	cloak	clock	clog	colic	colleague
	collect	colloquy	cowlick	Gaelic	galling	goalkeeper
	kilogram	kulak	aqualung			
758	cauliflower	clavate	cleave	cleft	clever	cliff
	culvert	clove	gallivant	glove	golf	gulf
	killifish	qualify	oakleaf	skillful		
759	calabash	caliber	calliope	calaboose	calypso	clip
	club	coalpit	collaborator	collapse	culpable	gallop
	glib	glob	gallbladder	globe	goalpost	gullible
	gulp	kelp	kelpie	kleptomaniac	eclipse	equilibrium
	eucalyptus					

760	cages gashes	caches gauges	catches gouges	cautious gushes	coaches gauchos	couches sagacious
761	cached gadget	cashed gashed	caged gushed	coached actuate	couched	cogitate
762	cash-in auction	catch-on equation	caution occasion	cushion hog-chain	kitchen section	action
763	cashmere	catchman	catchment	coachman	accouchement	Scotchman
764	cashier gasher	catcher gouger	catchword gusher	cochairman kosher	cojuror actuary	gauger hung-jury
765	cageling cudgel	cajole actual	casual eggshell	catchall	catch-letter	catchlight
766	cachucha	cage-shaped	co-judge			
767	caoutchouc gashing	cashing gushing	catching hogchoker	coaching	gauging	gouging
768	cageful	catchfly	catchphrase	cochief		
769	cashbook ketchup	cashboy key-job	cashbox	catchpenny	catchpole	catch-up

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770	cakes gags quacks	caucus gangster quickset	coax kicks quickstep	coexist kings	cox quicksand	cooks quicksilver
771	cockatoo cockeyed kingdom	cocotte cooked aquacade	cocowood cookout quick-tempered	cactus coquette quick-witted	cocktail gaycat	cockade go-getter
772	cocaine Cockaigne cognomen	cocona cockney hog-cane	coconut cow-cane	cockiness kuchen	cocoon quickness	cognate cognizance
773	cocomat kingmaker	cogman quagmire	cuckoo-meat hockey-game	cucumber	gingham	kakemono
774	caulker cokery Quaker	cockerel cookery quicker	cockhorse cougar	cockroach gangrene	cocorico go-cart	cograil kicker
775	cackle coequal googol	coagulate coggle kingly	cockle cogwheel quicklime	cocklaird gangland aquaculture	cocklight giggle	cockalorum goggle
776	cockcnafer quackish	cockshut	cockshy	cocksure	coction	cowcatcher
777	caulking kicking	cooking HongKong	cogging quaking	gagging	gawking	go-go-girl
778	cacophony kingfish	cock-feathered	cockfight	cow-calf	cuckooflower	kickoff
779	co-captain cocobolo kickup	cockboat cocopan kakapo	cockpaddle cookbook kingpin	cockpit cuckoo-bee kookaburra	cockup gangplank eggcup	cock-a-hoop kickback

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780	cafes coves	calves coveys	cavascope cuffs	caves gaffs	coffeehouse goofs	coifs kvass
781	caftan coughweed	cavate covet	cavatina gavotte	caveat gift	cavity goofed	coffeeweed aquavit
782	caffeine given	coven give-in	cavendish goofiness	cave-in	coffin	covenant
783	caveman	cavum	coffee-maker			
784	cavern cougher Kafir	caviar gaffer quaver	coiffure giver quiver	coffer gopher hangover	cover goofer	covert governor
785	cavalcade coffeeleaf scaffold	cavalier coeval scuffle	cavalla coffle	cavalry gavel	cavil equivalent	cafe-au-lait wakeful
786	coefficient	hagfish	hogfish	gavage		
787	coughing giving	cavaquinho kaffeeklatsch	caving kavika	cofactors quaffing	goofing equivocal	gaffing
788	coffee-flavored	give-off	goof-off	qui-vive		
789	coffeeberry	coffeebush	coffeepot	cough-up	give-up	goofball

790	cabs capstone keepsake	caboose capsule hiccups	capacity copious	capes copies	capsize cowboys	capstan gaps
791	capital cop-out	capitol cupid	capitulate kept	captain equipped	captivate sickbed	copied
792	cabana cow-pony occupant	cabin cowpuncher hogpen	cabinet Cuban	capon keypunch	coupon gibbon	cow-pen gaboon
793	cabman	capmint	capomo	equipment	kaboom	cubmaster
794	caper copyreader keyboard	caprice copyright kipper	cooper cooper aquapura	cooperate gabardine skipper	copper gabber	copra keeper
795	cabal cobblestone cubbyhole aquaplane	cable cobalt cueball quibble	capelet copilot cupel	capelin couple gable	Capella copula goblet	cobbler cowbell goblin
796	cabbage capture	cabochon capuchin	capacious kibosh	Cape-ash equipage	caption occupation	captious
797	capcase kapok scapegoat	caping keeping walkback	copycat kopeck	copying hockeypuck	cubic hogback	cupcake hokey-pokey
798	Cape-ivy gabfest	cap-flash	capful	cupful	keep-off	cubiform
799	capable cubeb	cap-a-pie cupbearer	capybara kabob	cobweb keep-up	copybook key-pipe	copyboy

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800	faces vases off-season	face-saving vices	fezzes vicissitude	fusses voices	phases half-size	physicist offices
801	facade feast vast avocet	facet festival vest eavesdropper	facedown festoon vestibule heavyset	fast fiesta vestige offset	faucet fist vestry offside	Faust fusty visit offstage
802	face-nail physiognomy off-center	fascinate vasoneurosis waveson	fasten vicenary	fussiness vicinity	fuzziness V-sign	pheasant aficionado
803	facemaking Ophism	faceman	vasomotion	vasomotor	vice-master	effacement
804	facer visor	facework vizier	fossorial officer	fusser	vasoreflex	viceroy
805	faceless fusillade vessel	face-lift fuselage viceless	face-wall foozie voiceless	facility vacillate half-slip	fizzle vaseline half-sole	fossil vassal officeholder
806	facetious	physician	vice-chairman	vice-chamberlain	vice-chancellor	visage
807	facing fussing vice-god	fiasco fuzzing vice-governor	fiscal physic vice-king	fizzing physique effacing	fuscous vesicle	fusing viceconsul
808	face-off voiceful	phosphor effusive	vis-a-vis evasive	visive halfcivilized	Vesuvian	vociferous
809	facepiece vesper	faceplate vice-president	faceup visible	feasible office-boy	fusible offspring	fussbudget

810	fatso footsore votes	fates footstep	feats footstool	feuds photostat	fights vats	fetus vets
811	faded vedette off-duty	fade-out voted	fated phototube	fetid half-tide	fitted half-witted	fuddyduddy heavy-duty
812	fade-in avoidance	fatten evidence	fitness halftone	footnote offtune	photonegative ophidian	photoengraving softness
813	fathom vitamin	fat-man half-dime	fitment halftime	footman offtime	footmark	photomap
814	father fetter footwork aviator	fatter fitter veteran evader	feather fighter veterinarian heavy-water	fedora fodder vetoer	feeder footrest voter	federal footwear after
815	fatal foothill fuddle	fee-tail foothold futile	fettle footlights vital	feudal footlocker vittles	fiddle footloose photoelectric	fidelity footwall half-dollar
816	fattish food-show	fat-chewing photogenic	fat-choy wattage	fetish	fiduciary	footage
817	fading footing	fatigue photograph	feeding voting	feuding vodka	fighting voiding	fitting evading
818	fateful photoflood	faithful photo-offset	fitful vaudeville	footfall votive	photofinishing affidavit	photoflash
819	fatback footpace	fed-up footpath	feedback footprint	football avoidable	footbridge	footpad

820	fancy vainness evince	fence veins heaviness	fiance vines hyphenize	finesse Venus offense	fines venison waviness	fins venous
821	faint find fountain vaunt affinity	fanatic fond fund vend event	fantastic fondu phantom vendetta have-not	fantail found phonetic vent hyphenated	fend foundation vanity vindicate offend	fiend foundry vantage vintage offhand
822	faineant evenhanded	fanion venenate	Fenian	finance	funniness	phoniness
823	funambulist ovenman	funnyman halfnumb	phenomenon	phoneme	venom	euphonium
824	fawner vineyard	finery evener	funeral heavenward	vanward	veneer	venerable
825	fanlight vanilla	final vainly	finely vinyl	finial heavenly	funnel	phenol
826	finch V-engine	finish avenge	Finnish evangel	vanish eventual	Venetian	vengeance
827	fawning Phoenix avionics	feigning phonic avuncular	finagle phonograph V-neck	finicky vanguard evening	fink vanquish euphonic	funicular vinegar
828	fanfare	fun-filled	viniferous	half-inferior	half-nephew	evenfall
829	fanback oven-baked	finable ovenbird	finback	phenobarbital	evenup	heaven-born

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830	fames fumes	fameuse fumous	famous vamoose	famose vomicine	femcee euphemism	foams half-mast
831	famed fumid	foamed vomit	fomites wavemeter	fumatorium half-made	fumed	fumet
832	famine vimana	feminine effeminate	foaminess half-minded	foeman half-monitor	foment half-moon	vehement if-money
833	foam-maker	off-moment	half-memorized			
834	femoral vomer	foamier Ave	femur Maria	foumart ephemeral	fumarole halfmarrow	fumer wavemark
835	family foamlike	fameless fumily	familiar fumulus	famulus half-mile	female	foamless
836	famish	fumagillin	fumagine	fumish	halfmeasure	
837	faming off-mike	femic wave-mechanical	fimicolous	foaming	fumigate	fuming
838	fameflower	fameful	foamflower	save-move	vimful	
839	fimble vimpa	fumble Phomopsis	fimbriate half-implied	vamp	vampire	vambrace

840	farce	fierce	firehouse	first	force	forest
	forsake	forswear	freeze	frost	foursome	foresee
	four-star	forsythia	furious	furze	frustrate	Pharisee
	phrase	various	varsity	veracity	verse	affairs
	avarice	averse	aphorism	aviarist	overseas	
841	farthing	ferret	fertile	firth	ford	fort
	forte	fortify	fortress	fourteen	forty	fraternal
	fraud	freedom	freight	fret	Friday	fright
	froth	fruit	further	furtive	fjord	variety
	vertebra	verdict	vertigo	afford	afraid	avert
	effort	everything	overdo	overdue	overtake	overthrow
	overturn					
842	farina	fairness	fern	foreign	forename	forenoon
	forehand	France	franchise	frantic	friend	front
	frown	furnace	furnish	variance	varnish	veranda
	effront	everyone	overnight			
843	farm	ferment	fireman	firm	form	formal
	foreman	former	formica	formula	forum	frame
	freemason	from	pharmacy	vermicelli	vermin	vermouth
	affirm	evermore				
844	farrier	farer	firearm	fireworks	forearm	forerunner
	forward	freer	friar	fryer	furor	furrier
	everywhere	overhear	over-here	overwork	offerer	wayfarer
	sufferer					
845	fairly	farewell	firelock	firewall	ferrule	forehand
	foreleg	forlorn	frail	Fraulein	freelancer	free-living
	freeloader	freewheeler	frill	frolic	furl	furlong
	furlough	everlasting	overall	overland	overlay	overlook
	overwhelm	virile				
846	ferocious	forage	forge	forejudge	foreshadow	foreshorten
	fourragere	fragile	fresh	frigid	frosh	variation
	veracious	verge	version	virgin	virtue	voracious
	avaricious	average	aversion	overcharge	overshoe	
847	farrago	firecracker	firing	forego	forget	forgive
	fork	fracas	fragrance	fraction	fragment	freak
	freckle	frequent	friction	frigate	frock	frog
	frug	frugal	virago	evergreen	offering	overact
	overcoat	overcome	overgrow	overkill		
848	farfetched	far-flung	far-off	fervor	foreface	forefather
	forefinger	forefoot	forfeit	forfend	forehoof	forefront
	forever	free-for-all	frivolity	verify	verve	effervescent
	overflow	overfreight	overview			
849	fireball	firebox	fireboat	firebrand	firebug	fireplace
	fireplug	firepower	fireproof	forbear	forbid	freebooter
	freeborn	four-poster	variable	verb	everybody	overpay
	overpower	overburden				

850	fallacy	FALSE	felicitate	fail-safe	fleece	fleas
	fogies	philosophy	valise	velocity	avulse	civilize
851	fallout	falter	fault	felt	field	filter
	filth	flat	flatter	fleet	flight	flitter
	float	flood	flotsam	fluid	flute	flutter
	fold	philodendron	philatelist	valet	valid	vault
	violet					
852	feline	felon	fall-in	flank	flannel	flinch
	flint	flaunt	flounce	fluent	flounder	fullness
	philanderer	philanthropist	valence	valentine	valiant	villain
	violence	violin	volunteer	vulnerable	avalanche	
853	fellowman	filament	film	flam	flame	flambeau
	flamboyant	flamenco	flamingo	flimsy	flume	flummery
	folium	fulham	fulminate	vellum	velometer	volume
	velum					
854	failure	flair	flare	filler	flirt	flier
	florin	floor	flour	flower	flurry	follower
	fuller	phalarope	philharmonic	valor	velour	valerian
	volar					
855	fallal	filial	flail	flawless	flyleaf	flywheel
	fuel-oil	full-length	philologist	valueless		
856	filch	flash	flesh	flush	fellowship	fledgling
	fletch	flowage	fly-ash	foliage	foolish	valuation
	village	violation	volition	vulture	affiliation	avulsion
	effulgence	evolution	ovalshaped			
857	falcon	fallguy	falling	felucca	filigree	filing
	flag	flake	flax	flex	flick	fling
	flog	flock	fluke	flying	follicle	following
	fulcrum	phlegmatic	phlox	volcano	vulgar	afflict
	havelock	Ivy-League				
858	fallfish	falloff	filefish	file-off	flavin	flavor
	flivver	fluvial	flyover	fluff	fulfill	fulvous
	valve	velvet	vilify	vulva	effluvium	evolve
	half-life	ivy-leaf				
859	fallible	flab	flap	flip	flop	flub
	fillip	filibuster	Filipino	flyboy	flypaper	followup
	foul-up	fullback	full-blooded	full-blown	valuable	voluble
	volplane	voluptuous	available			

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860	fascist voyages	fatuous vichyssoise	fishes effigies	fish-house officious	vicious savages	vouchsafe
861	fidget fitchet half-sheet	fished fugitive half-shot	fish-day vegetable officiate	fishtail vegetarian offshoot	fishweed vitiate	fishwood vouched
862	fagine Vishnu oafishness	fashion vision officiant	fission aviation offishness	fishiness efficient ovation	fishnet effusion sufficient	fusion evasion
863	fishman	fish-maw	fishmonger	fishmouth	avouchment	
864	feature future savagery	fisher voucher	fishweir voyager	fishworm voyageur	fissure officiary	fudger offshore
865	facial fishwell halfshell	fishhold fudgewheel official	fishily vagile	fishline vetchling	fishoil vigil	fishowl visual
866	fish-joint	vitiation	voyage-charter	fasciation		
867	fetching vouching	fishgig voyaging	fishhawk half-jug	fishhook	fishing	fetching
868	fishfall half-shaved	fish-farming	fishify	fish-flour	fishwife	fish-finder
869	fetch-up fish-bone off-job	fishable fishpond wave-shape	fish-bellied fishpound	fish-blooded fishplate	fishbowl fishy-back	fishberry vouchsafe

870	facsimile fogs vex	fakes folks efficacy	fix fox evokes	figs fucose half-caste	focsle focus offcast	focus vaccine
871	fact fictitious vocative wave-cut	factory fogeater affect	factotum folktale avocado	faggot vacate effect	fagott victim evacuate	fecket victor offcut
872	fecund viscount	fogginess half-cone	afghan safe-conduct	vacant	vagueness	vicuna
873	fagmaster off-camera	figment off-campus	focometer phocomelia	folkmoot	vacuum	offcome
874	faker foggara vicar safeguard	fakir fogger vigor	fiacre foghorn evoker	ficaria folkright half	figure vagrant acre	figwort vaquero safecracker
875	fackeltanz fogle vaguely off-color	faculty fogless vehicle	fecal folkland vocal	feckless folklore aviculture	fickle fugal half-eagle	focal fugleman if-clause
876	faction fogyish avocation	factual folkish efficacious	feature fugacious eviction	fiction vacation	figshell vaguish	foggage vocation
877	fagging evoking	faking havocking	fogging heavy-going	folkcraft offgoing	Viking	half-cock
878	fake-off	feckful	fogfruit	folkfest	halfcovered	ivy-covered
879	fogbank evocable	fogbow half-cap	fogbound huffcap	vagabond safekeeping	vocable	vocabulary

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880	fauvism frvestones half-faced	fives fiefs	fivesome fifes	fivesome vavasor	five-spot vivacity	five-star vivisection
881	fifth half-fed	fifty heavy-footed	fifteen hoof-foot	fiefdom	five-day	vivid
882	favonian avifauna	viewfinder	vivente	vivandier	ivy-vine	Vivien
883	five-man	five-minute	feoffment	half-feminine	vivamente	
884	favor fiferail	favorite five-year	fever foofaraw	feoffor viverrine	fifer half-fare	fiver half-evergreen
885	faveolus half-oval	favilla half-volley	five-leaf half-vowel	fiveling off-flavor	foveal V-value	half-full
886	fifeshire	five-shooter	foveation	vivacious		
887	fifing	five-corners	five-corns	half-vacant	ophiophagous	
888	faviform foveiform	fee-faw-fum vivificate	five-figure vivify	fivefold	five-finger	five-footer
889	fife-player viviparous	five-piece	fivepins	five-ply	fivepence	five-percenter

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890	fibs fubsy	fobs phobias	fibster VIPs	fops ovipositor	phoebes halfpace	Phoebus wave-base
891	vapid half-boot	if-bet half-paid	fibbed hoofbeat	fobbed offbeat	fopped Phi-Beta	half-bat V-bottom
892	Fabian half-pint	viewpoint hoofbound	half-bent	half-bound	halfopen	halfpenny
893	fob-maker	half-beam	off-beam	V-bomb		
894	fabric vapor ivyberry	febrile vibrate offprint	February viper	fiber evaporate	fibber halfbreed	foppery hoofprint
895	fable viable V-belt	fabulous affable V-block	feeble half-blooded safe-pledge	fibula half-bull	fipple ivybells	foible off-balance
896	fabaceous ivy-bush	fob-chain	foppish	half-page	halfpitch	off-pitch
897	fibbing halfback phobic	fobbing half-box vapography	fopping half-pack	ephebic half-pike	halfback hoof-pick	half-baked off-peak
898	fabiform	fob-off	half-paved	half-payoff		
899	fob-pocket	half-bib	half-bob	V-bob		

900	paces poses obsess	passes basis	pauses bosses	pieces buses	Pisces apsis	possess abscess
901	past piston base-hit best abstain upset	paste pistol bast bestow abstract upstairs	pastor post bastard bistro apostrophe upstart	pastry positive baste boast obstacle	peseta postiche beast boisterous obstruct	pest postillion beside bust opposite
902	peasant bison	poison business	baseness absent	basin epicene	bassinet obeisance	bassoon obscene
903	pacemaker besmear opossum	peacemaker besmirch spasm	pessimist besom specimen	baseman bosom	basement aposematic	beseem appeasement
904	pacer buzzer upsurge	passer absorb	piecework absurd	bazaar abuser	bizarre observe	boozer opposer
905	passel absolute	pusillanimous absolve	puzzle apostle	pussywillow epsilon	baseless obsolete	basil upswell
906	passage beseech	pasture besiege	position obsession	possession abscission	posture	bee-sage
907	passkey bascule bisque upswing	passing bask boscage	pesky basket abscond	pussycat bazooka episcopal	base-court bicycle obscure	basic biscuit opposing
908	Pacific abusive	pacifier specific	passive	peaceful	piassava	pussyfoot
909	passable passe-partout possible busybody	passback pass-up baseball upsweep	passband peaceable bespatter	passbook peace-pipe besprinkle	pass-by peace-plan biceps	passport pea-soup busboy

910	patsy	peau-de-soie	pedestal	pedestrian	poetess	pizza
	batsman	bathhouse	bathyscaphe	beads	beside	bodice
	boots	hypothesis	appetizer	obtest		
911	padded	petite	pas-de-deux	potato	beatitude	bedtime
	apathetic	appetite	aptitude	epithet	ebbtide	habitat
	hypothetical	update	uptight	up-to-date		
912	patent	patina	petunia	piton	potency	put-on
	python	badness	badinage	baton	beatnik	bedouin
	between	botany	butane	button	obedient	obtain
	uptown					
913	peatman	pitman	podium	badminton	betimes	bottom
	by-time	abdomen	apothegm	epidemic	epitome	hebdomad
	optimist	September	speedometer			
914	patriot	patron	patter	pattern	petroleum	pewter
	poetry	pother	potter	powder	putter	bather
	batter	battery	bedroom	betray	better	betroth
	bidder	bitter	bother	boudoir	butter	obtrusive
	upturn					
915	paddle	padlock	pedal	pitiless	poodle	pothole
	potluck	puddle	badly	battle	bedlam	beetle
	bodily	bootleg	bottle	butler	aptly	
916	patche	patagium	pedogenic	petition	potage	potash
	potiche	potsherd	potshot	pottage	potty-chair	beautician
	bootjack	habitation	optician			
917	padding	paddock	patting	pedicure	pedigree	petticoat
	petting	pudding	bathing	bedding	bedeck	betting
	biting	boating	bodega	bodkin	bodyguard	boutique
	abdicate	abduct	apothecary	heptagon	optical	uptake
918	pay-TV	pettifogger	pitfall	pitiful	pot-aufeu	potful
	put-off	baitfish	batfish	batfowl	beatify	beautiful
	bedfast	epitaph	spiteful			
919	paid-up	petit-pois	potable	potbelly	potpie	put-up
	bad-boy	batboy	beat-up	bedbug	bedpan	habitable
	hospitable					

920	panacea	pans	pains	pansy	pence	pencil
	pensive	pianist	pincers	pounce	puns	banister
	benzine	bonus	bounce	bunnies	buoyancy	abeyance
	happiness	hypnosis				
921	paint	panda	pant	panther	pendant	penetrate
	penitent	penthouse	pint	point	pond	pontoon
	pound	pundit	band	banter	bayonet	behind
	bend	beneath	bent	beyond	bind	bond
	bonito	bonnet	bound	bounty	bundle	abandon
	append	appoint	hypnotist			
922	panhandle	penance	peninsula	penknife	pennant	pennon
	pinane	pinion	puniness	banana	benign	bonanza
	bunion	open-end	opinion	opponent		
923	panama	Pan-American	panman	penman	penumbra	pinman
	benumb	bonhomie	byname	eponym	open-minded	openmouthed
924	panorama	peignoir	penury	pioneer	banner	beanery
	binary	binaural	boner	abnormal	habanera	open-air
	opener	weaponry				
925	panel	penal	penholder	penniless	penile	penult
	pianola	pinhole	pinwheel	banal	hobnail	openly
	spaniel					
926	panache	panjandrum	pawnshop	penchant	pension	penuche
	pinch	poncho	pongee	punch	pungent	punish
	banish	banjo	banshee	bench	bunch	appanage
	espionage	Spanish	spinach			
927	pancake	pancreas	panic	penguin	pincushion	pink
	pinkeye	Pinkerton	pinko	pinnacle	punctual	punk
	bank	banquet	bunk	bunker	abnegate	happening
	opening					
928	painful	panfish	panfried	pen-friend	pianoforte	pinafore
	pinfall	pinfeather	pinfire	pinfish	pinfold	benefit
	benevolent	bonefish	bonfire	boniface	bowie-knife	
929	panoply	panpipe	panplane	pawnbroker	penny-pincher	penny-plain
	pennyprick	pinball	pinbone	pineapple	pinpoint	pinprick
	pinniped	pinup	bonapartist	bonebreaker	hobnob	open-up

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930	palmist Bahamas	paymaster beams	poems bemuse	pomace bombs	pomes booms	pumice bums
931	pomade bimetal by-motive	poematic biometrics hop-o-my-thumb	pommetty bombed	beamed bomb-out	beamtrawl boomed	behemoth boomtown
932	payment bemoan hypomania	pieman bimanual	pimento bimonthly	pimiento bowman	Pomona abominable	bemman ape-man
933	palm-marten	baum-marten	boom-man			
934	palmer boomerang	Pomeranian bummer	beamer submarine	bemire	bomber	boomer
935	palm-oil boomless	pimelate boomlet	pomelo	pommel	pomologist	pummel
936	bombshell submission	boomage	beamish	pomaceous	pumiceous	bumptious
937	palming booming	pemmican boomkin	pomegranate bumkin	pumex bumming	beaming	bombing
938	palmification opium-fiend	palmful	pamphlet	pemphigus	pomfret	pomiform
939	pampas pompadour bamboozle bumper	pamper pom-pom bimbo	pimp pompon bimeby	pimpernel pump bombard	pimple Bambi bombast	pomp bamboo bumblebee

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940	parasite	parasol	parcel	Paris	parson	perceive
	percentage	persecute	persist	person	persuade	pierce
	porcelain	porous	praise	precede	precise	prescription
	presence	preserve	president	press	prestige	presume
	price	prison	proceed	process	prospect	prosper
	purse	barrister	bracelet	brass	brazen	breast
	breeze	bristle	oppress	hobbyhorse	upper-house	
941	parade	paradise	pardon	part	party	period
	pert	pirate	port	portrait	predict	pretense
	pretext	pretzel	pride	product	protect	protein
	protest	proud	prude	purity	bard	baritone
	barter	beard	berth	bird	birdie	birth
	board	braid	bread	breath	breed	bride
	bridle	British	broad	broth	brother	brutal
	burden	aboard	abroad	apart	apparatus	Opart
	operate					
942	parent	paranoid	paranymph	parentheses	prank	principle
	print	pronoun	prune	barn	barren	baron
	boron	born	brain	bran	branch	brand
	brandy	brawn	brink	bronco	bronze	brown
	burn	abhorrent	apparent	apprehend	apprentice	apron
	auburn					
943	paramount	paramour	preempt	permanent	permeate	permit
	premier	prime	primitive	primo	primrose	prominent
	promise	promote	prompt	pyramid	pyromaniac	barmaid
	brim	broom	barometer	brome		
944	pairer	parer	poorer	prairie	prayer	prerequisite
	prerogative	prewar	prior	priority	prorated	barrier
	bearer	borer	borrower	brewer	briar	uproar

945	parallel	paralyze	parlay	parley	parliament	parlor
	parole	payroll	pearl	peerless	peril	poorly
	prelate	prelude	prologue	prolong	prowl	purloin
	barley	barrel	brawl	braille	beer-hall	broil
	burial	burlesque	burlap	apparel	whippoorwill	April
946	parachute	parchment	parish	peerage	perch	perjury
	Persia	porch	porridge	portion	preach	precious
	prejudice	pressure	project	purchase	purge	barge
	barrage	biracial	birch	breach	breeches	brewage
	bridge	brioche	broach	brochure	brooch	brush
	burgeoning	aborigine	abortion	apparition	apportion	aperture
	appreciate	approach	operation			
947	park	paring	parquet	perk	pork	practice
	pregnant	prick	proclaim	prognosis	program	progress
	proxy	purgatory	purring	bargain	bark	barkeeper
	barracks	barricade	bearing	brag	brake	break
	breakfast	brick	brig	bring	briquette	broker
	brook	burglar	abracadabra	abrogate	approximate	uppercut
948	paraffin	paraphernalia	paraphrase	paravane	parvenu	perfect
	perfidious	perforate	perform	perfume	periphery	pervert
	pervious	powerful	prefab	preface	prefer	prevent
	preview	previous	privacy	profane	profession	proficiency
	profile	profit	proof	prophecy	prove	providence
	province	provision	provocation	purify	barfly	brave
	brief	abbreviate	approve			
949	parable	parapet	parboil	perhaps	peripatetic	perpetrate
	perpetuate	perplex	porpoise	prepare	probable	probate
	problem	prohibit	prop	propeller	propter	proportion
	proposal	proprietor	purple	purpose	barbarian	barber
	barbell	barbecue	bourbon	brob	burp	abrupt
	approbate	appropriation	apropos	hyperbole		

950	palace	pals	place	placebo	placid	plasma
	plaza	plaster	please	plus	police	policy
	polestar	pollster	pulse	ballast	blaspheme	blast
	blaze	blessing	bliss	blister	blizzard	blouse
	bluster	bolster	bluestocking	bylaws	bulls-eye	applause
	hapless	upholstery	hopeless			
951	paladin	pallet	palliative	pelt	pilot	plaid
	plait	plate	platform	platinum	plead	pleat
	plight	plot	poultry	polite	ballad	bald
	ballot	behold	belt	bewilder	billet	bladder
	blade	bleed	bloat	blood	blot	bold
	bolt	build	bulldog	bullet	bulletin	ability
	applaud	uphold	obliterate			
952	paleness	plain	plaint	plane	plank	plankton
	plant	plenty	plunder	pollen	balance	balloon
	baloney	beeline	bland	blank	blanket	blend
	blind	blink	blonde	bluenose	blunt	bologna
	bowline	bullion	by-line	appliance	opulent	upland
953	palmetto	palominos	pall-mall	playmate	plum	plume
	plumb	plumber	plump	polemic	polymer	blame
	blemish	blimp	bloom	bloomers	boloman	bolometer
	aplomb	sublime	supplement			
954	pallor	peeler	pillar	pillory	pliers	player
	plower	plural	polar	puller	bailor	billiards
	blower	boiler	bowler	bullring	bulrush	bulwark
	applier	supplier				
955	palolo	playland	playlet	pool-hall	balalaika	blue-law
	boll-hull	bullwheel				
956	pillage	pleasure	pledge	plush	polish	polution
	belch	bleach	bilge	blotch	bludgeon	bluejay
	blush	bulge	bullish	ablution	abolish	appellation
	apology	applejack	oblige	whiplash	splash	
957	paling	pelican	pilgrim	playing	placard	placate
	plague	plaque	playground	pluck	polecat	polygamy
	polyglot	polygon	bailiwick	balcony	belong	billycock
	black	block	bloke	blowgun	bulk	bull-oak
	bullock	bullying	applicant	appealing	spelling	
958	paleface	palfrey	palaver	pelf	pelvis	pilaf
	playful	plover	pullover	pulverize	bayleaf	bailiff
	belfry	belief	beloved	billfold	bullfinch	bluefish
	bluff	bolivar	boulevard	bullfight	bullfrog	bull-voiced
	bullyhuff	oblivion	uplift			
959	pallbearer	palpitate	pillbox	playboy	playpen	plebe
	pliable	plop	polyp	pulp	pulpit	ballplayer
	bellboy	blip	blooper	blowup	blueberry	bluebird
	bulb	bluebonnet	bullpen	bullpup	bullwhip	bullyboy
	applepie	epileptic				

960	pachisi pushes bushes	patches badges Apaches	peaches beaches	pitches beeches	pouches beaugeste	poaches bitches
961	peashooter bewitched upshot	pitch-dark bowshot habituate	pitchout budget	poached bushed	pushed bush-out	beachhead bush-tea
962	pageant potion apogean	passion pudginess option	patient pushiness	pidgin beechnut	pigeon biogenic	pitch-in bitchiness
963	pajamas bigeminal	pitchman bushman	pushmobile bushmaster	beachman bushhammer	beachmaster boojum	bechamel abashment
964	patchwork badger obituary	pejorative batcher	pitcher bushranger	poacher bushrope	pochard butcher	pusher abjure
965	patchouli bitchily habitual	petulance bushel	pitchhole bushland	pugilist bushleague	bachelor special	Beaujolais speechless
966	beach-shoe	pushchair	habituation			
967	patching beachcomber object	paycheck bushcraft upcheck	pishogue bushgrass upchuck	pitching bush-hook subject	pushing bushing	pushcart abject
968	pitch-faced bushveld	pitchfork by-Jove	pushful upshift	pushover	bashful	bushfighter
969	page-boy pushball bush-pea	patch-up push-button bushpig	peach-pie push-up heebie-jeebies	pishposh bishop hobby-shop	pie-shape bushberry	pitchblende bushboy

970	packs	pax	PX	Picasso	pikes	pigskin
	pixie	pox	Bacchus	backs	bags	bangs
	because	bequest	bogs	bogus	books	box
	boxer	abacus	apex	epics	epoxy	
971	packet	pact	pagoda	picket	pigtail	pocket
	bacteria	beget	bequeath	big-idea	bigot	bucket
	buckwheat	bugout	hepcat	ubiquitous	aspect	spectator
972	pagan	pecan	pecuniary	picayune	pickaninny	piquant
	pignut	bacchanal	bacon	beacon	beckon	begin
	begonia	beguine	big-name	bigness	bugginess	bygone
	epigone	hypochondria				
973	pigment	pygmy	Pygmalion	bakemeat	becalm	become
	bigamy	bigmouth	boogeyman	bookmaker	bookmark	book-match
	bookmobile	buggyman	hippocampus			
974	packer	picaroon	picker	pickerel	piker	poker
	pucker	backer	backward	baker	beggar	beaker
	bigger	biography	bicarbonate	buck-ram	Apocrypha	epicure
	epigram	hypocrite	Hippocratic			
975	peculiar	piccalilli	piccolo	pickle	picklock	backlash
	backlog	bagel	beagle	becloud	biweekly	beguile
	bogland	booklet	buckle	bugle	bugloss	by
	golly	apiculture				
976	package	picture	backwash	baggage	bakeshop	baksheesh
	big-jaw	bocaccio	bookish	bookshelf	bookshop	buckish
	buckjump	buckshot				
977	peacock	packing	pecking	pegging	picking	poking
	backache	background	backing	begging	bigwig	boogie-woogie
	bookcase	bookkeeper	bugging	buckwagon		
978	peccavi	pickoff	pigfish	backfield	backfire	back-formation
	bagful	big-if	bogfern	bugfish		
979	peekaboo	pickup	pickpocket	piggyback	pigpen	backbencher
	backboard	backbone	back-pay	back-up	bagpipe	beekeeper
	bookbinder	bugaboo	bugbane	bugbear	bugbite	buckplate
	buckpot	buck-up	hubcap	upkeep		

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980	paves beefsteak	pavestone beehives	payoffs behaves	peeves obfuscate	puffs obvious	beefs
981	paved beef-tea befuddle	pavid beeftongue behaved	pivot befit abaft	puffed buffet obviate	beefeater buff-tip webfoot	beefheaded buvette
982	pavane buffoon	puffin Epiphany	puffiness hopvine	bovine	bouffant	bowfin
983	pavement	bivium				
984	paver before boyfriend	pay-for befriend buffer	poverty behavior obverse	puffer beverage	beaver biform	beefer bowfront
985	pavilion beeflower bivalve	piffle bee-fly buffalo	baffle befool bufflehead	B-flat bevel hopeful	befall bifold upflow	beefless bivalence
986	pavane	peevish	bifacial	bufagin		
987	paving bivouac	puffing buffcoat	beefcake buffing	beef-cow by-effect	behaving aboveground	bifocals
988	puff-off	puff-fish				
989	puffball buffable	puffbird buffbar	puff-up aboveboard	beefburger	beef-pie	beef-up

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990	papacy PPS	papistry babies	papoose biopsy	pip-squeak bobbysoxer	poppies bobsled	puppies bypass
991	pipe-die babyhood hippopotamus	pipit baptism upbeat	pipette biped	Poppy-Day bobtail	puppet booby-trap	babbitt bypath
992	peppiriess bobbinet	pipe-in hipbone	pippin	popinjay	baboon	bobbin
993	Papiamento	pipeman	bee-balm	baby-minder		
994	paper piper	paprika popery	pauper potpourri	peeper puberty	pepper by-product	peppermint bipartisan
995	papal pipeline publisher bobolink	payable poplin pupil bubble	pebble poplar Pueblo byplay	peephole populace babble apoplexy	people popular Bible	piebald public biplane
996	pea-patch babushka	peep-show babyish	pipage baby-show	popish booby-hatch	poppy-ash	poppy-show
997	payback pipkin	paybook popcorn	pipecoline popgun	pipe-clay poppycock	pipe-key babying	piping bobcat
998	papaver poppyfish	pepful baby-faced	pipefish bobfly	pipeful bubby-flower	pop-off pupfish	popover pupiform
999	pep-up baby-boy	pipable baobab	pipe-band bebop	pipe-opener bubbybush	poppy-bee	pop-up
1000	dioceses	diseases	disuses			

16.2 Picture Words from John D. Cook

I added these to the master tables above. Many come from contemporary science fiction.

Adidas	Aquinas	Bilbo			
Bones	bump	cabasa	Chekhov	Chewbacca	chimp
cobra	Cupid	Darth	dashiki		
Douthat (Ross)	Dudley	Ents	floss		
Frodo	Gimli	Gollum	Gorn	Han Solo	
Harry Mudd	Icarus	Jabba (the) Hutt	jacks	Kirk	Lando
Linus	Lysol				
parrot	Pepsi	phaser	Pluto	possum	
Sauron	sawzall	sax	Scottie	Shelob	shofar
socks	Spock	sweets			
Vader	wheelbarrow				